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HWANG et al.(10) **Pub. No.: US 2025/0282845 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **FUSION PROTEINS COMPRISING
CHIMERIC ANTIGEN RECEPTORS AND
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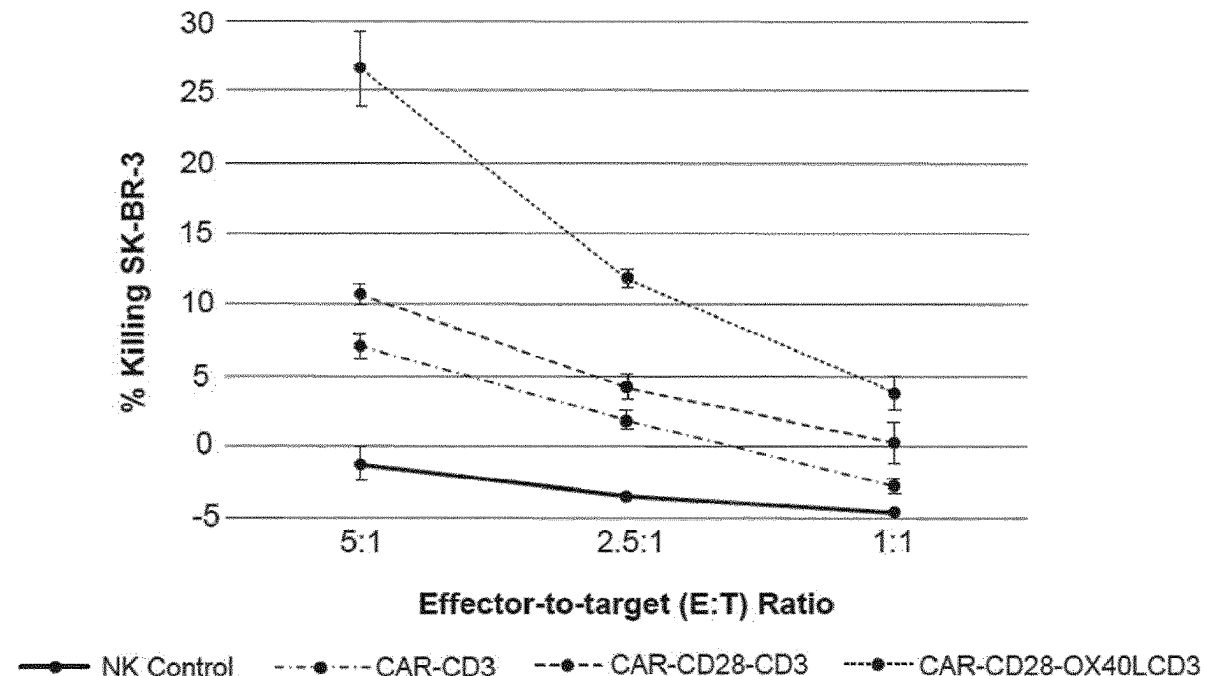
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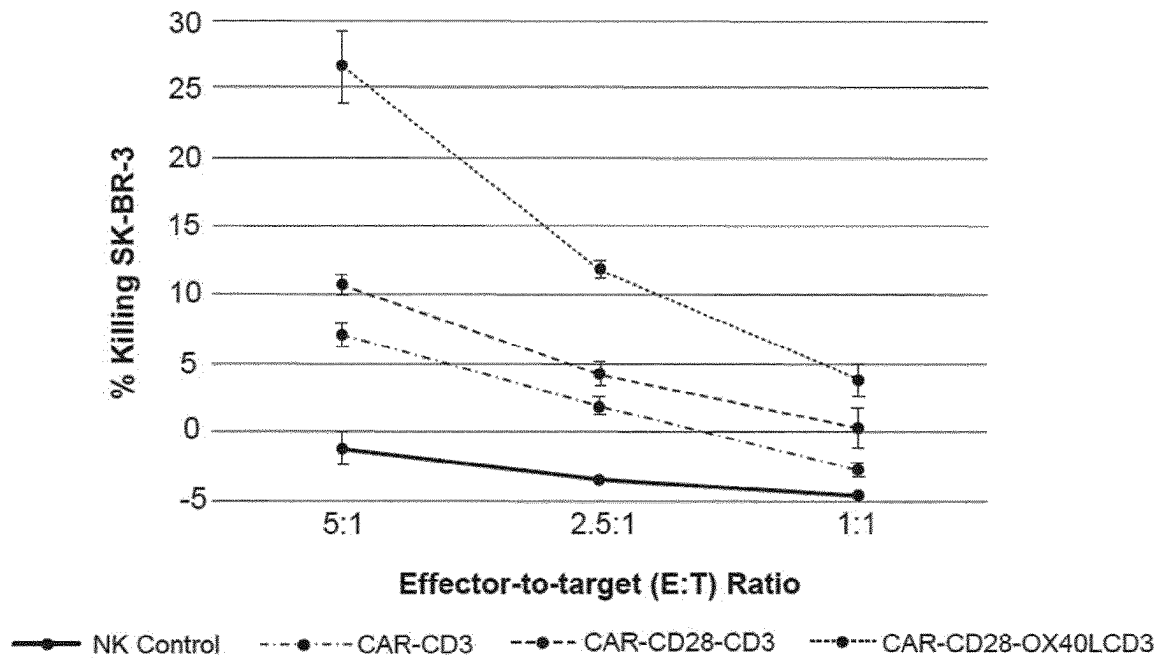
(57)

ABSTRACT

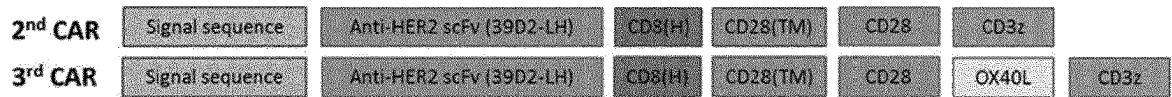
Provided herein, amongst other things, are polynucleotides comprising a) a chimeric antigen receptor (CAR) comprising: an extracellular antigen binding domain comprising an antibody or antigen binding fragment thereof; a transmembrane region; and an intracellular signaling region comprising an OX40L intracellular signaling domain; and b) a nucleic acid encoding an IL-15.

Specification includes a Sequence Listing.

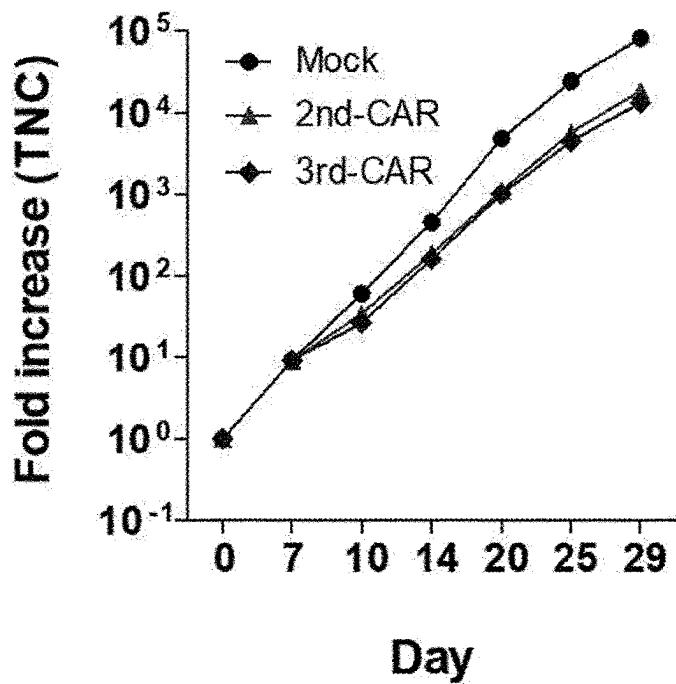
[Fig. 1]



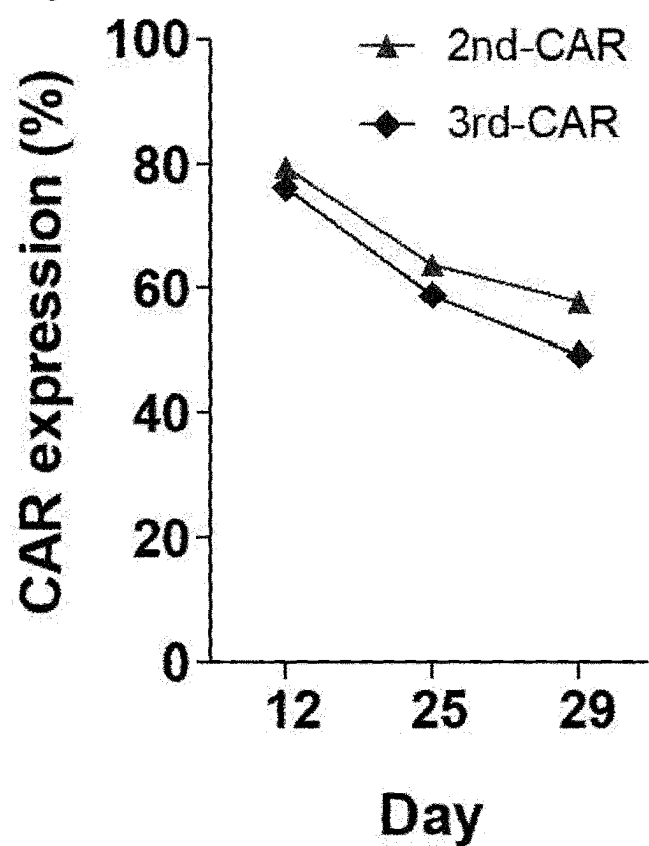
[Fig. 2]



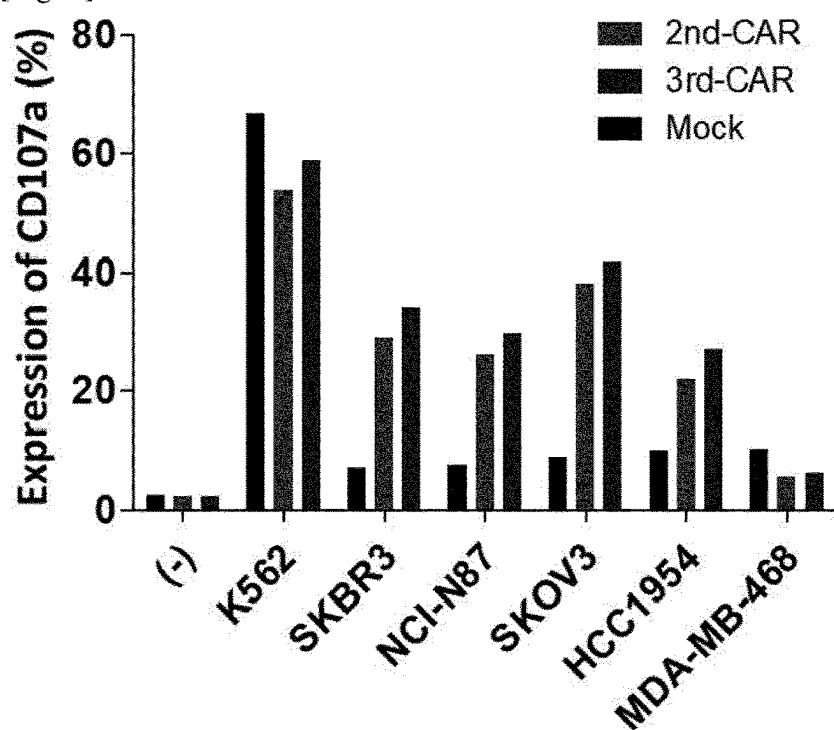
[Fig. 3]



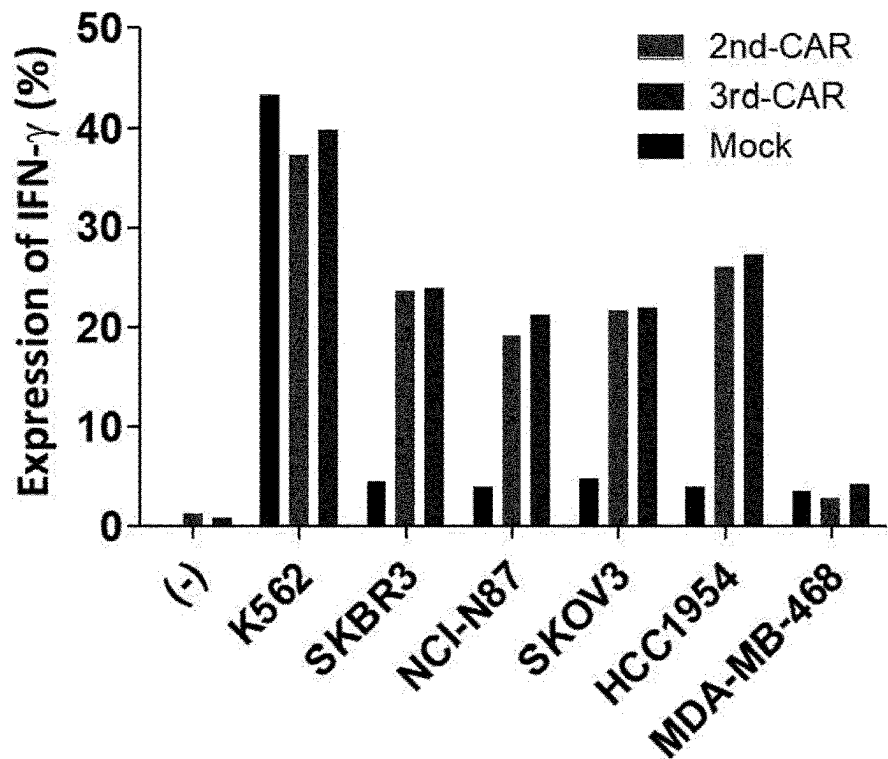
[Fig. 4]



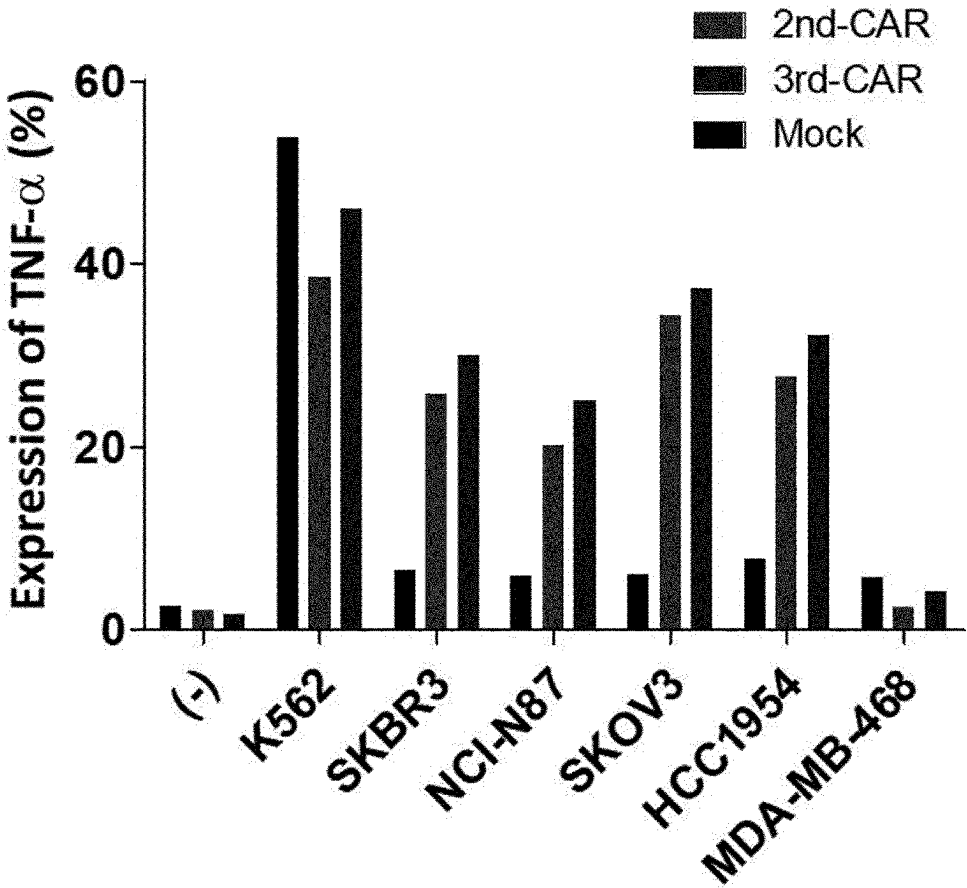
[Fig. 5]



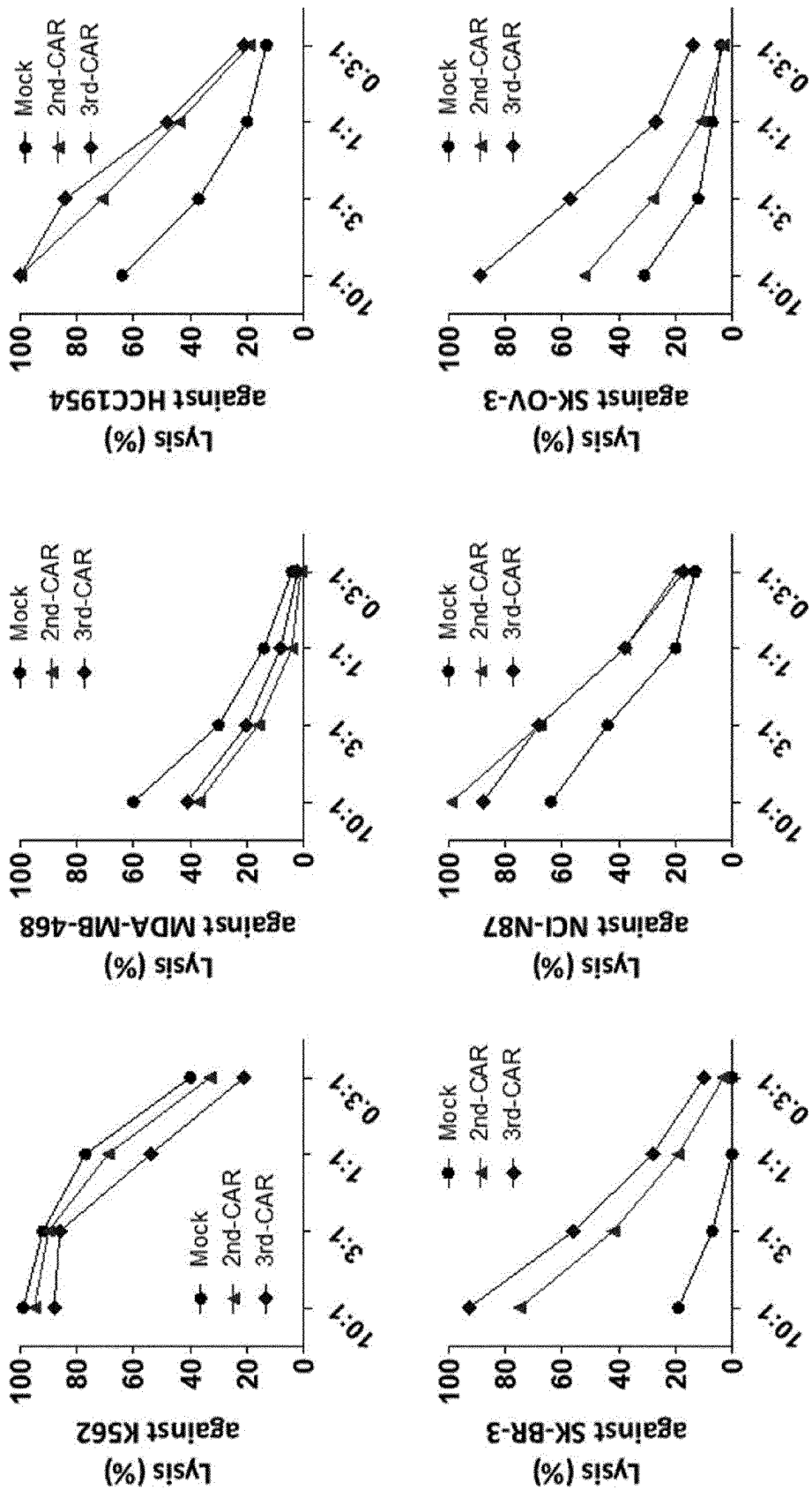
[Fig. 6]



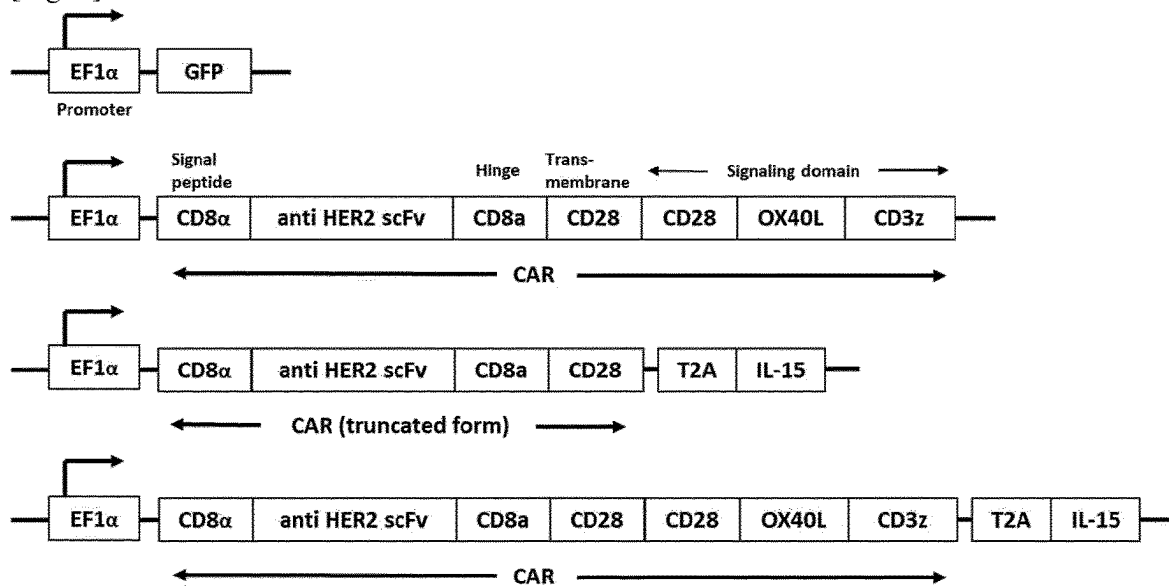
[Fig. 7]



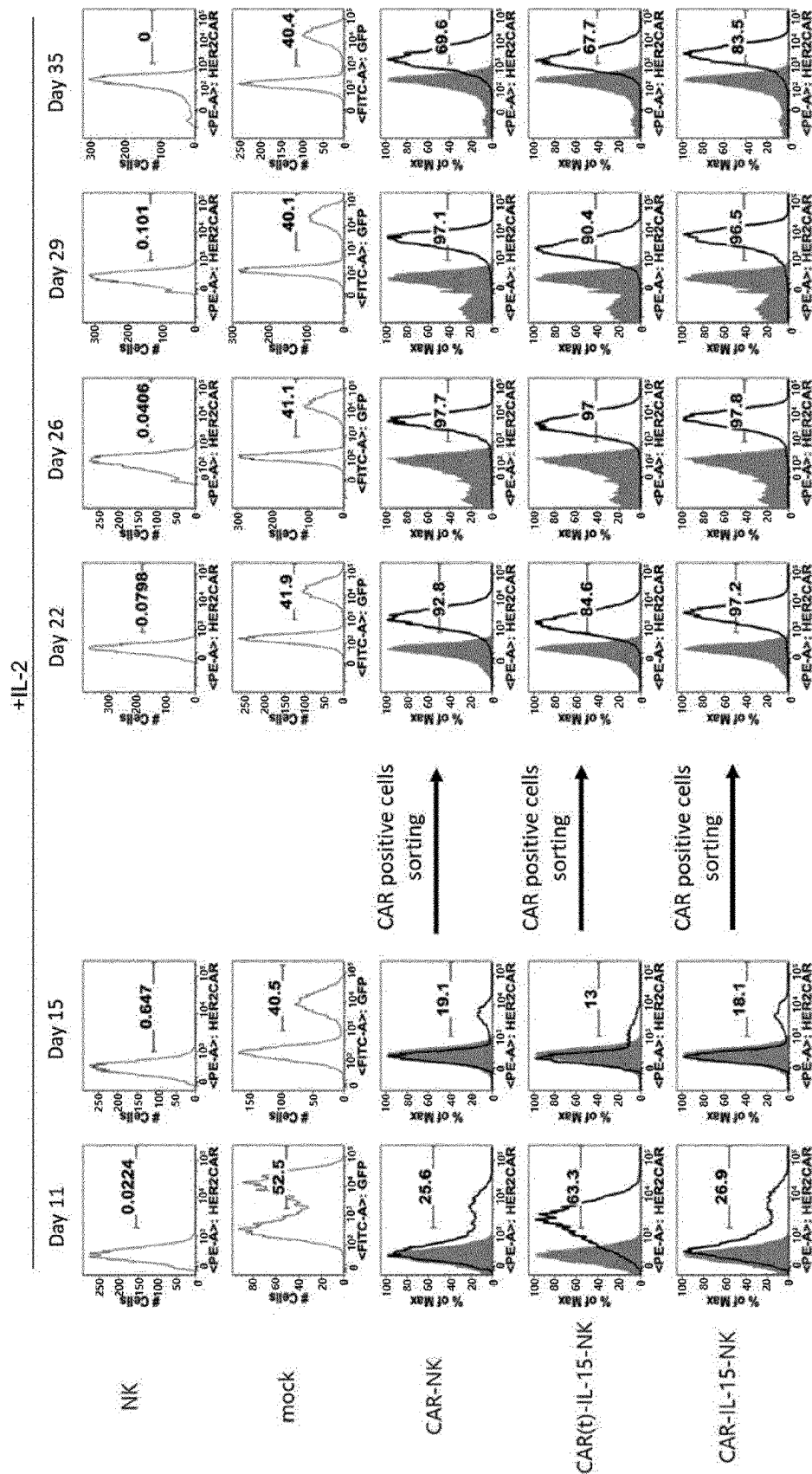
[Fig. 8]



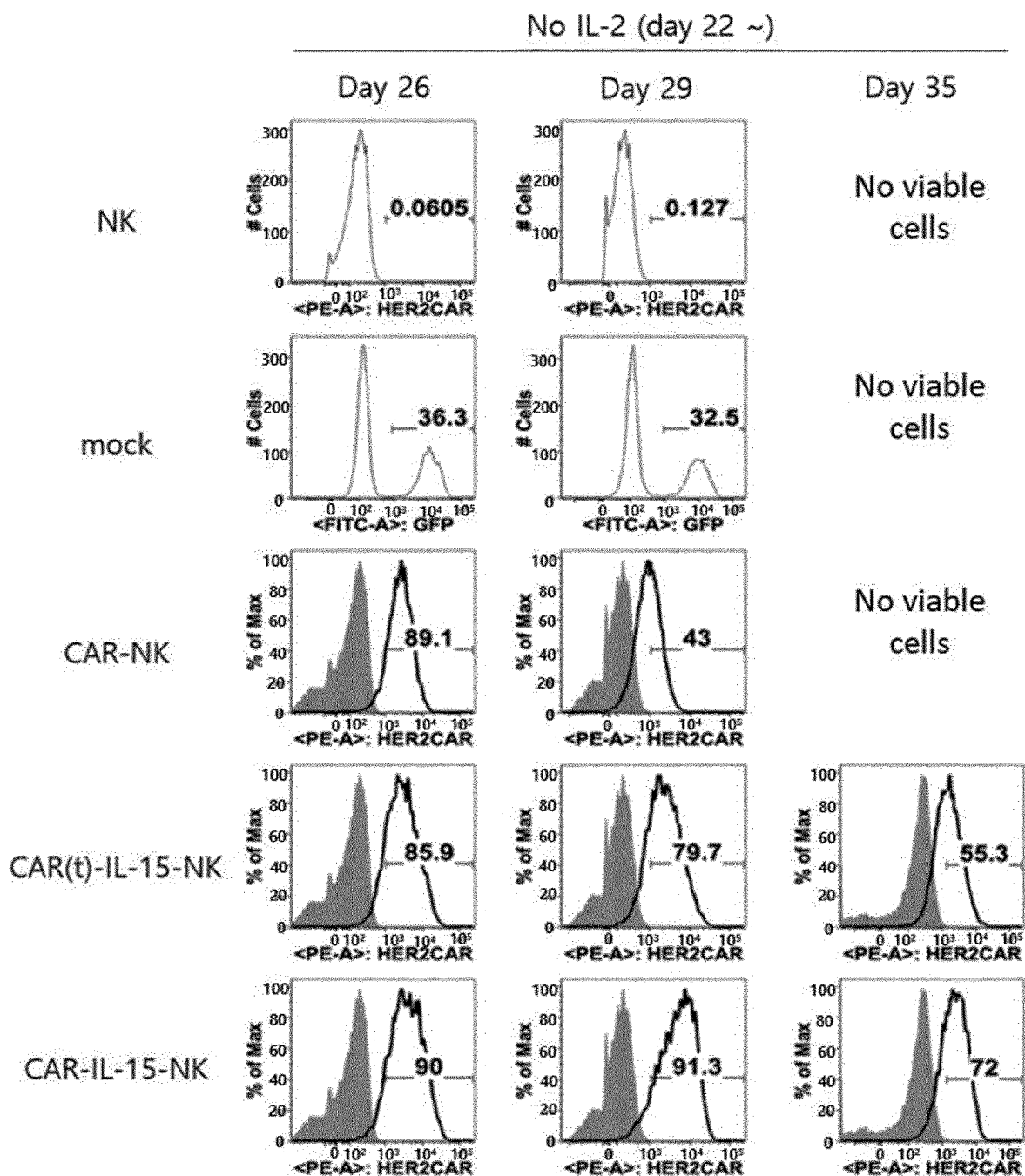
[Fig. 9]



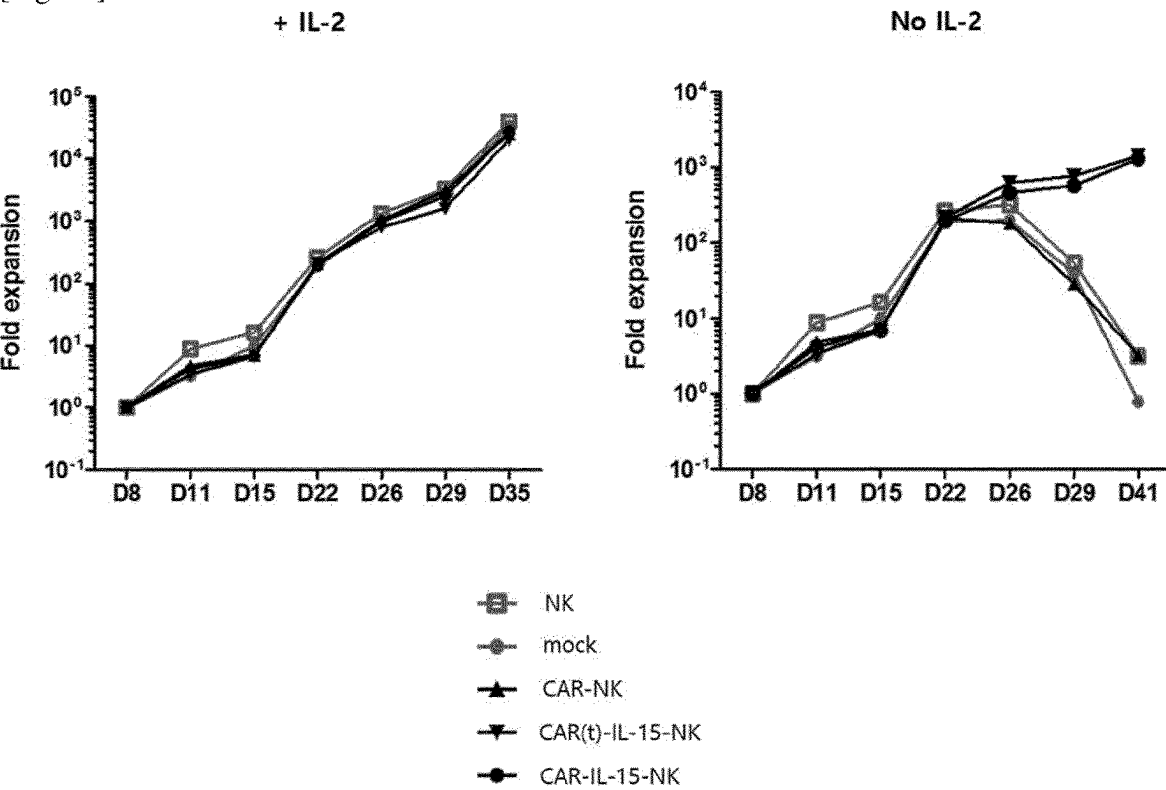
[Fig. 10]



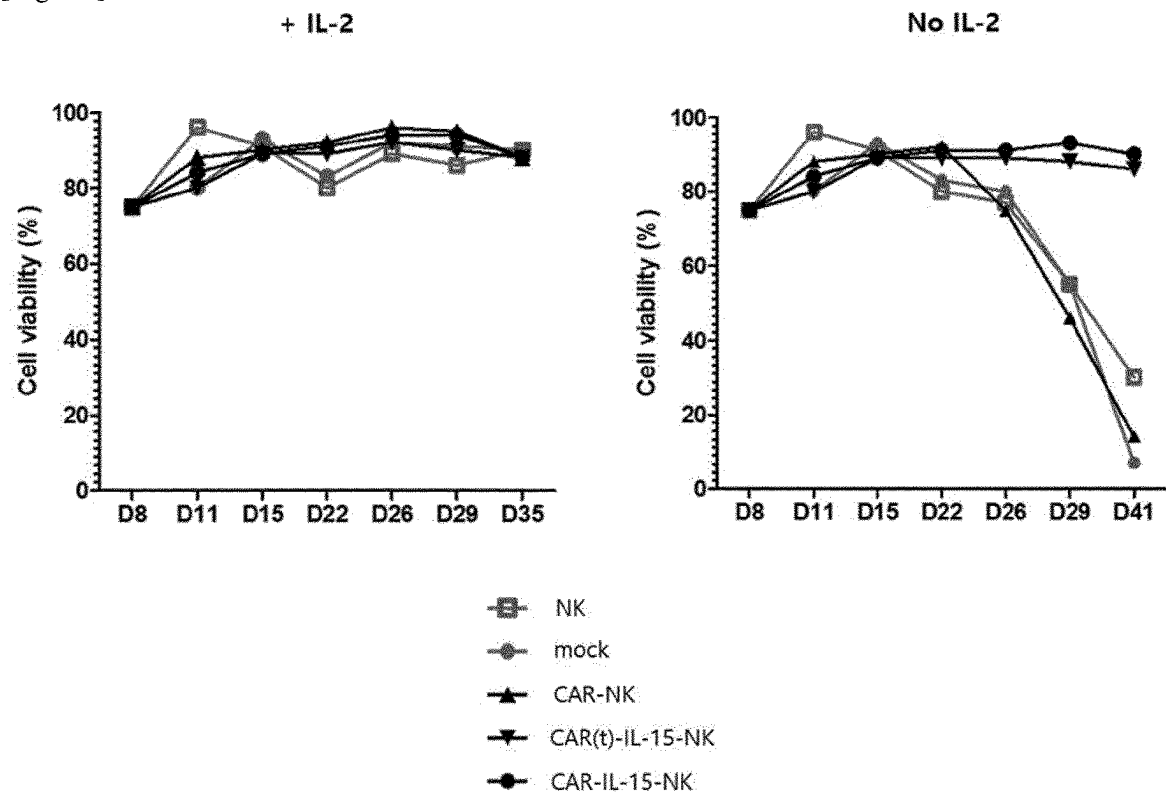
[Fig. 11]



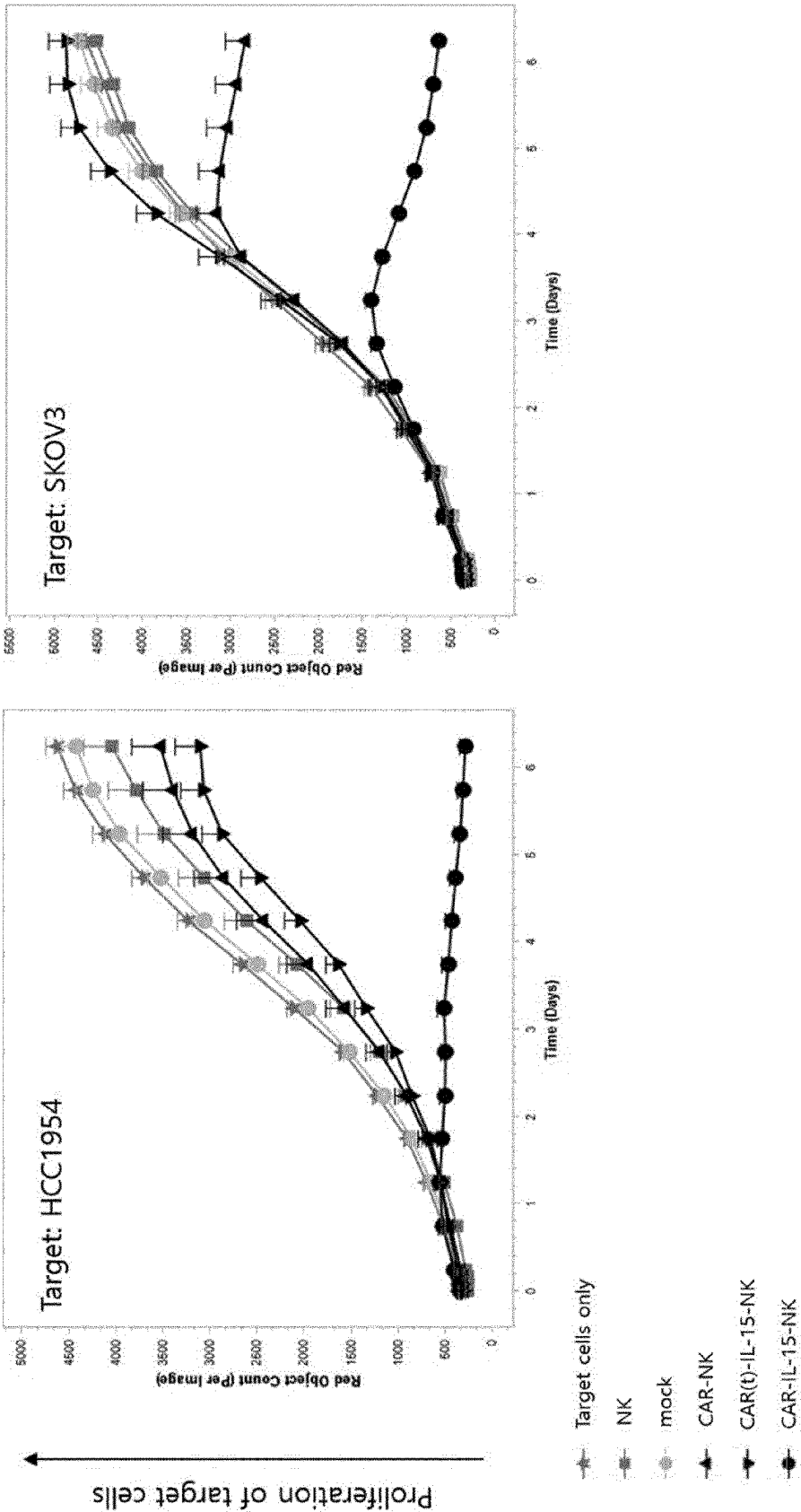
[Fig. 12]



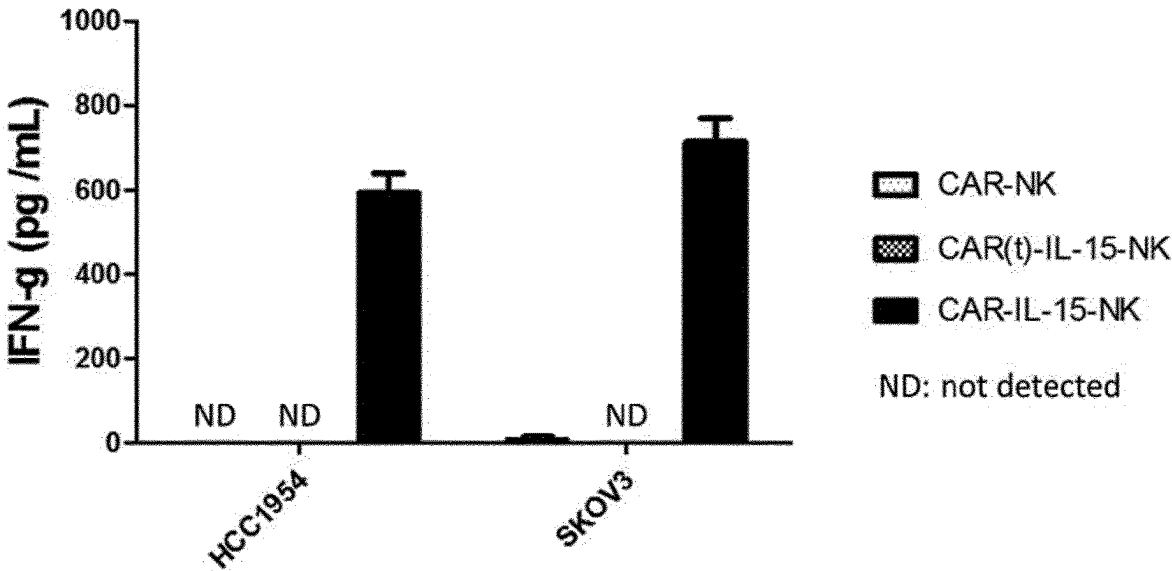
[Fig. 13]



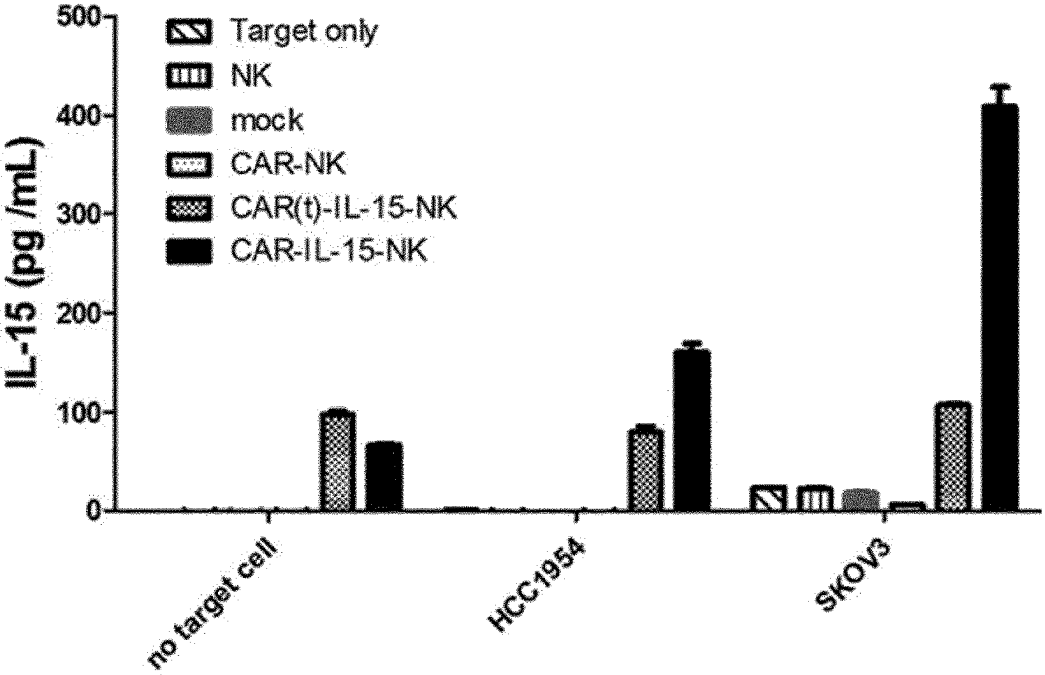
[Fig. 14]



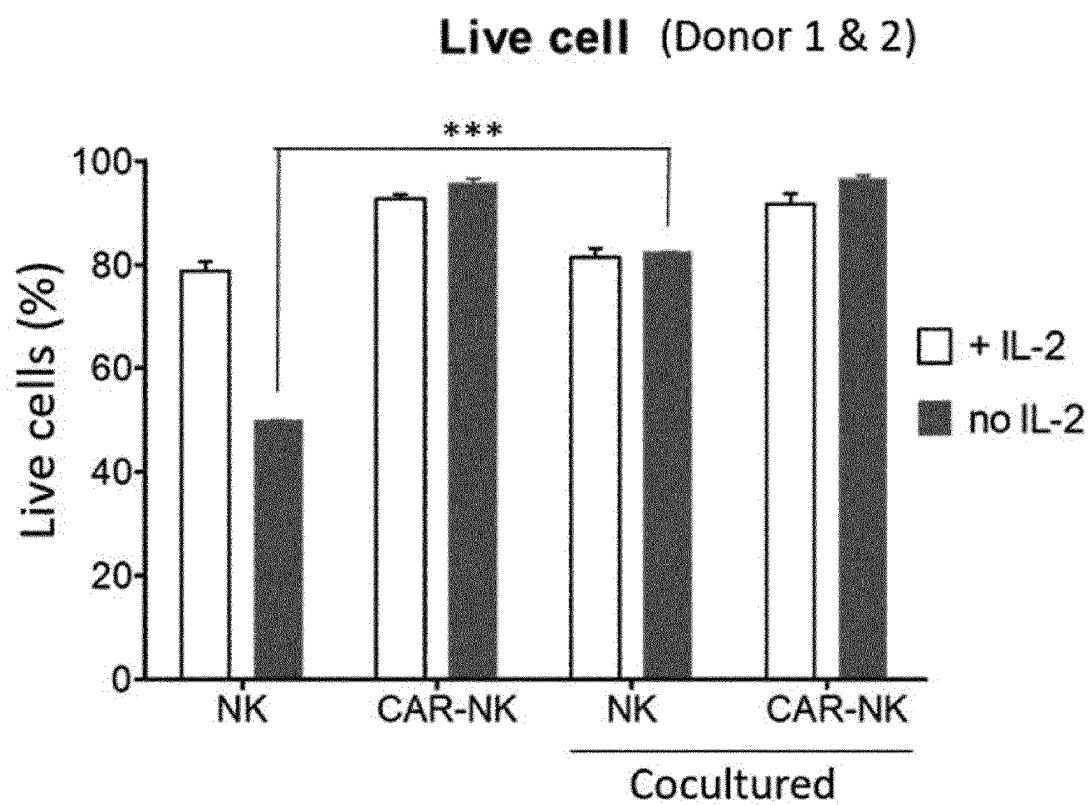
[Fig. 15]



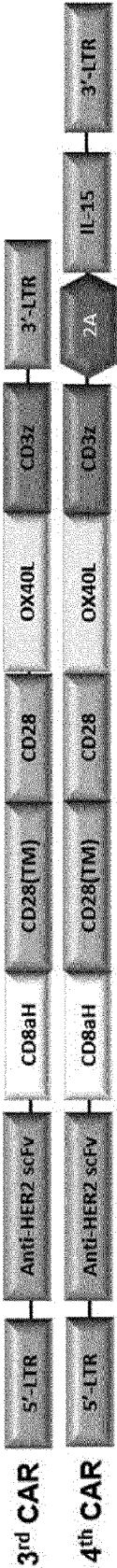
[Fig. 16]



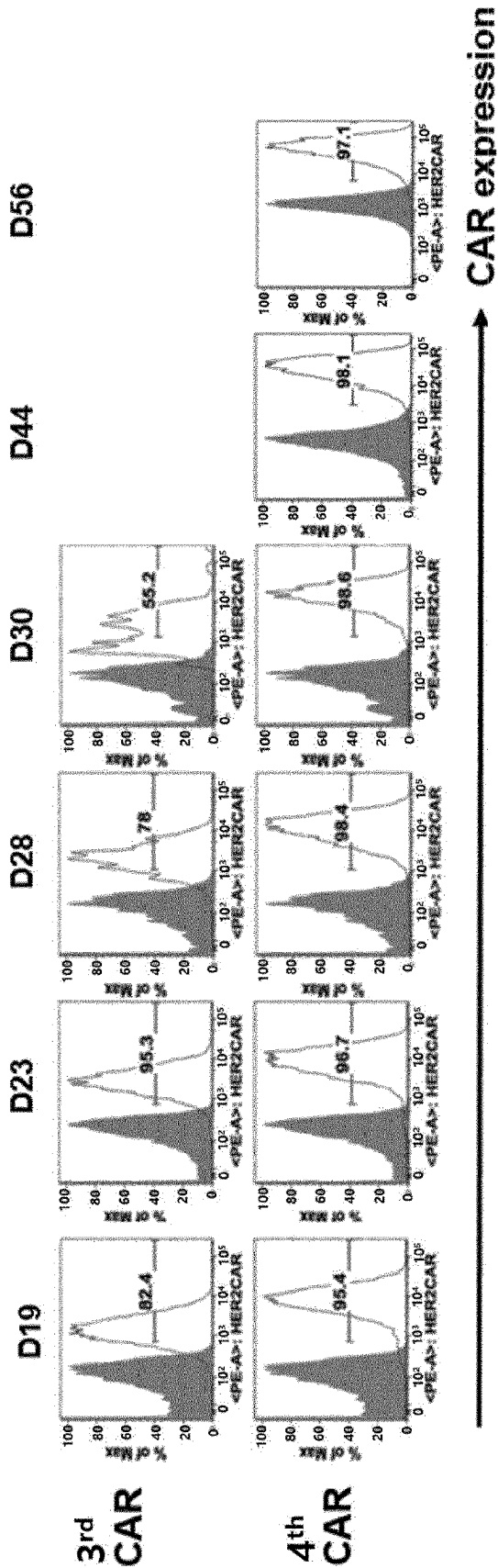
[Fig. 17]



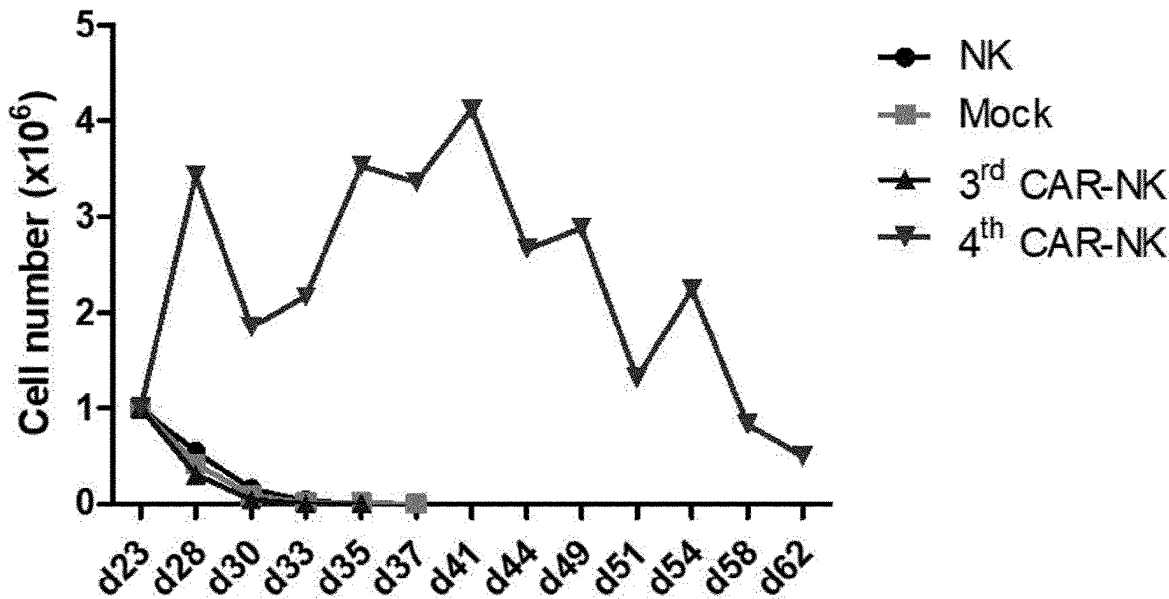
[Fig. 18]



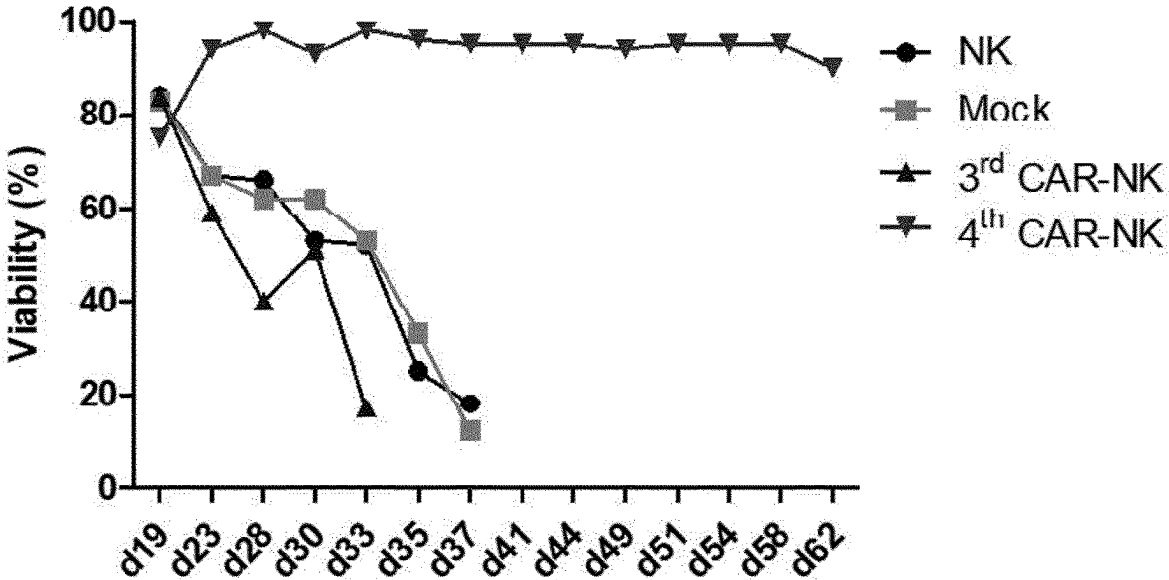
[Fig. 19]



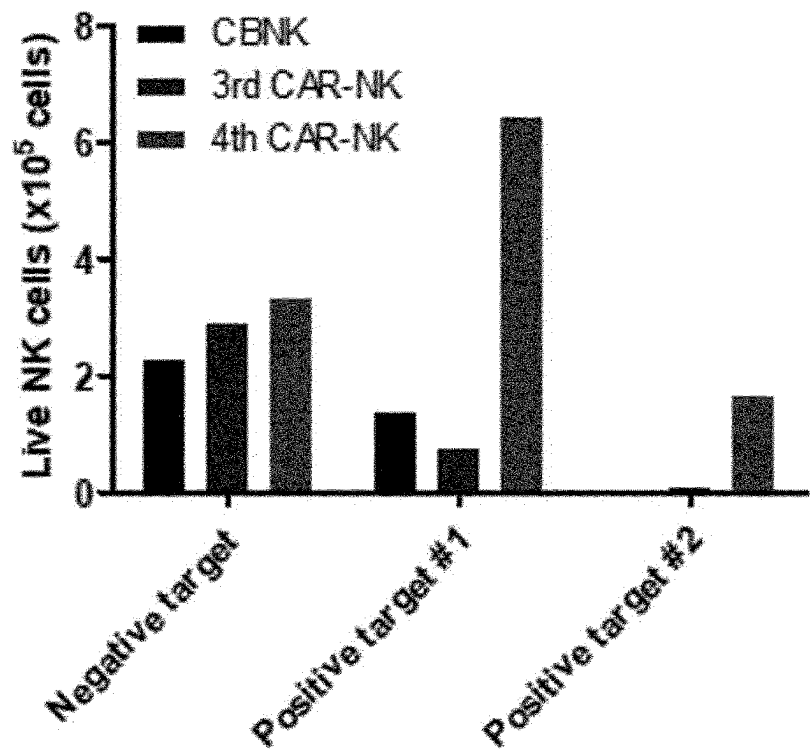
[Fig. 20]



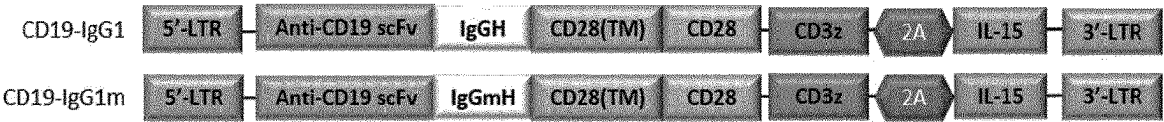
[Fig. 21]



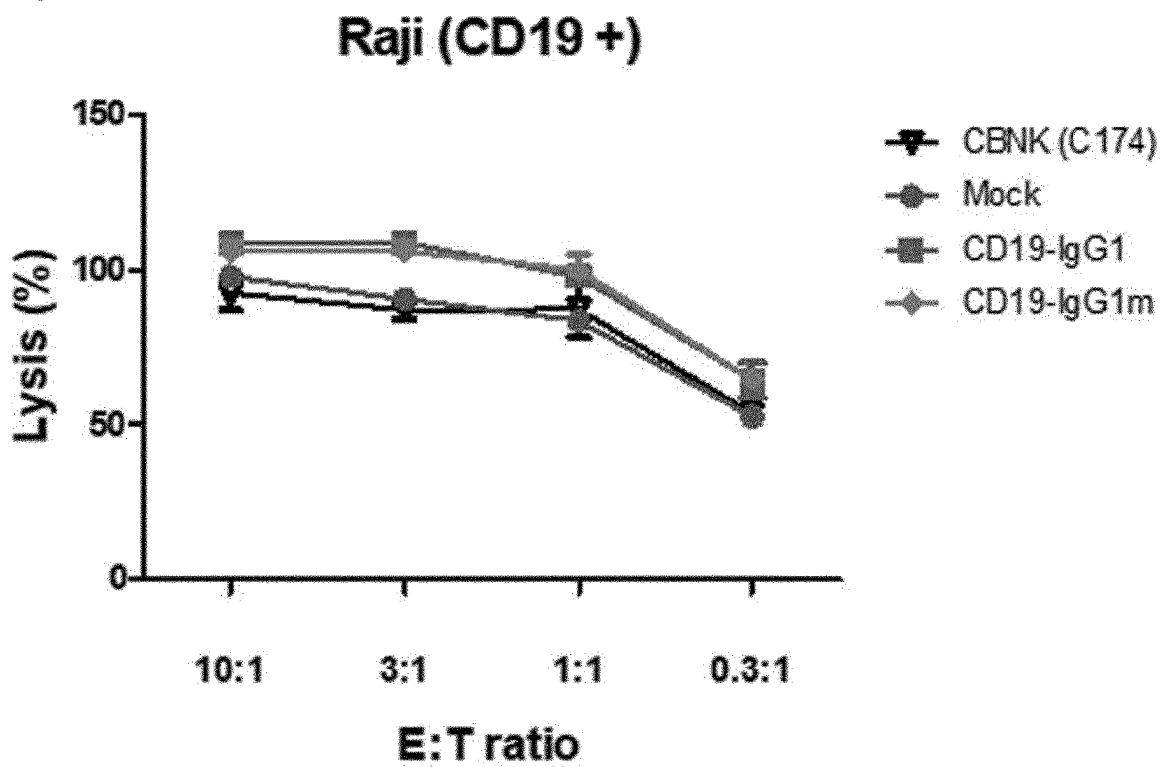
[Fig. 22]



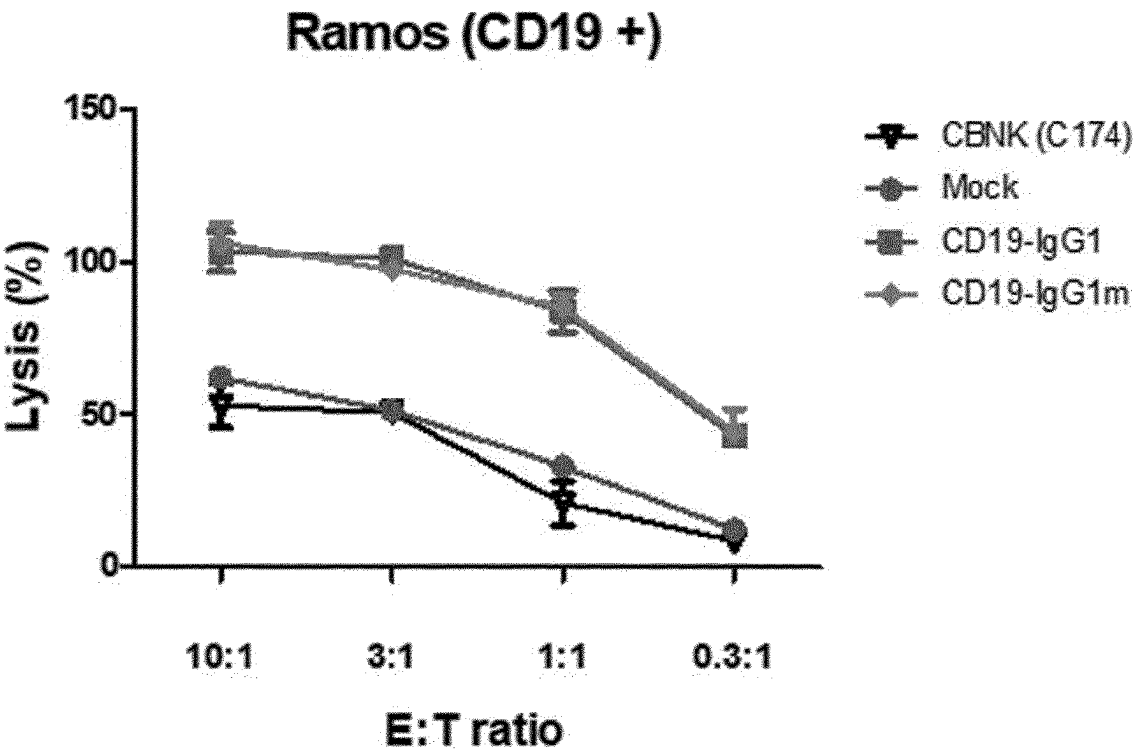
[Fig. 23]



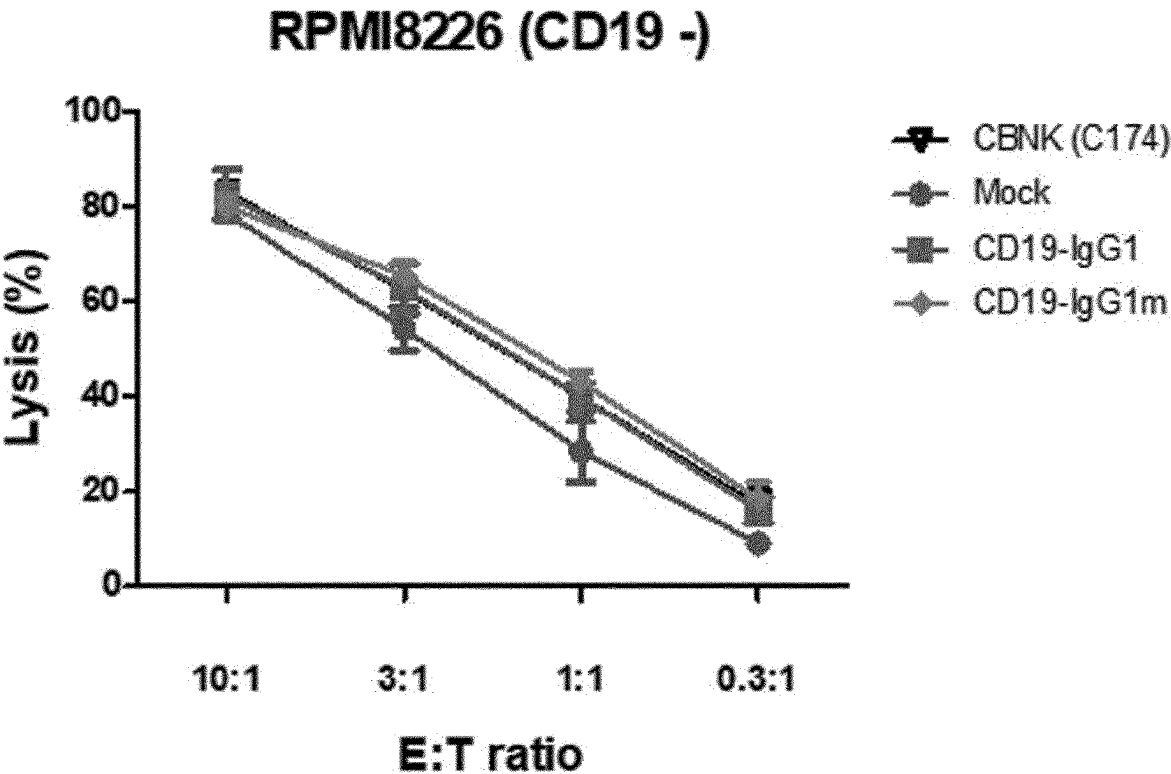
[Fig. 24]



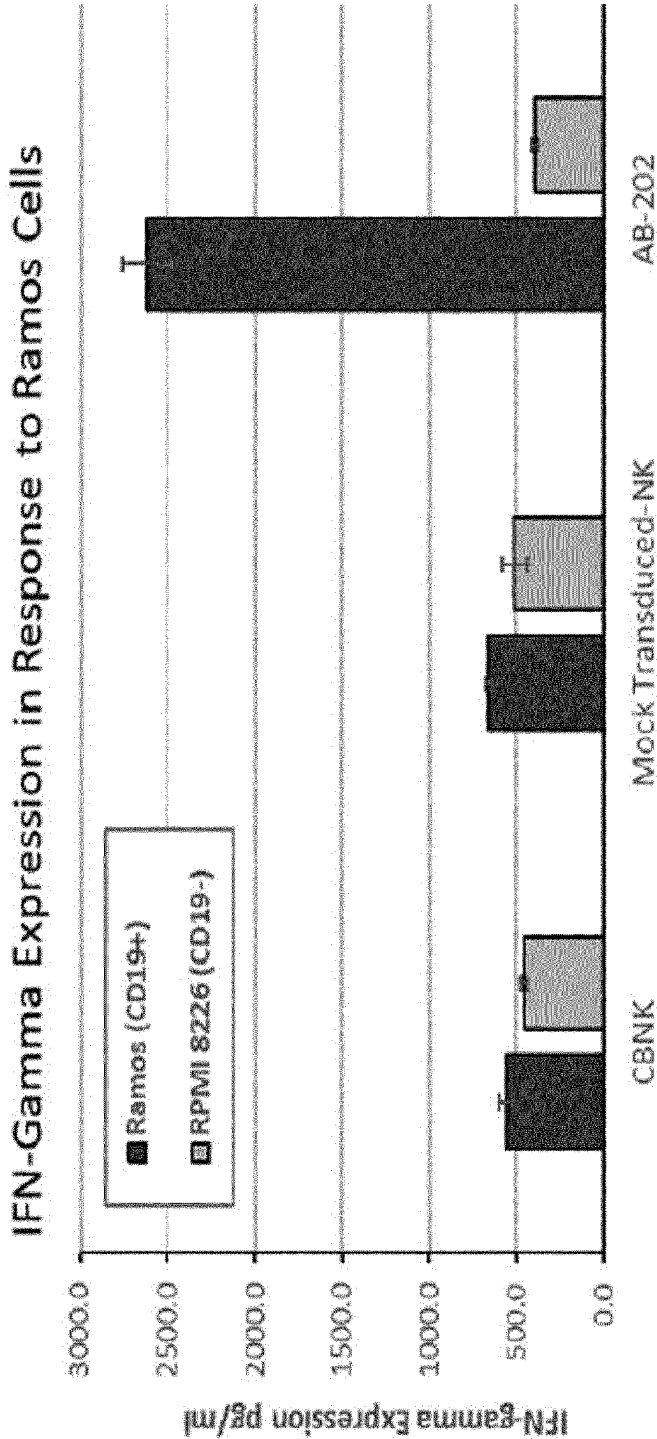
[Fig. 25]



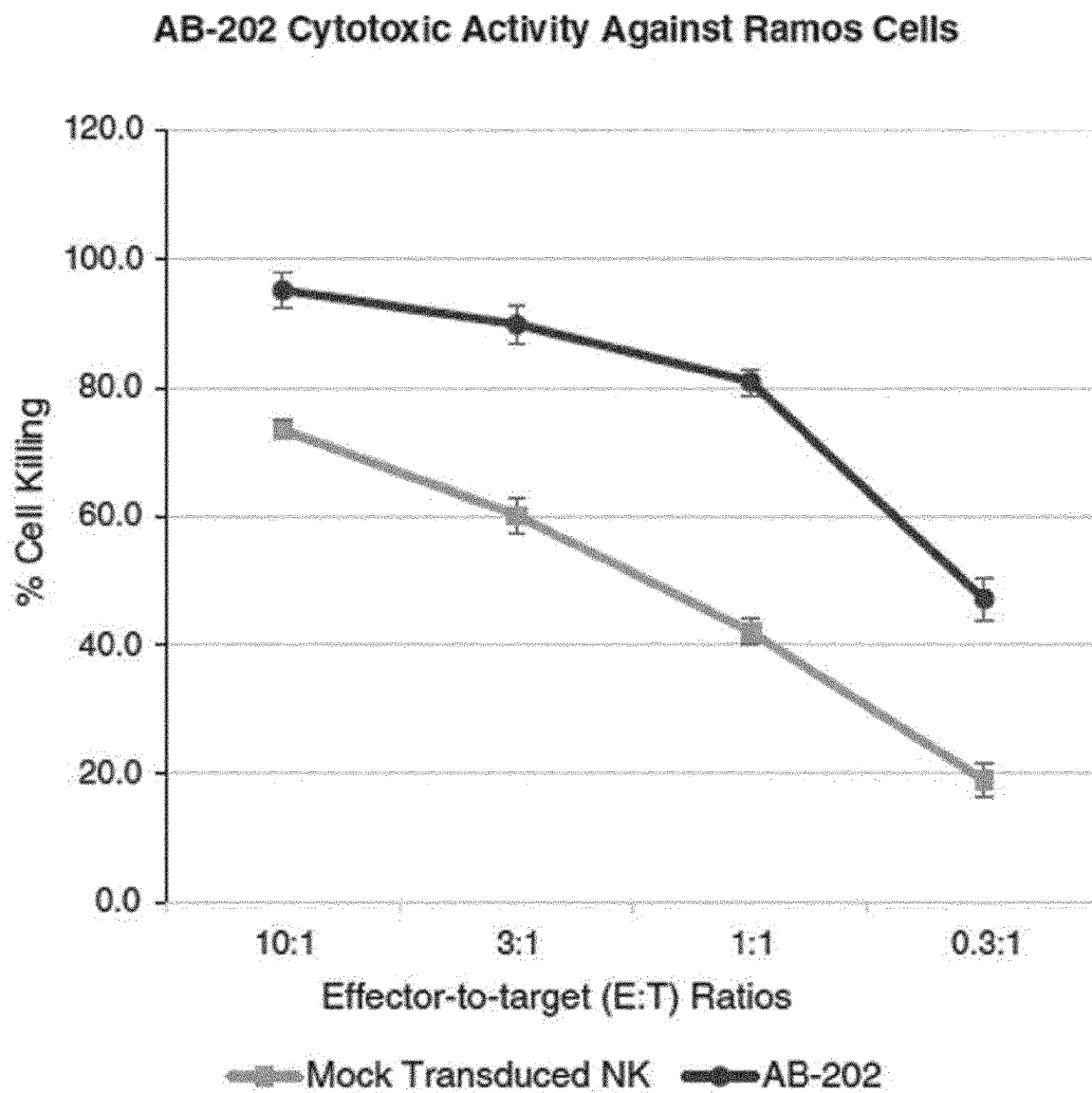
[Fig. 26]



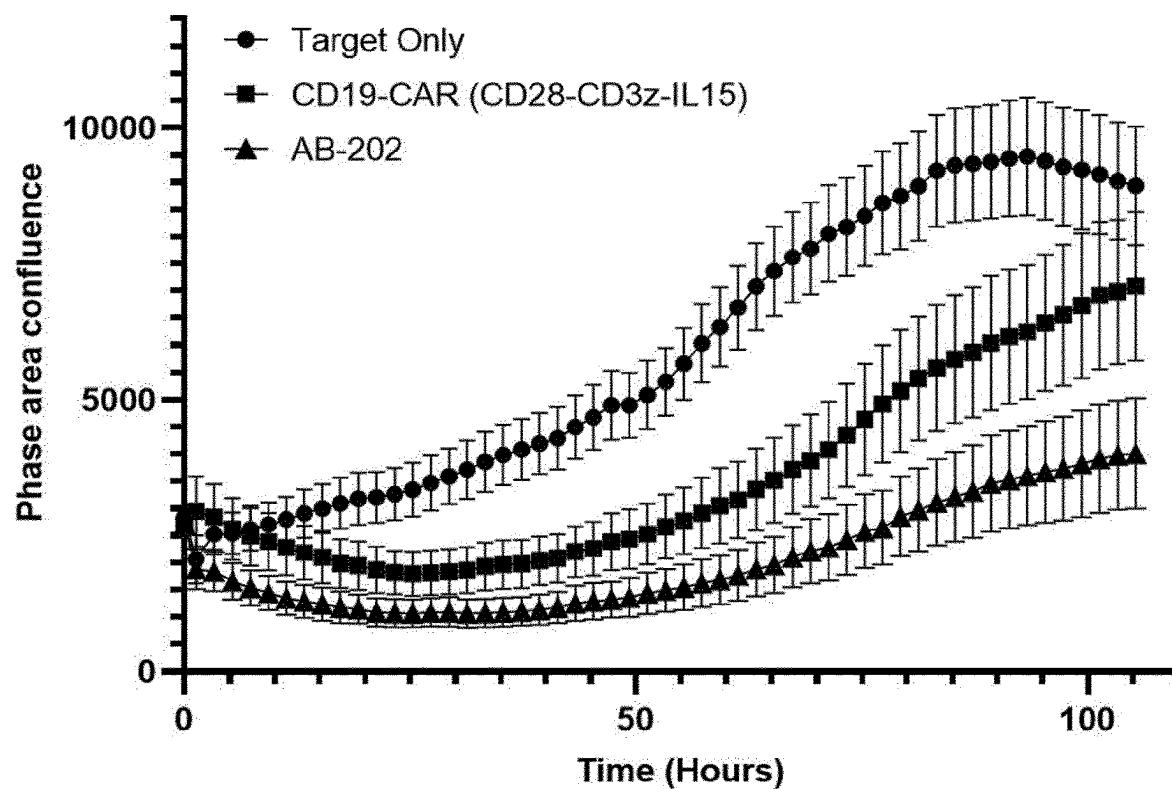
[Fig. 27]



[Fig. 28]



[Fig. 29]



**FUSION PROTEINS COMPRISING
CHIMERIC ANTIGEN RECEPTORS AND
IL-15**

CLAIM OF PRIORITY

[0001] This application is a 371 National Stage application no. PCT/KR2022/095077, filed on Apr. 7, 2022, which application claims the benefit of Korean Patent Application No. 10-2021-0045765, filed on Apr. 8, 2021. The entire contents of the foregoing are incorporated herein by reference.

SEQUENCE LISTING

[0002] This application contains a Sequence Listing that has been submitted electronically as an ASCII text file named 49755-0018US1_SL_ST25.txt. The ASCII text file, created on May 14, 2024, is 94,254 bytes in size. The material in the ASCII text file is hereby incorporated by reference in its entirety.

BACKGROUND

[0003] Targeted therapies, including the use of adoptive cell therapies such as chimeric antigen receptor T cells (CAR Ts), have revolutionized cancer treatment. These cell therapies may be autologous (CAR T cells manufactured using a patient's own T cells) or allogeneic (CAR T cells manufactured using T cells from healthy donors. Currently, there are no autologous CAR T therapies approved to treat HER2-specific cancers.

[0004] Whilst transformative in their ability to treat targeted hematologic cancers, challenging obstacles have arisen in the clinic since 2017, when the first CAR-T therapies targeting CD19 B-cell malignancies were approved by the United States Food and Drug Administration, with the use of autologous CAR-T cell products. CAR T cell manufacturing is a resource-intensive process that can result in failure to produce a viable autologous cell therapy for some patients. The average manufacturing time of 3 weeks that is needed for autologous CAR T cell products may be too long for critically ill patients. Finally, due to the complex nature of the manufacture and delivery of CAR-T cell product, which require close monitoring at top-tier cancer and medical centers, access to this treatment option may be out of reach, both financially and geographically, for most patients. Importantly, even for those patients who have access to this innovative treatment, CAR-T cell products confer a risk of serious and potentially deadly adverse effects. These adverse effects include cytokine release syndrome (CRS) and neurotoxicity, which can be difficult to manage or control.

[0005] Allogeneic CAR-T cell therapies, which utilize cells from healthy donors, may overcome some of the manufacturing and logistical challenges of autologous CAR-T cell therapies. However, these "off-the-shelf" CAR T cell therapies also have issues that include a potentially higher risk of graft-versus-host disease (GVHD) and ineffectiveness due to rapid clearance by the patient's immune system.

[0006] Similar to T cells, allogeneic NK cells engineered to express CARs with anti-tumor activity may provide an important treatment option for cancer patients. NK cells do not suffer from some of the shortcomings of allogeneic CAR-T cells, which often retain expression of endogenous

T cell receptors in addition to engineered chimeric antigen receptors. As a result, allogeneic CAR-NK cell treatments can be administered safely to patients without many of the risks associated with allogeneic T cell therapies, including GVHD. However, CAR-NK cells face many of the same challenges as other allogeneic cell therapies, including product sourcing, scalability, persistence, and dose-to-dose variability. For example, the signaling and activation pathways in NK cells can differ from those of T cells, which may require different combinations of co-stimulatory molecules. In another example, NK cell therapies often require administration of cytokines to enhance the ability of those cells to persist inside of a patient. Thus, there is a need for CAR-NK cells that are engineered to enhance the persistence and cytotoxicity of NK cells.

[0007] The present invention addresses these and other deficiencies in the art.

SUMMARY

[0008] NK cells are immune cells that can engage tumor cells through a complex array of receptors on their cell surface, as well as through antibody-dependent cellular cytotoxicity (ADCC). NK cells may have an advantage over other immune cells, such as the T cells used in CAR-T cell therapy and other cell therapies. Autologous CAR-T cells must be engineered from a patient's own cells. Such engineering can take time, during which period the patient's disease may progress significantly. Such patients may require a bridging therapy to sustain them until their autologous CAR-T cells are ready. Not all patients qualify for autologous CAR-T therapy. For example, some patients may be too sick or may not have sufficient numbers of T cells suitable for engineering purposes. Not all manufacturing runs of autologous CAR-T cells result in sufficient cell numbers or sufficiently active cells to be therapeutically effective. When such manufacturing runs are successful, patients typically only receive a single dose of autologous CAR-T treatment. Because the risk of acute side effects like ICANS and CRS are greatest immediately after administering CAR-T cells, repeat dosing is potentially too risky if the patient will only see marginal benefit from a second, third, or further dose. Additionally, because autologous CAR-T treatments must be unique for each patient, the costs of such treatments can make them unaffordable for many patients who would otherwise benefit from them.

[0009] In an exemplary advantage, NK cells can be used as allogeneic therapies, meaning that NK cells from one donor can be safely used in one or many patients without the requirement for HLA matching, gene editing, or other genetic manipulations. As a result, allogeneic CAR-NK cells can be manufactured in bulk, cryopreserved, shipped throughout the world, and administered on demand at the point of care. Thus, the allogeneic cell therapies can be administered to a patient immediately, without the need to wait for the patient's own cells to be engineered and administered and without the need for a bridging therapy. Because the allogeneic therapies described herein can be manufactured in bulk using campaign-manufacturing methods, the costs associated with manufacturing and delivering the allogeneic therapies described herein has the promise to be significantly lower than those of autologous CAR-T therapies. Campaign manufacturing also reduces variability between batches and allows a patient to receive multiple

doses of CAR-NK cells made from a single batch derived from a single donor where preferable.

[0010] The ability to offer repeat dosing may allow patients to experience or maintain a deeper or prolonged response from the therapy. For example, patients can receive response-based dosing, during which the patient continues to receive doses of CAR-NK cell therapy for as long as the patient derives a benefit. The number of doses and the number of cells administered in each dose can also be tailored to the individual patient. In such cases, the patient is not limited by the number of cells he or she can provide during the cell harvests associated with autologous CAR-T therapy. Thus, the CAR-NK cell therapies described herein can be tailored to each patient based on that patient's own response. In some cases, the therapy can also be reinitiated if the patient relapses.

[0011] Allogeneic NK cells may provide an important treatment option for cancer patients. In one exemplary advantage, NK cells have been well tolerated without evidence of graft-versus-host disease, neurotoxicity or cytokine release syndrome associated with other cell-based therapies. In another exemplary advantage, NK cells do not require prior antigen exposure to antigens to identify and lyse tumor cells. In another exemplary advantage, NK cells have the inherent ability to bridge between innate immunity and engender a multi-clonal adaptive immune response resulting in long-term anticancer immune memory. All of these features contribute to the potential for NK cell efficacy as cancer treatment options.

[0012] For example, NK cells can recruit and activate other components of the immune system. Activated NK cells secrete cytokines and chemokines, such as interferon gamma (IFN γ); tumor necrosis factor alpha (TNF α); and macrophage inflammatory protein 1 (MIP1) that signal and recruit T cells to tumors. Through direct killing of tumor cells, NK cells also expose tumor antigens for recognition by the adaptive immune system.

[0013] Additionally, umbilical cord blood units with preferred characteristics for enhanced clinical activity (e.g., high-affinity CD16 and Killer cell Immunoglobulin-like Receptor (KIR) B-haplotype) can be selected by utilizing a diverse umbilical cord blood bank as a source for NK cells.

[0014] Engineered NK cells, e.g., the CAR-NK cells described herein, have an advantage over autologous cell therapies, e.g., T cells used in CAR-T cell therapy, because the NK cells can be used as allogeneic therapies. Thus, NK cells from one donor can be safely used in one or many patients.

[0015] Provided herein, amongst other things, are polynucleotides comprising a) a chimeric antigen receptor (CAR) comprising: an extracellular antigen binding domain comprising an antibody or antigen binding fragment thereof; a transmembrane region; and an intracellular signaling region comprising an OX40L intracellular signaling domain; and b) a nucleic acid encoding an IL-15.

[0016] Provided herein are polynucleotide(s) comprising: a) a nucleic acid encoding a chimeric antigen receptor (CAR) comprising an intracellular signaling region comprising an OX40L intracellular signaling domain; and b) a nucleic acid encoding an IL-15.

[0017] In some embodiments, the intracellular signaling region comprises a CD28 intracellular signaling domain and a CD3-zeta signaling domain.

[0018] In some embodiments, the OX40L intracellular signaling domain comprises an amino acid sequence set forth in SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.

[0019] In some embodiments, the OX40L intracellular signaling domain comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.

[0020] In some embodiments, the OX40L intracellular signaling domain is encoded by a nucleic acid comprising SEQ ID NO: 11 or SEQ ID NO: 12.

[0021] In some embodiments, the intracellular signaling region further comprises a CD28 intracellular signaling domain.

[0022] In some embodiments, the CD28 intracellular signaling domain comprises an amino acid sequence set forth in SEQ ID NO: 5.

[0023] In some embodiments, the CD28 intracellular signaling domain comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 5.

[0024] In some embodiments, the CD28 intracellular signaling domain is encoded by a nucleic acid comprising SEQ ID NO: 6 or SEQ ID NO: 7.

[0025] In some embodiments, the intracellular signaling region further comprises an CD3-zeta intracellular signaling domain.

[0026] In some embodiments, the CD3-zeta intracellular signaling domain comprises an amino acid sequence set forth in SEQ ID NO: 13.

[0027] In some embodiments, the CD3-zeta intracellular signaling domain comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 13.

[0028] In some embodiments, the CD3-zeta intracellular signaling domain is encoded by a nucleic acid comprising SEQ ID NO: 14 or SEQ ID NO: 15.

[0029] In some embodiments, the IL-15 comprises the amino acid sequence set forth in SEQ ID NO: 22.

[0030] In some embodiments, the IL-15 comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 22.

[0031] In some embodiments, the IL-15 is encoded by a nucleic acid comprising SEQ ID NO: 17 or SEQ ID NO: 18.

[0032] In some embodiments, the polynucleotide encodes a polyprotein comprising the CAR and the IL-15.

[0033] In some embodiments, the polynucleotide further comprises a nucleic acid encoding a self-cleaving peptide.

[0034] In some embodiments, the CAR is joined to the IL-15 by the self-cleaving peptide.

[0035] In some embodiments, the self-cleaving peptide is a T2A self-cleaving peptide.

[0036] In some embodiments, the T2A self-cleaving peptide comprises an amino acid sequence set forth in SEQ ID NO: 14 or SEQ ID NO: 18.

[0037] In some embodiments, the T2A self-cleaving peptide is capable of inducing ribosomal skipping between the CAR and the IL-15.

[0038] In some embodiments, the chimeric antigen receptor further comprises a transmembrane domain.

[0039] In some embodiments, the transmembrane domain is a CD28 transmembrane domain.

[0040] In some embodiments, the transmembrane domain comprises an amino acid sequence set forth in SEQ ID NO: 32.

[0041] In some embodiments, the transmembrane domain comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 32.

[0042] In some embodiments, the transmembrane domain is encoded by a nucleic acid comprising SEQ ID NO: 33 or SEQ ID NO: 34.

[0043] In some embodiments, the chimeric antigen receptor further comprises a hinge domain.

[0044] In some embodiments, the hinge domain is a CD8a hinge domain.

[0045] In some embodiments, the hinge domain comprises an amino acid sequence set forth in SEQ ID NO: 29.

[0046] In some embodiments, the hinge domain comprises an amino acid sequence having at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to SEQ ID NO: 29.

[0047] In some embodiments, the hinge domain is encoded by a nucleic acid comprising SEQ ID NO: 30 or SEQ ID NO: 31.

[0048] In some embodiments, the chimeric antigen receptor further comprises an antigen binding domain.

[0049] In some embodiments, the antigen binding domain comprises an antigen binding fragment.

[0050] In some embodiments, the antigen binding fragment is a single chain Fv (scFv).

[0051] In some embodiments, the antigen binding domain comprises a V_L region and a V_H region.

[0052] In some embodiments, the V_L region is amino terminal to the V_H region.

[0053] In some embodiments, the V_L region is carboxy terminal to the V_H region.

[0054] Also provided herein are vector(s) comprising the polynucleotide(s) described herein.

[0055] In some embodiments, the vector is a viral vector.

[0056] In some embodiments, the viral vector is a retroviral vector or a lentiviral vector.

[0057] Also provided herein are cell(s) comprising the polynucleotide(s) and/or vector(s) described herein.

[0058] Also provided herein are cell(s) expressing the chimeric antigen receptor(s) and the IL-15 encoded by the polynucleotide(s) and/or vector(s) described herein.

[0059] In some embodiments, the cell is a lymphocyte.

[0060] In some embodiments, the lymphocyte is a natural killer (NK) cell.

[0061] In some embodiments, the lymphocyte is a T cell.

[0062] In some embodiments, the cell is a human cell.

[0063] In some embodiments, the cell is a primary cell obtained from a subject.

[0064] In some embodiments, the cell is a primary cell obtained from cord blood.

[0065] In some embodiments, the cell comprises a KIR-B haplotype.

[0066] In some embodiments, the cell express CD16 having the V/V polymorphism at F158.

[0067] Also provided herein are population(s) of cells comprising a plurality of the cells described herein.

[0068] In some embodiments, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%,

98%, or 99% of the cells comprise the polynucleotide(s) and/or vector(s) described herein.

[0069] In some embodiments, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% of the cells express the chimeric antigen receptor and the IL-15 encoded by the polynucleotide(s) and/or vector(s) described herein.

[0070] Also provided herein are pharmaceutical composition(s) comprising the population(s) of cells described herein.

[0071] In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable excipient.

[0072] In some embodiments, the pharmaceutical composition further comprises: (a) human albumin; (b) dextran; (c) glucose; (d) DMSO; and (e) a buffer.

[0073] In some embodiments, the pharmaceutical composition comprises from 30 to 50 mg/mL human albumin.

[0074] In some embodiments, the pharmaceutical composition comprises 50 mg/mL human albumin.

[0075] In some embodiments, the pharmaceutical composition comprises 20 to 30 mg/mL dextran.

[0076] In some embodiments, the pharmaceutical composition comprises 25 mg/mL dextran.

[0077] In some embodiments, the dextran is Dextran 40.

[0078] In some embodiments, the pharmaceutical composition comprises from 12 to 15 mg/mL glucose.

[0079] In some embodiments, the pharmaceutical composition comprises 12.5 mg/mL glucose.

[0080] In some embodiments, the pharmaceutical composition comprises less than 27.5 g/L glucose.

[0081] In some embodiments, the pharmaceutical composition comprises from 50 to 60 mL/mL DMSO.

[0082] In some embodiments, the pharmaceutical composition comprises 55 mg/mL DMSO.

[0083] In some embodiments, the pharmaceutical composition comprises 40 to 60% v/v buffer.

[0084] In some embodiments, the buffer is phosphate buffered saline.

[0085] In some embodiments, the pharmaceutical composition further comprises water.

[0086] In some embodiments, the pharmaceutical composition comprises: (a) about 40 mg/mL human albumin; (b) about 25 mg/mL Dextran 40; (c) about 12.5 mg/mL glucose; (d) about 55 mg/mL DMSO; and (e) about 0.5 mL/mL phosphate buffered saline.

[0087] In some embodiments, the pharmaceutical composition further comprises 0.5 mL/mL water.

[0088] Also provided herein are frozen vial(s) comprising the composition(s) described herein.

[0089] Also provided herein are methods of treatment comprising administering the cell(s), population(s) of cells, and/or composition(s) described herein to a subject having a disease or condition.

[0090] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in

their entirety. In case of conflict, the present specification, including definitions, will control.

[0091] Other features and advantages of the invention will be apparent from the following detailed description and FIGURES, and from the claims.

INCORPORATION BY REFERENCE

[0092] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

[0093] The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which:

[0094] Error! Reference source not found. shows that CAR-NKs comprising a co-stimulatory domain comprising OX40L exhibited greater cytotoxic potential than those without OX40L.

[0095] Error! Reference source not found. shows schematics of CAR constructs.

[0096] Error! Reference source not found. shows proliferation of the CAR constructs of Error! Reference source not found.

[0097] Error! Reference source not found. shows CAR expression of the CAR constructs of Error! Reference source not found.

[0098] Error! Reference source not found. shows CD107a expression of the CAR constructs of Error! Reference source not found. Bars, from left to right: Mock, 2nd-CAR, 3rd-CAR.

[0099] Error! Reference source not found. shows IFN- γ expression of the CAR constructs of Error! Reference source not found. Bars, from left to right: Mock, 2nd-CAR, 3rd-CAR.

[0100] Error! Reference source not found. shows TNF- α expression of the CAR constructs of Error! Reference source not found. Bars, from left to right: Mock, 2nd-CAR, 3rd-CAR.

[0101] Error! Reference source not found. shows short term cytotoxicity of the CAR constructs of Error! Reference source not found.

[0102] Error! Reference source not found. shows schemes of CAR constructs. From top to bottom: mock GFP expressing NK (mock-NK); CAR without IL-15 (CAR-NK); truncated CAR with IL-15 (CAR (t)-IL-15-NK); CAR with IL-15 (CAR-IL-15-NK, AB-201).

[0103] Error! Reference source not found. shows CAR expression on NK cells cultured in the presence of IL-2.

[0104] Error! Reference source not found. shows CAR expression on NK cells cultured in the absence of IL-2.

[0105] Error! Reference source not found. shows proliferation of NK cells cultured in the presence of IL-2 and in the absence of IL-2.

[0106] Error! Reference source not found. shows viability of NK cells cultured in the presence of IL-2 and in the absence of IL-2.

[0107] Error! Reference source not found. shows cytotoxicity of NK cells.

[0108] Error! Reference source not found. shows IFN γ production of NK cells.

[0109] Error! Reference source not found. shows IL-15 production of NK cells.

[0110] Error! Reference source not found. demonstrates that secretion of IL-15 Maintains the Survival of Bystander NK Cells.

[0111] Error! Reference source not found. is a schematic of two different CAR structures.

[0112] Error! Reference source not found. shows CAR expression of the CAR structures shown in Error! Reference source not found. over time (in days).

[0113] Error! Reference source not found. shows survival of cells expressing the CAR structures shown in Error! Reference source not found. (cell numbers).

[0114] Error! Reference source not found. shows survival of cells expressing the CAR structures shown in Error! Reference source not found. (percent viability).

[0115] Error! Reference source not found. shows the number of viable NK cells after 7 days of co-culture of cells expressing the CAR structures shown in Error! Reference source not found. with target cells without IL-2 support. Bars, from left to right: cord blood NK cells (CBNK); 3rd CAR-NK; 4th CAR-NK.

[0116] Error! Reference source not found. is a schematic of two different CAR structures.

[0117] Error! Reference source not found. shows short term cytotoxicity of NK cells expressing the CAR structures shown in Error! Reference source not found. against Raji (CD19+) cells).

[0118] Error! Reference source not found. shows short term cytotoxicity of NK cells expressing the CAR structures shown in Error! Reference source not found. against Ramos (CD19+) cells).

[0119] Error! Reference source not found. shows short term cytotoxicity of NK cells expressing the CAR structures shown in Error! Reference source not found. against RPMI8226 (CD19-) cells).

[0120] Error! Reference source not found. shows AB-202 in Vitro Cytotoxicity.

[0121] Error! Reference source not found. shows AB-202 in Vitro Cytotoxicity.

[0122] Error! Reference source not found. shows AB-202 in Vitro Cytotoxicity.

DETAILED DESCRIPTION

[0123] Provided herein are, amongst other things, Natural Killer (NK) cells, e.g., CAR-NK cells, methods for producing the NK cells, pharmaceutical compositions comprising the NK cells, and methods of treating patients suffering, e.g., from cancer, with the NK cells.

I. EXPANSION AND STIMULATION OF NATURAL KILLER CELLS

[0124] In some embodiments, natural killer cells are expanded and stimulated, e.g., by culturing and stimulation with feeder cells.

[0125] NK cells can be expanded and stimulated as described, for example, in US 2020/0108096 or WO 2020/101361, both of which are incorporated herein by reference in their entirety. Briefly, the source cells can be cultured on

modified HuT-78 (ATCC® TIB-161™) cells that have been engineered to express 4-1BBL, membrane bound IL-21, and a mutant TNF α as described in US 2020/0108096.

[0126] Suitable NK cells can also be expanded and stimulated as described herein.

[0127] In some embodiments, NK cells are expanded and stimulated by a method comprising: (a) providing NK cells, e.g., a composition comprising NK cells, e.g., CD3(−) depleted cells; and (b) culturing in a medium comprising feeder cells and/or stimulation factors, thereby producing a population of expanded and stimulated NK cells.

A. Natural Killer Cell Sources

[0128] In some embodiments, the NK cell source is selected from the group consisting of peripheral blood, peripheral blood lymphocytes (PBLs), peripheral blood mononuclear cells (PBMCs), bone marrow, umbilical cord blood (cord blood), isolated NK cells, NK cells derived from induced pluripotent stem cells, NK cells derived from embryonic stem cells, and combinations thereof.

[0129] In some embodiments, the NK cell source is a single unit of cord blood.

[0130] In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises from or from about 1×10^7 to or to about 1×10^9 total nucleated cells. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises from or from about 1×10^8 to or to about 1.5×10^8 total nucleated cells. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises 1×10^8 total nucleated cells. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises about 1×10^8 total nucleated cells. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises 1×10^9 total nucleated cells. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises about 1×10^9 total nucleated cells.

[0131] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises from about 20% to about 80% CD16+ cells. In some embodiments, the NK cell source, e.g., the cord blood unit, comprises from or from about 20% to or to about 80%, from about 20% to or to about 70%, from about 20% to or to about 60%, from about 20% to or to about 40%, from about 20% to or to about 30%, from about 30% to or to about 80%, from about 30% to or to about 70%, from about 30% to or to about 60%, from about 30% to or to about 50%, from about 30% to or to about 40%, from about 40% to or to about 80%, from about 40% to or to about 70%, from about 40% to or to about 60%, from about 40% to or to about 50%, from about 40% to or to about 30%, from about 50% to or to about 80%, from about 50% to or to about 70%, from about 50% to or to about 60%, from about 60% to or to about 80%, from about 60% to or to about 70%, or from about 70% to or to about 80% CD16+ cells. In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 80% CD16+ cells. Alternately, some NK cell sources may comprise CD16+ cells at a concentration of greater than 80%.

[0132] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% MLG2A+ cells.

[0133] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% NKG2C+ cells.

[0134] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% NKG2D+ cells.

[0135] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% NKp46+ cells.

[0136] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% NKp30+ cells.

[0137] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% DNAM-1+ cells.

[0138] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% NKp44+ cells.

[0139] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% CD25+ cells.

[0140] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% CD62L+ cells.

[0141] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% CD69+ cells.

[0142] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% CXCR3+ cells.

[0143] In some embodiments, the NK cell source, e.g., the cord blood unit, comprises less than 40%, e.g., less than 30%, e.g., less than 20%, e.g., less than 10%, e.g., less than 5% CD57+ cells.

[0144] In some embodiments, NK cells in the NK cell source comprise a KIR B allele of the KIR receptor family. See, e.g., Hsu et al., “The Killer Cell Immunoglobulin-Like Receptor (KIR) Genomic Region: Gene-Order, Haplotypes and Allelic Polymorphism,” *Immunological Review* 190:40-52 (2002); and Pyo et al., “Different Patterns of Evolution in the Centromeric and Telomeric Regions of Group A and B Haplotypes of the Human Killer Cell Ig-like Receptor Locus,” *PLoS One* 5: e15115 (2010).

[0145] In some embodiments, NK cells in the NK cell source comprise the 158 V/V variant of CD16 (i.e. homozygous CD16 15V polymorphism). See, e.g., Koene et al., “Fc γ RIIIa-158V/F Polymorphism Influences the Binding of IgG by Natural Killer Cell Fc γ RIIIa, Independently of the Fc γ RIIIa-48L/R/H Phenotype,” *Blood* 90:1109-14 (1997).

[0146] In some embodiments, NK cells in the cell source comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16.

[0147] In some embodiments, the NK cells in the cell source are not genetically engineered.

[0148] In some embodiments, the NK cells in the cell source do not comprise a CD16 transgene.

[0149] In some embodiments, the NK cells in the cell source do not express an exogenous CD16 protein.

[0150] In some embodiments, the NK cell source is CD3(+) depleted. In some embodiments, the method comprises depleting the NK cell source of CD3(+) cells. In some embodiments, depleting the NK cell source of CD3(+) cells comprises contacting the NK cell source with a CD3 binding antibody or antigen binding fragment thereof. In some embodiments, the CD3 binding antibody or antigen binding fragment thereof is selected from the group consisting of OKT3, UCHT1, and HIT3a, and fragments thereof. In some embodiments, the CD3 binding antibody or antigen binding fragment thereof is OKT3 or an antigen binding fragment thereof. In some embodiments, the antibody or antigen binding fragment thereof is attached to a bead, e.g., a magnetic bead. In some embodiments, the depleting the composition of CD3(+) cells comprises contacting the composition with a CD3 targeting antibody or antigen binding fragment thereof attached to a bead and removing the bead-bound CD3(+) cells from the composition. The composition can be depleted of CD3 cells by immunomagnetic selection, for example, using a CliniMACS T cell depletion set ((LS Depletion set (162-01) Miltenyi Biotec).

[0151] In some embodiments, the NK cell source CD56+ enriched, e.g., by gating on CD56 expression.

[0152] In some embodiments, the NK cell source is both CD56+ enriched and CD3(+) depleted, e.g., by selecting for cells with CD56+CD3- expression.

[0153] In some embodiments, the NK cell source comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and is + enriched and CD3(+) depleted, e.g., by selecting for cells with CD56+CD3- expression.

B. Feeder Cells

[0154] Disclosed herein are feeder cells for the expansion of NK cells. These feeder cells advantageously allow NK cells to expand to numbers suitable for the preparation of a pharmaceutical composition as discussed herein. In some cases, the feeder cells allow the expansion of NK cells without the loss of CD16 expression, which often accompanies cell expansion on other types of feeder cells or using other methods. In some cases, the feeder cells make the expanded NK cells more permissive to freezing such that a higher proportion of NK cells remain viable after a freeze/thaw cycle or such that the cells remain viable for longer periods of time while frozen. In some cases, the feeder cells allow the NK cells to retain high levels of cytotoxicity, including ADCC, extend survival, increase persistence, and enhance or retain high levels of CD16. In some cases, the feeder cells allow the NK cells to expand without causing significant levels of exhaustion or senescence.

[0155] Feeder cells can be used to stimulate the NK cells and help them to expand more quickly, e.g., by providing substrate, growth factors, and/or cytokines.

[0156] NK cells can be stimulated using various types of feeder cells, including, but not limited to peripheral blood mononuclear cells (PBMC), Epstein-Barr virus-transformed

B-lymphoblastoid cells (e.g., EBV-LCL), myelogenous leukemia cells (e.g., K562), and CD4(+) T cells (e.g., HuT), and derivatives thereof.

[0157] In some embodiments, the feeder cells are inactivated, e.g., by γ -irradiation or mitomycin-c treatment.

[0158] Suitable feeder cells for use in the methods described herein are described, for example, in US 2020/0108096, which is hereby incorporated by reference in its entirety.

[0159] In some embodiments, the feeder cell(s) are inactivated CD4(+) T cell(s). In some embodiments, the inactivated CD4(+) T cell(s) are HuT-78 cells (ATCC® TIB-161™) or variants or derivatives thereof. In some embodiments, the HuT-78 derivative is H9 (ATCC® HTB-176™).

[0160] In some embodiments, the inactivated CD4(+) T cell(s) express OX40L. In some embodiments, the inactivated CD4(+) T cell(s) are HuT-78 cells or variants or derivatives thereof that express OX40L (SEQ ID NO: 4) or a variant thereof.

[0161] In some embodiments, the feeder cells are HuT-78 cells engineered to express at least one gene selected from the group consisting of 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and mutant TNFalpha (SEQ ID NO: 3) ("eHuT-78 cells"), or variants thereof.

[0162] In some embodiments, the inactivated CD4(+) T cell(s) are HuT-78 (ATCC® TIB-161™) cells or variants or derivatives thereof that express an ortholog of OX40L, or variant thereof. In some embodiments, the feeder cells are HuT-78 cells engineered to express at least one gene selected from the group consisting of an 4-1BBL ortholog or variant thereof, a membrane bound IL-21 ortholog or variant thereof, and mutant TNFalpha ortholog, or variant thereof.

[0163] In some embodiments, the feeder cells are HuT-78 cell(s) that express OX40L (SEQ ID NO: 4) and are engineered to express 4-1BBL (SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and mutant TNFalpha (SEQ ID NO: 3) ("eHuT-78 cells") or variants or derivatives thereof.

[0164] In some embodiments, the feeder cells are expanded, e.g., from a frozen stock, before culturing with NK cells.

C. Stimulating Factors

[0165] NK cells can also be stimulated using one or more stimulation factors other than feeder cells, e.g., signaling factors, in addition to or in place of feeder cells.

[0166] In some embodiments, the stimulating factor, e.g., signaling factor, is a component of the culture medium, as described herein. In some embodiments, the stimulating factor, e.g., signaling factor, is a supplement to the culture medium, as described herein.

[0167] In some embodiments, the stimulation factor(s) are cytokine(s). In some embodiments, the cytokine(s) are selected from the group consisting of IL-2, IL-12, IL-15, IL-18, IL-21, IL-23, IL-27, IFN- α , IFN β , and combinations thereof.

[0168] In some embodiments, the cytokine is IL-2.

[0169] In some embodiments, the cytokines are a combination of IL-2 and IL-15.

[0170] In some embodiments, the cytokines are a combination of IL-2, IL-15, and IL-18.

[0171] In some embodiments, the cytokines are a combination of IL-2, IL-18, and IL-21.

D. Culturing

[0172] The NK cells can be expanded and stimulated by co-culturing an NK cell source and feeder cells and/or other stimulation factors. Suitable NK cell sources, feeder cells, and stimulation factors are described herein.

[0173] In some cases, the resulting population of expanded natural killer cells is enriched and/or sorted after expansion. In some cases, the resulting population of expanded natural killer cells is not enriched and/or sorted after expansion.

[0174] Also described herein are compositions comprising the various culture compositions described herein, e.g., comprising NK cells. For example, a composition comprising a population of expanded cord blood-derived natural killer cells comprising a KIR-B haplotype and homozygous for a CD16 158V polymorphism and a plurality of engineered HuT78 cells.

[0175] Also described herein are vessels, e.g., vials, cryobags, and the like, comprising the resulting populations of expanded natural killer cells. In some cases, a plurality of vessels comprising portions of the resulting populations of expanded natural killer cells, e.g., at least 10, e.g., 20, 30, 40, 50, 60, 70, 80, 90, 100, 125, 150, 175, 200, 250, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, or 1200 vessels.

[0176] Also described herein are bioreactors comprising the various culture compositions described herein, e.g., comprising NK cells. For example, a culture comprising natural killer cells from a natural killer cell source, e.g., as described herein, and feeder cells, e.g., as described herein. Also described herein are bioreactors comprising the resulting populations of expanded natural killer cells.

1. Culture Medium

[0177] Disclosed herein are culture media for the expansion of NK cells. These culture media advantageously allow NK cells to expand to numbers suitable for the preparation of a pharmaceutical composition as discussed herein. In some cases, the culture media allows NK cells to expand without the loss of CD16 expression that often accompanies cell expansion on other helper cells or in other media.

[0178] In some embodiments, the culture medium is a basal culture medium, optionally supplemented with additional components, e.g., as described herein.

[0179] In some embodiments, the culture medium, e.g., the basal culture medium, is a serum-free culture medium. In some embodiments, the culture medium, e.g., the basal culture medium, is a serum-free culture medium supplemented with human plasma and/or serum.

[0180] Suitable basal culture media include, but are not limited to, DMEM, RPMI 1640, MEM, DMEM/F12, SCGM (CellGenix®, 20802-0500 or 20806-0500), LGM-3™ (Lonza, CC-3211), TexMACS™ (Miltenyi Biotec, 130-097-196), ALyS™ 505NK-AC (Cell Science and Technology Institute, Inc., 01600P02), ALyS™ 505NK-EX (Cell Science and Technology Institute, Inc., 01400P10), CTS™ AIM-V™ SFM (ThermoFisher Scientific, A3830801), CTST™ OpTmizer™ (ThermoFisher Scientific, A1048501, ABS-001, StemXxVivo and combinations thereof).

[0181] The culture medium may comprise additional components, or be supplemented with additional components,

such as growth factors, signaling factors, nutrients, antigen binders, and the like. Supplementation of the culture medium may occur by adding each of the additional component or components to the culture vessel either before, concurrently with, or after the medium is added to the culture vessel. The additional component or components may be added together or separately. When added separately, the additional components need not be added at the same time.

[0182] In some embodiments, the culture medium comprises plasma, e.g., human plasma. In some embodiments, the culture medium is supplemented with plasma, e.g., human plasma. In some embodiments, the plasma, e.g., human plasma, comprises an anticoagulant, e.g., trisodium citrate.

[0183] In some embodiments, the medium comprises and/or is supplemented with from or from about 0.5% to or to about 10% v/v plasma, e.g., human plasma. In some embodiments, the medium is supplemented with from or from about 0.5% to or to about 9%, from or from about 0.5% to or to about 8%, from or from about 0.5% to or to about 7%, from or from about 0.5% to or to about 6%, from or from about 0.5% to or to about 5%, from or from about 0.5% to or to about 4%, from or from about 0.5% to or to about 3%, from or from about 0.5% to or to about 2%, from or from about 0.5% to or to about 1%, from or from about 1% to or to about 10%, from or from about 1% to or to about 9%, from or from about 1% to or to about 8%, from or from about 1% to or to about 7%, from or from about 1% to or to about 6%, from or from about 1% to or to about 5%, from or from about 1% to or to about 4%, from or from about 1% to or to about 3%, from or from about 1% to or to about 2%, from or from about 2% to or to about 10%, from or from about 2% to or to about 9%, from or from about 2% to or to about 8%, from or from about 2% to or to about 7%, from or from about 2% to or to about 6%, from or from about 2% to or to about 5%, from or from about 2% to or to about 4%, from or from about 2% to or to about 3%, from or from about 3% to or to about 10%, from or from about 3% to or to about 9%, from or from about 3% to or to about 8%, from or from about 3% to or to about 7%, from or from about 3% to or to about 6%, from or from about 3% to or to about 5%, from or from about 3% to or to about 4%, from or from about 4% to or to about 10%, from or from about 4% to or to about 9%, from or from about 4% to or to about 8%, from or from about 4% to or to about 7%, from or from about 4% to or to about 6%, from or from about 4% to or to about 5%, from or from about 5% to or to about 10%, from or from about 5% to or to about 9%, from or from about 5% to or to about 8%, from or from about 5% to or to about 7%, from or from about 5% to or to about 6%, from or from about 6% to or to about 10%, from or from about 6% to or to about 9%, from or from about 6% to or to about 8%, from or from about 6% to or to about 7%, from or from about 6% to or to about 7%, from or from about 7% to or to about 10%, from or from about 7% to or to about 9%, from or from about 7% to or to about 8%, from or from about 8% to or to about 10%, from or from about 8% to or to about 9%, or from or from about 9% to or to about 10% v/v plasma, e.g., human plasma. In some embodiments, the culture medium comprises and/or is supplemented with from 0.8% to 1.2% v/v human plasma. In some embodiments, the culture medium comprises and/or is supplemented with 1.0% v/v human plasma. In some embodiments, the culture medium comprises and/or is supplemented with about 1.0% v/v human plasma.

[0184] In some embodiments, the culture medium comprises serum, e.g., human serum. In some embodiments, the culture medium is supplemented with serum, e.g., human serum. In some embodiments, the serum is inactivated, e.g., heat inactivated. In some embodiments, the serum is filtered, e.g., sterile-filtered.

[0185] In some embodiments, the culture medium comprises glutamine. In some embodiments, the culture medium is supplemented with glutamine. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 2.0 to or to about 6.0 mM glutamine. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 2.0 to or to about 5.5, from or from about 2.0 to or to about 5.0, from or from about 2.0 to or to about 4.5, from or from about 2.0 to or to about 4.0, from or from about 2.0 to or to about 3.5, from or from about 2.0 to or to about 3.0, from or from about 2.0 to or to about 2.5, from or from about 2.5 to or to about 6.0, from or from about 2.5 to or to about 5.5, from or from about 2.5 to or to about 5.0, from or from about 2.5 to or to about 4.5, from or from about 2.5 to or to about 4.0, from or from about 2.5 to or to about 3.5, from or from about 2.5 to or to about 3.0, from or from about 3.0 to or to about 6.0, from or from about 3.0 to or to about 5.5, from or from about 3.0 to or to about 5.0, from or from about 3.0 to or to about 4.5, from or from about 3.0 to or to about 4.0, from or from about 3.0 to or to about 3.5, from or from about 3.5 to or to about 6.0, from or from about 3.5 to or to about 5.5, from or from about 3.5 to or to about 5.0, from or from about 3.5 to or to about 4.5, from or from about 3.5 to or to about 4.0, from or from about 4.0 to or to about 6.0, from or from about 4.0 to or to about 5.5, from or from about 4.0 to or to about 5.0, from or from about 4.0 to or to about 4.5, from or from about 4.5 to or to about 6.0, from or from about 4.5 to or to about 5.5, from or from about 4.5 to or to about 5.0, from or from about 4.5 to or to about 4.0, from or from about 5.0 to or to about 6.0, from or from about 5.0 to or to about 5.5, or from or from about 5.5 to or to about 6.0 mM glutamine. In some embodiments, the culture medium comprises and/or is supplemented with from 3.2 mM glutamine to 4.8 mM glutamine. In some embodiments, the culture medium comprises and/or is supplemented with 4.0 mM glutamine. In some embodiments, the culture medium comprises and/or is supplemented with about 4.0 mM glutamine.

[0186] In some embodiments, the culture medium comprises one or more cytokines. In some embodiments, the culture medium is supplemented with one or more cytokines.

[0187] In some embodiments, the cytokine is selected from IL-2, IL-12, IL-15, IL-18, and combinations thereof.

[0188] In some embodiments, the culture medium comprises and/or is supplemented with IL-2. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 150 to or to about 2,500 IU/mL IL-2. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 200 to or to about 2,250, from or from about 200 to or to about 2,000, from or from about 200 to or to about 1,750, from or from about 200 to or to about 1,500, from or from about 200 to or to about 1,250, from or from about 200 to or to about 1,000, from or from about 200 to or to about 750, from or from about 200 to or to about 500, from or from about 200 to or to about 250, from or from about 250 to or to about 2,500, from or from about 250 to or to about 2,250, from or from about 250 to or to about 2,000, from or from about 250

to or to about 1,750, from or from about 250 to or to about 1,500, from or from about 250 to or to about 1,250, from or from about 250 to or to about 1,000, from or from about 250 to or to about 750, from or from about 250 to or to about 500, from or from about 500 to or to about 2,500, from or from about 500 to or to about 2,250, from or from about 500 to or to about 2,000, from or from about 500 to or to about 1,750, from or from about 500 to or to about 1,500, from or from about 500 to or to about 1,250, from or from about 500 to or to about 1,000, from or from about 500 to or to about 750, from or from about 750 to or to about 2,250, from or from about 750 to or to about 2,000, from or from about 750 to or to about 1,750, from or from about 750 to or to about 1,500, from or from about 750 to or to about 1,250, from or from about 750 to or to about 1,000, from or from about 1,000 to or to about 2,500, from or from about 1,000 to or to about 2,250, from or from about 1,000 to or to about 2,000, from or from about 1,000 to or to about 1,750, from or from about 1,000 to or to about 1,500, from or from about 1,000 to or to about 1,250, from or from about 1,250 to or to about 2,500, from or from about 1,250 to or to about 2,250, from or from about 1,250 to or to about 2,000, from or from about 1,250 to or to about 1,750, from or from about 1,250 to or to about 1,500, from or from about 1,500 to or to about 2,500, from or from about 1,500 to or to about 2,250, from or from about 1,500 to or to about 2,000, from or from about 1,500 to or to about 1,750, from or from about 1,750 to or to about 2,500, from or from about 1,750 to or to about 2,250, from or from about 1,750 to or to about 2,000, from or from about 2,000 to or to about 2,500, from or from about 2,000 to or to about 2,250, or from or from about 2,250 to or to about 2,500 IU/mL IL-2.

[0189] In some embodiments, the culture medium comprises and/or is supplemented with from 64 µg/L to 96 µg/L IL-2. In some embodiments, the culture medium comprises and/or is supplemented with 80 µg/L IL-2 (approximately 1,333 IU/mL). In some embodiments, the culture medium comprises and/or is supplemented with about 80 µg/L.

[0190] In some embodiments, the culture medium comprises and/or is supplemented with a combination of IL-2 and IL-15.

[0191] In some embodiments, the culture medium comprises and/or is supplemented with a combination of IL-2, IL-15, and IL-18.

[0192] In some embodiments, the culture medium comprises and/or is supplemented with a combination of IL-2, IL-18, and IL-21.

[0193] In some embodiments, the culture medium comprises and/or is supplemented with glucose. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 0.5 to or to about 3.5 g/L glucose. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 0.5 to or to about 3.0, from or from about 0.5 to or to about 2.5, from or from about 0.5 to or to about 2.0, from or from about 0.5 to or to about 1.5, from or from about 0.5 to or to about 1.0, from or from about 1.0 to or to about 3.0, from or from about 1.0 to or to about 2.5, from or from about 1.0 to or to about 2.0, from or from about 1.0 to or to about 1.5, from or from about 1.5 to or to about 3.0, from or from about 1.5 to or to about 2.5, from or from about 1.5 to or to about 2.0, from or from about 2.0 to or to about 3.0, from or from about 2.0 to or to about 2.5, or from or from about 2.5 to or to about 3.0 g/L glucose. In some embodiments, the culture

or from about 3.5 to or to about 5, from or from about 3.5 to or to about 4.5, from or from about 3.5 to or to about 4, from or from about 4 to or to about 6, from or from about 4 to or to about 5.5, from or from about 4 to or to about 5, from or from about 4 to or to about 4.5, from or from about 4.5 to or to about 6, from or from about 4.5 to or to about 5.5, from or from about 4.5 to or to about 5, from or from about 5 to or to about 6, from or from about 5 to or to about 5.5, or from or from about 5.5 to or to about 6 g/L albumin, e.g., human albumin. In some embodiments, the culture medium comprises and/or is supplemented with from 3.2 to 4.8 g/L albumin, e.g., human albumin. In some embodiments, the culture medium comprises about 4 g/L albumin, e.g., human albumin

[0198] In some embodiments, the culture medium is supplemented with Poloxamer 188. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 0.1 to or to about 2.0 g/L Poloxamer 188. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 0.1 to or to about 1.8, from or from about 0.1 to or to about 1.6, from or from about 0.1 to or to about 1.4, from or from about 0.1 to or to about 1.2, from or from about 0.1 to or to about 1.0, from or from about 0.1 to or to about 0.8, from or from about 0.1 to or to about 0.6, from or from about 0.1 to or to about 0.4, from or from about 0.1 to or to about 0.2, from or from about 0.2 to or to about 2.0, from or from about 0.2 to or to about 1.8, from or from about 0.2 to or to about 1.6, from or from about 0.2 to or to about 1.4, from or from about 0.2 to or to about 1.2, from or from about 0.2 to or to about 1.0, from or from about 0.2 to or to about 0.8, from or from about 0.2 to or to about 0.6, from or from about 0.2 to or to about 0.4, from or from about 0.4 to or to about 2.0, from or from about 0.4 to or to about 1.8, from or from about 0.4 to or to about 1.6, from or from about 0.4 to or to about 1.4, from or from about 0.4 to or to about 1.2, from or from about 0.4 to or to about 1.0, from or from about 0.4 to or to about 0.8, from or from about 0.4 to or to about 0.6, from or from about 0.6 to or to about 2.0, from or from about 0.6 to or to about 1.8, from or from about 0.6 to or to about 1.6, from or from about 0.6 to or to about 1.4, from or from about 0.6 to or to about 1.2, from or from about 0.6 to or to about 1.0, from or from about 0.6 to or to about 0.8, from or from about 0.8 to or to about 2.0, from or from about 0.8 to or to about 1.8, from or from about 0.8 to or to about 1.6, from or from about 0.8 to or to about 1.4, from or from about 0.8 to or to about 1.2, from or from about 0.8 to or to about 1.0, from or from about 1.0 to or to about 2.0, from or from about 1.0 to or to about 1.8, from or from about 1.0 to or to about 1.6, from or from about 1.0 to or to about 1.4, from or from about 1.0 to or to about 1.2, from or from about 1.2 to or to about 2.0, from or from about 1.2 to or to about 1.8, from or from about 1.2 to or to about 1.6, from or from about 1.2 to or to about 1.4, from or from about 1.4 to or to about 2.0, from or from about 1.4 to or to about 1.8, from or from about 1.4 to or to about 1.6, from or from about 1.6 to or to about 2.0, from or from about 1.6 to or to about 1.8, or from or from about 1.8 to or to about 2.0 g/L Poloxamer 188. In some embodiments, the culture medium comprises from 0.8 to 1.2 g/L Poloxamer 188. In some embodiments, the culture medium comprises 1.0 g/L Poloxamer 188. In some embodiments, the culture medium comprises about 1.0 g/L Poloxamer 188.

[0199] In some embodiments, the culture medium comprises and/or is supplemented with one or more antibiotics.

[0200] A first exemplary culture medium is set forth in Table 1.

TABLE 1

Exemplary Culture Medium #1		
Component	Exemplary Concentration Range	Exemplary Concentration
CellgroSCGM liquid medium	undiluted	undiluted
Human Plasma	0.8-1.2% (v/v)	1.0% v/v
Glutamine	3.2-4.8 mM	4.0 mM
IL-2	64-96 μ g/L	80 μ g/L

[0201] A second exemplary culture medium is set forth in Table 2.

TABLE 2

Exemplary Culture Medium #2		
Component	Exemplary Concentration Range	Exemplary Concentration
RPMI1640	7.6-13.2 g/L	10.4 g/L
Human Plasma	0.8-1.2% (v/v)	1.0% v/v
Glucose	1.6-2.4 g/L	2.0 g/L
Glutamine	3.2-4.8 mM	4.0 mM
Sodium Pyruvate	0.8-1.2 mM	1.0 mM
Sodium Hydrogen Carbonate	1.6-2.4 g/L	2.0 g/L
IL-2	64-96 μ g/L	80 μ g/L
Albumin 20% solution	1.6-2.5% v/v	2.0% v/v
	(3.2 to 4.8 g/L)	(4.0 g/L)
Poloxamer 188	0.8-1.2 g/L	1.0 g/L

2. CD3 Binding Antibodies

[0202] In some embodiments, the culture medium comprises and/or is supplemented with a CD3 binding antibody or antigen binding fragment thereof. In some embodiments, the CD3 binding antibody or antigen binding fragment thereof is selected from the group consisting of OKT3, UCHT1, and HIT3a, or variants thereof. In some embodiments, the CD3 binding antibody or antigen binding fragment thereof is OKT3 or an antigen binding fragment thereof.

[0203] In some embodiments, the CD3 binding antibody or antigen binding fragment thereof and feeder cells are added to the culture vessel before addition of NK cells and/or culture medium.

[0204] In some embodiments, the culture medium comprises and/or is supplemented with from or from about 5 ng/ml to or to about 15 ng/mL OKT3. In some embodiments, the culture medium comprises and/or is supplemented with from or from about 5 to or to about 12.5, from or from about 5 to or to about 10, from or from about 5 to or to about 7.5, from or from about 7.5 to or to about 15, from or from about 7.5 to or to about 12.5, from or from about 7.5 to or to about 10, from or from about 10 to or to about 15, from or from about 10 to or to about 12.5, or from or from about 12.5 to or to about 15 ng/mL OKT3. In some embodiments, the culture medium comprises and/or is supplemented with 10 ng/mL OKT3. In some embodiments, the culture medium comprises and/or is supplemented with about 10 ng/ml OKT3.

3. Culture Vessels

[0205] A number of vessels are consistent with the disclosure herein. In some embodiments, the culture vessel is selected from the group consisting of a flask, a bottle, a dish, a multiwall plate, a roller bottle, a bag, and a bioreactor.

[0206] In some embodiments, the culture vessel is treated to render it hydrophilic. In some embodiments, the culture vessel is treated to promote attachment and/or proliferation. In some embodiments, the culture vessel surface is coated with serum, collagen, laminin, gelatin, poy-L-lysine, fibronectin, extracellular matrix proteins, and combinations thereof.

[0207] In some embodiments, different types of culture vessels are used for different stages of culturing.

[0208] In some embodiments, the culture vessel has a volume of from or from about 100 mL to or to about 1,000 L. In some embodiments, the culture vessel has a volume of or about 125 mL, of or about 250 mL, of or about 500 mL, of or about 1 L, of or about 5 L, of about 10 L, of or or about 20 L.

[0209] In some embodiments, the culture vessel is a bioreactor.

[0210] In some embodiments, the bioreactor is a rocking bed (wave motion) bioreactor. In some embodiments, the bioreactor is a stirred tank bioreactor. In some embodiments, the bioreactor is a rotating wall vessel. In some embodiments, the bioreactor is a perfusion bioreactor. In some embodiments, the bioreactor is an isolation/expansion automated system. In some embodiments, the bioreactor is an automated or semi-automated bioreactor. In some embodiments, the bioreactor is a disposable bag bioreactor.

[0211] In some embodiments, the bioreactor has a volume of from about 100 mL to about 1,000 L. In some embodiments, the bioreactor has a volume of from about 10 L to about 1,000 L. In some embodiments, the bioreactor has a volume of from about 100 L to about 900 L. In some embodiments, the bioreactor has a volume of from about 10 L to about 800 L. In some embodiments, the bioreactor has a volume of from about 10 L to about 700 L, about 10 L to about 600 L, about 10 L to about 500 L, about 10 L to about 400 L, about 10 L to about 300 L, about 10 L to about 200 L, about 10 L to about 100 L, about 10 L to about 90 L, about 10 L to about 80 L, about 10 L to about 70 L, about 10 L to about 60 L, about 10 L to about 50 L, about 10 L to about 40 L, about 10 L to about 30 L, about 10 L to about 20 L, about 20 L to about 1,000 L, about 20 L to about 900 L, about 20 L to about 800 L, about 20 L to about 700 L, about 20 L to about 600 L, about 20 L to about 500 L, about 20 L to about 400 L, about 20 L to about 300 L, about 20 L to about 200 L, about 20 L to about 100 L, about 20 L to about 90 L, about 20 L to about 80 L, about 20 L to about 70 L, about 20 L to about 60 L, about 20 L to about 50 L, about 20 L to about 40 L, about 20 L to about 30 L, about 30 L to about 1,000 L, about 30 L to about 900 L, about 30 L to about 800 L, about 30 L to about 700 L, about 30 L to about 600 L, about 30 L to about 500 L, about 30 L to about 400 L, about 30 L to about 300 L, about 30 L to about 200 L, about 30 L to about 100 L, about 30 L to about 90 L, about 30 L to about 80 L, about 30 L to about 70 L, about 30 L to about 60 L, about 30 L to about 50 L, about 30 L to about 40 L, about 40 L to about 1,000 L, about 40 L to about 900 L, about 40 L to about 800 L, about 40 L to about 700 L, about 40 L to about 600 L, about 40 L to about 500 L, about 40 L to about 400 L, about 40 L to about 300 L, about 40 L

to about 200 L, about 40 L to about 100 L, about 40 L to about 90 L, about 40 L to about 80 L, about 40 L to about 70 L, about 40 L to about 60 L, about 40 L to about 50 L, about 50 L to about 1,000 L, about 50 L to about 900 L, about 50 L to about 800 L, about 50 L to about 700 L, about 50 L to about 600 L, about 50 L to about 500 L, about 50 L to about 400 L, about 50 L to about 300 L, about 50 L to about 200 L, about 50 L to about 100 L, about 50 L to about 90 L, about 50 L to about 80 L, about 50 L to about 70 L, about 50 L to about 60 L, about 60 L to about 1,000 L, about 60 L to about 900 L, about 60 L to about 800 L, about 60 L to about 700 L, about 60 L to about 600 L, about 60 L to about 500 L, about 60 L to about 400 L, about 60 L to about 300 L, about 60 L to about 200 L, about 60 L to about 100 L, about 60 L to about 90 L, about 60 L to about 80 L, about 60 L to about 70 L, about 70 L to about 1,000 L, about 70 L to about 900 L, about 70 L to about 800 L, about 70 L to about 700 L, about 70 L to about 600 L, about 70 L to about 500 L, about 70 L to about 400 L, about 70 L to about 300 L, about 70 L to about 200 L, about 70 L to about 100 L, about 70 L to about 90 L, about 70 L to about 80 L, about 80 L to about 1,000 L, about 80 L to about 900 L, about 80 L to about 800 L, about 80 L to about 700 L, about 80 L to about 600 L, about 80 L to about 500 L, about 80 L to about 400 L, about 80 L to about 300 L, about 80 L to about 200 L, about 80 L to about 100 L, about 80 L to about 90 L, about 90 L to about 1,000 L, about 90 L to about 900 L, about 90 L to about 800 L, about 90 L to about 700 L, about 90 L to about 600 L, about 90 L to about 500 L, about 90 L to about 400 L, about 90 L to about 300 L, about 90 L to about 200 L, about 90 L to about 100 L, about 100 L to about 1,000 L, about 100 L to about 900 L, about 100 L to about 800 L, about 100 L to about 700 L, about 100 L to about 600 L, about 100 L to about 500 L, about 100 L to about 400 L, about 100 L to about 300 L, about 100 L to about 200 L, about 200 L to about 1,000 L, about 200 L to about 900 L, about 200 L to about 800 L, about 200 L to about 700 L, about 200 L to about 600 L, about 200 L to about 500 L, about 200 L to about 400 L, about 200 L to about 300 L, about 300 L to about 1,000 L, about 300 L to about 900 L, about 300 L to about 800 L, about 300 L to about 700 L, about 300 L to about 600 L, about 300 L to about 500 L, about 300 L to about 400 L, about 400 L to about 1,000 L, about 400 L to about 900 L, about 400 L to about 800 L, about 400 L to about 700 L, about 400 L to about 600 L, about 400 L to about 500 L, about 500 L to about 1,000 L, about 500 L to about 900 L, about 500 L to about 800 L, about 500 L to about 700 L, about 500 L to about 600 L, about 600 L to about 1,000 L, about 600 L to about 900 L, about 600 L to about 800 L, about 600 L to about 700 L, about 700 L to about 1,000 L, about 700 L to about 900 L, about 700 L to about 800 L, about 800 L to about 1,000 L, about 800 L to about 900 L, or about 900 L to about 1,000 L. In some embodiments, the bioreactor has a volume of about 50 L.

[0212] In some embodiments, the bioreactor has a volume of from 100 mL to 1,000 L. In some embodiments, the bioreactor has a volume of from 10 L to 1,000 L. In some embodiments, the bioreactor has a volume of from 100 L to 900 L. In some embodiments, the bioreactor has a volume of from 10 L to 800 L. In some embodiments, the bioreactor has a volume of from 10 L to 700 L, 10 L to 600 L, 10 L to 500 L, 10 L to 400 L, 10 L to 300 L, 10 L to 200 L, 10 L to 100 L, 10 L to 90 L, 10 L to 80 L, 10 L to 70 L, 10 L to

60 L, 10 L to 50 L, 10 L to 40 L, 10 L to 30 L, 10 L to 20 L, 20 L to 1,000 L, 20 L to 900 L, 20 L to 800 L, 20 L to 700 L, 20 L to 600 L, 20 L to 500 L, 20 L to 400 L, 20 L to 300 L, 20 L to 200 L, 20 L to 100 L, 20 L to 90 L, 20 L to 80 L, 20 L to 70 L, 20 L to 60 L, 20 L to 50 L, 20 L to 40 L, 20 L to 30 L, 30 L to 1,000 L, 30 L to 900 L, 30 L to 800 L, 30 L to 700 L, 30 L to 600 L, 30 L to 500 L, 30 L to 400 L, 30 L to 300 L, 30 L to 200 L, 30 L to 100 L, 30 L to 90 L, 30 L to 80 L, 30 L to 70 L, 30 L to 60 L, 30 L to 50 L, 30 L to 40 L, 40 L to 1,000 L, 40 L to 900 L, 40 L to 800 L, 40 L to 700 L, 40 L to 600 L, 40 L to 500 L, 40 L to 400 L, 40 L to 300 L, 40 L to 200 L, 40 L to 100 L, 40 L to 90 L, 40 L to 80 L, 40 L to 70 L, 40 L to 60 L, 40 L to 50 L, 50 L to 1,000 L, 50 L to 900 L, 50 L to 800 L, 50 L to 700 L, 50 L to 600 L, 50 L to 500 L, 50 L to 400 L, 50 L to 300 L, 50 L to 200 L, 50 L to 100 L, 50 L to 90 L, 50 L to 80 L, 50 L to 70 L, 50 L to 60 L, 60 L to 1,000 L, 60 L to 900 L, 60 L to 800 L, 60 L to 700 L, 60 L to 600 L, 60 L to 500 L, 60 L to 400 L, 60 L to 300 L, 60 L to 200 L, 60 L to 100 L, 60 L to 90 L, 60 L to 80 L, 60 L to 70 L, 70 L to 1,000 L, 70 L to 900 L, 70 L to 800 L, 70 L to 700 L, 70 L to 600 L, 70 L to 500 L, 70 L to 400 L, 70 L to 300 L, 70 L to 200 L, 70 L to 100 L, 70 L to 90 L, 70 L to 80 L, 80 L to 1,000 L, 80 L to 900 L, 80 L to 800 L, 80 L to 700 L, 80 L to 600 L, 80 L to 500 L, 80 L to 400 L, 80 L to 300 L, 80 L to 200 L, 80 L to 100 L, 80 L to 90 L, 90 L to 1,000 L, 90 L to 900 L, 90 L to 800 L, 90 L to 700 L, 90 L to 600 L, 90 L to 500 L, 90 L to 400 L, 90 L to 300 L, 90 L to 200 L, 90 L to 100 L, 100 L to 1,000 L, 100 L to 900 L, 100 L to 800 L, 100 L to 700 L, 100 L to 600 L, 100 L to 500 L, 100 L to 400 L, 100 L to 300 L, 100 L to 200 L, 200 L to 1,000 L, 200 L to 900 L, 200 L to 800 L, 200 L to 700 L, 200 L to 600 L, 200 L to 500 L, 200 L to 400 L, 200 L to 300 L, 300 L to 1,000 L, 300 L to 900 L, 300 L to 800 L, 300 L to 700 L, 300 L to 600 L, 300 L to 500 L, 300 L to 400 L, 400 L to 1,000 L, 400 L to 900 L, 400 L to 800 L, 400 L to 700 L, 400 L to 600 L, 400 L to 500 L, 500 L to 1,000 L, 500 L to 900 L, 500 L to 800 L, 500 L to 700 L, 500 L to 600 L, 600 L to 1,000 L, 600 L to 900 L, 600 L to 800 L, 600 L to 700 L, 700 L to 1,000 L, 700 L to 900 L, 700 L to 800 L, 800 L to 1,000 L, 800 L to 900 L, or 900 L to 1,000 L. In some embodiments, the bioreactor has a volume of 50 L.

4. Cell Expansion and Stimulation

[0213] In some embodiments, the natural killer cell source, e.g., single unit of cord blood, is co-cultured with feeder cells to produce expanded and stimulated NK cells.

[0214] In some embodiments, the co-culture is carried out in a culture medium described herein, e.g., exemplary culture medium #1 (Table 1) or exemplary culture medium #2 (Table 2).

[0215] In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises from or from about 1×10^7 to or to about 1×10^9 total nucleated cells prior to expansion. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises from or from about 1×10^8 to or to about 1.5×10^8 total nucleated cells prior to expansion. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises 1×10^8 total nucleated cells prior to expansion. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises about 1×10^8 total nucleated cells prior to expansion. In some embodiments, the natural killer

cell source, e.g., single unit of cord blood, comprises 1×10^9 total nucleated cells prior to expansion. In some embodiments, the natural killer cell source, e.g., single unit of cord blood, comprises about 1×10^9 total nucleated cells prior to expansion.

[0216] In some embodiments, cells from the co-culture of the natural killer cell source, e.g., single unit of cord blood and feeder cells are harvested and frozen, e.g., in a cryo-preservation composition described herein. In some embodiments, the frozen cells from the co-culture are an infusion-ready drug product. In some embodiments, the frozen cells from the co-culture are used as a master cell bank (MCB) from which to produce an infusion-ready drug product, e.g., through one or more additional co-culturing steps, as described herein. Thus, for example, a natural killer cell source can be expanded and stimulated as described herein to produce expanded and stimulated NK cells suitable for use in an infusion-ready drug product without generating any intermediate products. A natural killer cell source can also be expanded and stimulated as described herein to produce an intermediate product, e.g., a first master cell bank (MCB). The first MCB can be used to produce expanded and stimulated NK cells suitable for use in an infusion-ready drug product, or, alternatively, be used to produce another intermediate product, e.g., a second MCB. The second MCB can be used to produce expanded and stimulated NK cells suitable for an infusion-ready drug product, or alternatively, be used to produce another intermediate product, e.g., a third MCB, and so on.

[0217] In some embodiments, the ratio of feeder cells to cells of the natural killer cell source or MCB cells inoculated into the co-culture is from or from about 1:1 to or to about 4:1. In some embodiments, the ratio of feeder cells to cells of the natural killer cell source or MCB cells is from or from about 1:1 to or to about 3.5:1, from or from about 1:1 to or to about 3:1, from or from about 1:1 to or to about 2.5:1, from or from about 1.1 to or to about 2:1, from or from about 1:1 to or to about 1.5:1, from or from about 1.5:1 to or to about 4:1, from or from about 1.5:1 to or to about 3.5:1, from or from about 1.5:1 to or to about 3:1, from or from about 1.5:1 to or to about 2.5:1, from or from about 1.5:1 to or to about 2:1, from or from about 2:1 to or to about 4:1, from or from about 2:1 to or to about 3.5:1, from or from about 2:1 to or to about 3:1, from or from about 2:1 to or to about 2.5:1, from or from about 2.5:1 to or to about 4:1, from or from about 2.5:1 to or to about 3.5:1, from or from about 2.5:1 to or to about 3:1, from or from about 3:1 to or to about 4:1, from or from about 3:1 to or to about 3.5:1, or from or from about 3.5:1 to or to about 4:1. In some embodiments, the ratio of feeder cells to cells of the natural killer cell source or MCB inoculated into the co-culture is 2.5:1. In some embodiments, the ratio of feeder cells to cells of the natural killer cell source or MCB inoculated into the co-culture is about 2.5:1.

[0218] In some embodiments, the co-culture is carried out in a disposable culture bag, e.g., a 1L disposable culture bag. In some embodiments, the co-culture is carried out in a bioreactor, e.g., a 50L bioreactor. In some embodiments, culture medium is added to the co-culture after the initial inoculation.

[0219] In some embodiments, the co-culture is carried out for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or more days. In some embodiments, the co-culture is carried out for a maximum of 16 days.

[0220] In some embodiments, the co-culture is carried out at 37° C. or about 37° C.

[0221] In some embodiments, the co-culture is carried out at pH 7.9 or about pH 7.9.

[0222] In some embodiments, the co-culture is carried out at a dissolved oxygen (DO) level of 50% or more.

[0223] In some embodiments, exemplary culture medium #1 (Table 1) is used to produce a MCB and exemplary culture medium #2 (Table 2) is used to produce cells suitable for an infusion-ready drug product.

[0224] In some embodiments, the co-culture of the natural killer cell source, e.g., single unit of cord blood, with feeder cells yields from or from about 50×10^8 to or to about 50×10^{12} cells, e.g., MCB cells or infusion-ready drug product cells. In some embodiments, the expansion yields from or from about 50×10^8 to or to about 25×10^{10} , from or from about 10×10^8 to or to about 1×10^{10} , from or from about 50×10^8 to or to about 75×10^9 , from or from about 50×10^8 to or to about 50×10^9 , from or from about 50×10^8 to or to about 25×10^9 , from or from about 50×10^8 to or to about 1×10^9 , from or from about 50×10^8 to or to about 75×10^8 , from or from about 75×10^8 to or to about 50×10^{10} , from or from about 75×10^8 to or to about 25×10^{10} , from or from about 75×10^8 to or to about 1×10^{10} , from or from about 75×10^9 , from or from about 75×10^8 to or to about 50×10^9 , from or from about 75×10^8 to or to about 25×10^9 , from or from about 75×10^8 to or to about 1×10^9 , from or from about 1×10^9 to or to about 50×10^{10} , from or from about 1×10^9 to or to about 25×10^{10} , from or from about 1×10^9 to or to about 1×10^{10} , from or from about 1×10^9 to or to about 75×10^9 , from or from about 1×10^9 to or to about 50×10^9 , from or from about 1×10^9 to or to about 25×10^9 , from or from about 25×10^9 to or to about 50×10^{10} , from or from about 25×10^9 to or to about 25×10^{10} , from or from about 25×10^9 to or to about 1×10^{10} , from or from about 25×10^9 to or to about 75×10^9 , from or from about 25×10^9 to or to about 50×10^9 , from or from about 50×10^9 to or to about 50×10^{10} , from or from about 50×10^9 to or to about 25×10^{10} , from or from about 50×10^9 to or to about 1×10^{10} , from or from about 50×10^9 to or to about 75×10^9 , from or from about 75×10^9 to or to about 50×10^{10} , from or from about 75×10^9 to or to about 25×10^{10} , from or from about 75×10^9 to or to about 1×10^{10} , from or from about 1×10^{10} to or to about 50×10^{10} , from or from about 1×10^{10} to or to about 25×10^{10} , or from or from about 25×10^{10} to or to about 50×10^{10} cells, e.g., MCB cells or infusion-ready drug product cells.

[0225] In some embodiments, the expansion yields from or from about 60 to or to about 100 vials, each comprising from or from about 600 million to or to about 1 billion cells, e.g., MCB cells or infusion-ready drug product cells. In some embodiments, the expansion yields 80 or about 80 vials, each comprising or consisting of 800 million or about 800 million cells, e.g., MCB cells or infusion-ready drug product cells.

[0226] In some embodiments, the expansion yields from or from about a 100 to or to about a 500 fold increase in the number of cells, e.g., the number of MCB cells relative to the number of cells, e.g., NK cells, in the natural killer cell source. In some embodiments, the expansion yields from or from about a 100 to or to about a 500, from or from about a 100 to or to about a 400, from or from about a 100 to or to about a 300, from or from about a 100 to or to about a 200, from or from about a 200 to or to about a 450, from or from about a 200 to or to about a 400, from or from about a 100

to or to about a 350, from or from about a 200 to or to about a 300, from or from about a 200 to or to about a 250, from or from about a 250 to or to about a 500, from or from about a 250 to or to about a 450, from or from about a 200 to or to about a 400, from or from about a 250 to or to about a 350, from or from about a 250 to or to about a 300, from or from about a 300 to or to about a 500, from or from about a 300 to or to about a 450, from or from about a 300 to or to about a 400, from or from about a 300 to or to about a 350, from or from about a 350 to or to about a 500, from or from about a 350 to or to about a 450, from or from about a 350 to or to about a 400 fold increase in the number of cells, e.g., the number of MCB cells relative to the number of cells, e.g., NK cells, in the natural killer cell source.

[0227] In some embodiments, the expansion yields from or from about a 100 to or to about a 70,000 fold increase in the number of cells, e.g., the number of MCB cells relative to the number of cells, e.g., NK cells, in the natural killer cell source. In some embodiments, the expansion yields at least a 10,000 fold, e.g., 15,000 fold, 20,000 fold, 25,000 fold, 30,000 fold, 35,000 fold, 40,000 fold, 45,000 fold, 50,000 fold, 55,000 fold, 60,000 fold, 65,000 fold, or 70,000 fold increase in the number of cells, e.g., the number of MCB cells relative to the number of cells, e.g., NK cells, in the natural killer cell source.

[0228] In some embodiments, the co-culture of the MCB cells and feeder cells yields from or from about 500 million to or to about 1.5 billion cells, e.g., NK cells suitable for use in an MCB and/or in an infusion-ready drug product. In some embodiments, the co-culture of the MCB cells and feeder cells yields from or from about 500 million to or to about 1.5 billion, from or from about 500 million to or to about 1.25 billion, from or from about 500 million to or to about 1 billion, from or from about 500 million to or to about 750 million, from or from about 750 million to or to about 1.5 billion, from or from about 500 million to or to about 1.25 billion, from or from about 750 million to or to about 1 billion, from or from about 1 billion to or to about 1.5 billion, from or from about 1 billion to or to about 1.25 billion, or from or from about 1.25 billion to or to about 1.5 billion cells, e.g., NK cells suitable for use in an MCB and/or an infusion-ready drug product.

[0229] In some embodiments, the co-culture of the MCB cells and feeder cells yields from or from about 50 to or to about 150 vials of cells, e.g., infusion-ready drug product cells, each comprising from or from about 750 million to or to about 1.25 billion cells, e.g., NK cells suitable for use in an MCB and/or an infusion-ready drug product. In some embodiments, the co-culture of the MCB cells and feeder cells yields 100 or about 100 vials, each comprising or consisting of 1 billion or about 1 billion cells, e.g., NK cells suitable for use in an MCB and/or an infusion-ready drug product.

[0230] In some embodiments, the expansion yields from or from about a 100 to or to about a 500 fold increase in the number of cells, e.g., the number of NK cells suitable for use in an MCB and/or an infusion-ready drug product relative to the number of starting MCB cells. In some embodiments, the expansion yields from or from about a 100 to or to about a 500, from or from about a 100 to or to about a 400, from or from about a 100 to or to about a 300, from or from about a 100 to or to about a 200, from or from about a 200 to or to about a 450, from or from about a 200 to or to about a 400, from or from about a 100 to or to about a 350, from or from

about a 200 to or to about a 300, from or from about a 200 to or to about a 250, from or from about a 250 to or to about a 500, from or from about a 250 to or to about a 450, from or from about a 200 to or to about a 400, from or from about a 250 to or to about a 350, from or from about a 250 to or to about a 300, from or from about a 300 to or to about a 500, from or from about a 300 to or to about a 450, from or from about a 300 to or to about a 400, from or from about a 300 to or to about a 350, from or from about a 350 to or to about a 500, from or from about a 350 to or to about a 450, from or from about a 350 to or to about a 400 fold increase in the number of cells, e.g., the number of NK cells suitable for use in an MCB and/or an infusion-ready drug product relative to the number of starting MCB cells.

[0231] In some embodiments, the expansion yields from or from about a 100 to or to about a 70,000 fold increase in the number of cells, e.g., the number of NK cells suitable for use in an MCB and/or an infusion-ready drug product relative to the number of starting MCB cells. In some embodiments, the expansion yields at least a 10,000 fold, e.g., 15,000 fold, 20,000 fold, 25,000 fold, 30,000 fold, 35,000 fold, 40,000 fold, 45,000 fold, 50,000 fold, 55,000 fold, 60,000 fold, 65,000 fold, or 70,000 fold increase in the number of cells, e.g., the number of NK cells suitable for use in an MCB and/or an infusion-ready drug product relative to the number of starting MCB cells.

[0232] In embodiments where the cells are engineered during expansion and stimulation, as described herein, not all of the expanded and stimulated cells will necessarily be engineered successfully, e.g., transduced successfully, e.g., transduced successfully with a vector comprising a heterologous protein, e.g., a heterologous protein comprising a CAR and/or IL-15 as described herein. Thus, the methods described herein can further comprise sorting engineered cells, e.g., engineered cells described herein, away from non-engineered cells.

[0233] In some embodiments, the engineered cells, e.g., transduced cells, are sorted from the non-engineered cells, e.g., the non-transduced cells using a reagent specific to an antigen of the engineered cells, e.g., an antibody that targets an antigen of the engineered cells but not the non-engineered cells. In some embodiments, the antigen of the engineered cells is a component of a CAR, e.g., a CAR described herein.

[0234] Systems for antigen-based cell separation of cells are available commercially, e.g., the CliniMACS® sorting system (Miltenyi Biotec).

[0235] In some embodiments, the engineered cells, e.g., transduced cells, are sorted from the non-engineered cells, e.g., the non-transduced cells using flow cytometry.

[0236] In some embodiments, the sorted engineered cells are used as an MCB. In some embodiments, the sorted engineered cells are used as a component in an infusion-ready drug product.

[0237] In some embodiments, the engineered cells, e.g., transduced cells, are sorted from the non-engineered cells, e.g., the non-transduced cells using a microfluidic cell sorting method. Microfluidic cell sorting methods are described, for example, in Dalili et al., "A Review of Sorting, Separation and Isolation of Cells and Microbeads for Biomedical Applications: Microfluidic Approaches," *Analyst* 144:87 (2019).

[0238] In some embodiments, from or from about 1% to or to about 99% of the expanded and stimulated cells are engineered successfully, e.g., transduced successfully, e.g.,

transduced successfully with a vector comprising a heterologous protein, e.g., a heterologous protein comprising a CAR and/or IL-15 as described herein. In some embodiments, from or from about 1% to or to about 90%, from or from about 1% to or to about 80%, from or from about 1% to or to about 70%, from or from about 1% to or to about 60%, from or from about 1% to or to about 50%, from or from about 1% to or to about 40%, from or from about 1% to or to about 30%, from or from about 1% to or to about 20%, from or from about 1% to or to about 10%, from or from about 1% to or to about 5%, from or from about 5% to or to about 99%, from or from about 5% to or to about 90%, from or from about 5% to or to about 80%, from or from about 5% to or to about 70%, from or from about 5% to or to about 60%, from or from about 5% to or to about 50%, from or from about 5% to or to about 40%, from or from about 5% to or to about 30%, from or from about 5% to or to about 20%, from or from about 5% to or to about 10%, from or from about 10% to or to about 99%, from or from about 10% to or to about 90%, from or from about 10% to or to about 80%, from or from about 10% to or to about 70%, from or from about 10% to or to about 60%, from or from about 10% to or to about 50%, from or from about 10% to or to about 40%, from or from about 10% to or to about 30%, from or from about 10% to or to about 20%, from or from about 10% to or to about 99%, from or from about 20% to or to about 90%, from or from about 20% to or to about 80%, from or from about 20% to or to about 70%, from or from about 20% to or to about 60%, from or from about 20% to or to about 50%, from or from about 20% to or to about 40%, from or from about 20% to or to about 30%, from or from about 20% to or to about 99%, from or from about 30% to or to about 90%, from or from about 30% to or to about 80%, from or from about 30% to or to about 70%, from or from about 30% to or to about 60%, from or from about 30% to or to about 50%, from or from about 30% to or to about 40%, from or from about 30% to or to about 99%, from or from about 40% to or to about 90%, from or from about 40% to or to about 80%, from or from about 40% to or to about 70%, from or from about 40% to or to about 60%, from or from about 40% to or to about 50%, from or from about 40% to or to about 99%, from or from about 50% to or to about 90%, from or from about 50% to or to about 80%, from or from about 50% to or to about 70%, from or from about 50% to or to about 60%, from or from about 50% to or to about 99%, from or from about 60% to or to about 90%, from or from about 60% to or to about 80%, from or from about 60% to or to about 70%, from or from about 60% to or to about 99%, from or from about 70% to or to about 90%, from or from about 70% to or to about 80%, from or from about 70% to or to about 99%, from or from about 80% to or to about 90%, or from or from about 90% to or to about 99% of the expanded and stimulated cells are engineered successfully, e.g., transduced successfully, e.g., transduced successfully with a vector comprising a heterologous protein, e.g., a heterologous protein comprising a CAR and/or IL-15 as described herein.

[0239] In some embodiments, frozen cells of a first or second MCB are thawed and cultured. In some embodiments, a single vial of frozen cells of the first or second MCB e.g., a single vial comprising 800 or about 800 million cells, e.g., first or second MCB cells, are thawed and cultured. In some embodiments, the frozen first or second MCB cells are cultured with additional feeder cells to

produce cells suitable for use either as a second or third MCB or in an infusion-ready drug product. In some embodiments, the cells from the co-culture of the first or second MCB are harvested and frozen.

[0240] In some embodiments, the cells from the co-culture of the natural killer cell source, a first MCB, or a second MCB are harvested, and frozen in a cryopreservation composition, e.g., a cryopreservation composition described herein. In some embodiments, the cells are washed after harvesting. Thus, provided herein is a pharmaceutical composition comprising activated and stimulated NK cells, e.g., activated and stimulated NK cells produced by the methods described herein, e.g., harvested and washed activated and stimulated NK cells produced by the methods described herein and a cryopreservation composition, e.g., a cryopreservation composition described herein.

[0241] In some embodiments, the cells are mixed with a cryopreservation composition, e.g., as described herein, before freezing. In some embodiments, the cells are frozen in cryobags. In some embodiments, the cells are frozen in cryovials.

[0242] In some embodiments, the method further comprises isolating NK cells from the population of expanded and stimulated NK cells.

5. Engineering

[0243] In some embodiments, the method further comprises engineering NK cell(s), e.g., to express a heterologous protein, e.g., a heterologous protein described herein, e.g., a heterologous protein comprising a CAR and/or IL-15.

[0244] In some embodiments, engineering the NK cell(s) to express a heterologous protein described herein comprises transforming or transfecting, e.g., stably transforming or transfecting the NK cells with a vector comprising a polynucleic acid encoding a heterologous protein described herein. Suitable vectors are described herein.

[0245] In some embodiments, engineering the NK cell(s) to express a heterologous protein described herein comprises introducing the heterologous protein via gene editing (e.g., zinc finger nuclease (ZFN) gene editing, ARCUS gene editing, CRISPR-Cas9 gene editing, or megaTAL gene editing) combined with adeno-associated virus (AAV) technology.

[0246] In some embodiments, the NK cell(s) are engineered to express a heterologous protein described herein, e.g., during or after culturing the composition in a medium comprising feeder cells. For example, in some cases, engineering (e.g., transduction) occurs during the expansion and stimulation process described herein, e.g., during co-culturing NK cell source(s) and feeder cell(s) as described herein, e.g., at day 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18 of co-culturing.

[0247] In some embodiments, the method further comprises engineering NK cell(s), e.g., to express, over-express, knock-out, or knock-down gene(s) or gene product(s).

[0248] In some embodiments, the natural killer cells are not genetically engineered.

[0249] In some embodiments, the NK cell(s) are engineered (e.g., transduced) in a culture medium supplemented with a stimulating factor (e.g., as described herein). In some embodiments, the stimulation factor(s) are cytokine(s). In some embodiments, the cytokine(s) are selected from the group consisting of IL-2, IL-12, IL-15, IL-18, IL-21, IL-23, IL-27, IFN- α , IFN- β , and combinations thereof.

[0250] In some embodiments, the cytokine is IL-21. In some embodiments, the cytokine is IL-2. In some embodiments, the cytokines are a combination of IL-2 and IL-21. In some embodiments, the cytokines are a combination of IL-2, IL-18, and IL-21.

[0251] In some embodiments, the stimulating factor is added to the culture medium at the time of engineering (e.g., transduction). In some embodiments, the stimulating factor is added to the culture medium after the time of engineered (e.g., transducing), e.g., from 1 to 48 hours after engineering, e.g., from 1 to 36, 1 to 24, 1 to 12, 12 to 28, 12 to 36, 12 to 24, 24 to 48, 24 to 36, or 36 to 48 hours after engineering. In some embodiments, the stimulating factor is added to the culture medium both at the time of transduction and after the time of engineering (e.g., from 1 to 48 hours after transduction).

[0252] In some embodiments, the culture is supplemented with the stimulating factor after culturing in a medium comprising feeder cells. Thus, in some cases, the culture medium will contain feeder cells at the time of engineering (e.g., transduction). In some cases, the feeder cells are removed from the culture prior to supplementation with the stimulating factor. In some cases, the feeder cells are not removed from the culture prior to supplementation with the stimulating factor. In some cases, no additional feeder cells are added to the culture during engineering, whether or not any residual feeder cells are removed. In some cases, both additional feeder cells and a stimulating factor are added to the culture during engineering. In some cases, additional feeder cells are not added to the culture during engineering but stimulating factors are added to the culture during engineering.

E. Properties of Expanded and Stimulated NK Cells

[0253] After having been ex vivo expanded and stimulated, e.g., as described herein, the expanded and stimulated NK cell populations not only have a number/density (e.g., as described above) that could not occur naturally in the human body, but they also differ in their phenotypic characteristics, (e.g., gene expression and/or surface protein expression) with the starting source material or other naturally occurring populations of NK cells.

[0254] In some cases, the starting NK cell source is a sample derived from a single individual, e.g., a single cord blood unit that has not been ex vivo expanded. Therefore, in some cases, the expanded and stimulated NK cells share a common lineage, i.e., they all result from expansion of the starting NK cell source, and, therefore, share a genotype via clonal expansion of a population of cells that are, themselves, from a single organism. Yet, they could not occur naturally at the density achieved with ex vivo expansion and also differ in phenotypic characteristics from the starting NK cell source.

[0255] In some cases, the population of expanded and stimulated NK cells comprises at least 100 million expanded natural killer cells, e.g., 200 million, 250 million, 300 million, 400 million, 500 million, 600 million, 700 million, 750 million, 800 million, 900 million, 1 billion, 2 billion, 3 billion, 4 billion, 5 billion, 6 billion, 7 billion, 8 billion, 9 billion, 10 billion, 15 billion, 20 billion, 25 billion, 50 billion, 75 billion, 80 billion, 9 billion, 100 billion, 200 billion, 250 billion, 300 billion, 400 billion, 500 billion, 600 billion, 700 billion, 800 billion, 900 billion, 1 trillion, 2

trillion, 3 trillion, 4 trillion, 5 trillion, 6 trillion, 7 trillion, 8 trillion, 9 trillion, or 10 trillion expanded natural killer cells.

[0256] In some embodiments, the expanded and stimulated NK cells comprise at least 80%, e.g., at least 90%, at least 95%, at least 99%, or 100% CD56+CD3-cells.

[0257] In some embodiments, the expanded and stimulated NK cells are not genetically engineered.

[0258] In some embodiments, the expanded and stimulated NK cells do not comprise a CD16 transgene.

[0259] In some embodiments, the expanded and stimulated NK cells do not express an exogenous CD16 protein.

[0260] The expanded and stimulated NK cells can be characterized, for example, by surface expression, e.g., of one or more of CD16, CD56, CD3, CD38, CD14, CD19, NKG2D, NKp46, NKp30, DNAM-1, and NKp44.

[0261] The surface protein expression levels stated herein, in some cases are achieved without positive selection on the particular surface protein referenced. For example, in some cases, the NK cell source, e.g., a single cord unit, comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and is + enriched and CD3(+) depleted, e.g., by gating on CD56+CD3- expression, but no other surface protein expression selection is carried out during expansion and stimulation.

[0262] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% NKG2D+ cells.

[0263] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% NKp46+ cells.

[0264] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% NKp30+ cells.

[0265] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% DNAM-1+ cells.

[0266] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% NKp44+ cells.

[0267] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% CD94+ (KLRD1) cells.

[0268] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD3+ cells.

[0269] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises

less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD14+ cells.

[0270] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD19+ cells.

[0271] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CXCR+ cells.

[0272] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD122+ (IL2RB) cells.

[0273] As described herein, the inventors have demonstrated that, surprisingly, the NK cells expanded and stimulated by the methods described herein express CD16 at high levels throughout the expansion and stimulation process, resulting in a cell population with high CD16 expression. The high expression of CD16 obviates the need for engineering the expanded cells to express CD16, which is important for initiating ADCC, and, therefore, a surprising and unexpected benefit of the expansion and stimulation methods described herein. Thus, in some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise 50% or more, e.g., 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% CD16+ NK cells.

[0274] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and comprise 50% or more, e.g., 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% CD16+ NK cells.

[0275] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, expressing CD16 is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0276] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, expressing NKG2D is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0277] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, expressing NKp30 is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0278] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above,

expressing DNAM-1 is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0279] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, expressing Nkp44 is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0280] In some embodiments, the percentage of expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, expressing Nkp46 is the same or higher than the percentage of natural killer cells in the seed cells from umbilical cord blood.

[0281] As described herein, the inventors have also demonstrated that, surprisingly, the NK cells expanded and stimulated by the methods described herein express CD38 at low levels. CD38 is an effective target for certain cancer therapies (e.g., multiple myeloma and acute myeloid leukemia). See, e.g., Jiao et al., "CD38: Targeted Therapy in Multiple Myeloma and Therapeutic Potential for Solid Cancers," *Expert Opinion on Investigational Drugs* 29(11): 1295-1308 (2020). Yet, when an anti-CD38 antibody is administered with NK cells, because NK cells naturally express CD38, they are at risk for increased fratricide. The NK cells expanded and stimulated by the methods described herein, however, express low levels of CD38 and, therefore, overcome the anticipated fratricide. While other groups have resorted to engineering methods such as genome editing to reduce CD38 expression (see, e.g., Gurney et al., "CD38 Knockout Natural Killer Cells Expressing an Affinity Optimized CD38 Chimeric Antigen Receptor Successfully Target Acute Myeloid Leukemia with Reduced Effector Cell Fratricide," *Haematologica* doi: 10.3324/haematol.2020.271908 (2020), the NK cells expanded and stimulated by the methods described herein express low levels of CD38 without the need for genetic engineering, which provides a surprising and unexpected benefits, e.g., for treating CD38+ cancers with the NK cells expanded and stimulated as described herein, e.g., in combination with a CD38 antibody.

[0282] Thus, in some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprise less than or equal to 80% CD38+ cells, e.g., less than or equal to 75%, 70%, 65%, 60%, 55%, 50%, 45%, 40%, 35%, 30%, 25%, or 20% CD38+ cells.

[0283] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and comprise less than or equal to 80% CD38+ cells, e.g., less than or equal to 75%, 70%, 65%, 60%, 55%, 50%, 45%, 40%, 35%, 30%, 25%, or 20% CD38+ cells.

[0284] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and comprise less than or equal to 80% CD38+ cells, e.g., less than or equal to 75%, 70%, 65%, 60%, 55%, 50%, 45%, 40%, 35%, 30%, 25%, or 20% CD38+ cells, and 50% or more, e.g., 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% CD16+ NK cells.

[0285] In some embodiments, the expanded and stimulated NK cells, e.g., from expansion and stimulation of a single cord blood unit, e.g., as described above, comprises both the KIR B allele of the KIR receptor family and the 158 V/V variant of CD16 and comprise: i) 50% or more, e.g., 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% CD16+ NK cells; and/or ii) less than or equal to 80% CD38+ cells, e.g., less than or equal to 75%, 70%, 65%, 60%, 55%, 50%, 45%, 40%, 35%, 30%, 25%, or 20% CD38+ cells; and/or iii) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% NKG2D+ cells; and/or iv) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% Nkp46+ cells; and/or v) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% Nkp30+ cells; and/or vi) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% DNAM-1+ cells; and/or vii) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% Nkp44+ cells; and/or viii) at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% CD94+ (KLRD1) cells; and/or ix) less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD3+ cells; and/or x) less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD14+ cells; and/or xi) less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD19+ cells; and/or xii) less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CXCR+ cells; and/or xiii) less than or equal to 20%, e.g., less than or equal to 10%, less than or equal to 5%, less than or equal to 1% or 0% CD122+ (IL2RB) cells.

[0286] In some embodiments, feeder cells do not persist in the expanded and stimulated NK cells, though, residual signature of the feeder cells may be detected, for example, by the presence of residual cells (e.g., by detecting cells with a particular surface protein expression) or residual nucleic acid and/or proteins that are expressed by the feeder cells.

[0287] For example, in some cases, the methods described herein include expanding and stimulating natural killer cells using engineered feeder cells, e.g., eHuT-78 feeder cells described above, which are engineered to express sequences that are not expressed by cells in the natural killer cell source, including the natural killer cells. For example, the engineered feeder cells can be engineered to express at least one gene selected from the group consisting of 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and mutant TNFalpha (SEQ ID NO: 3) ("eHuT-78 cells"), or variants thereof.

[0288] While these feeder cells may not persist in the expanded and stimulated NK cells, the expanded and stimulated NK cells may retain detectable residual amounts of cells, proteins, and/or nucleic acids from the feeder cells. Thus, their residual presence in the expanded and stimulated NK cells may be detected, for example, by detecting the cells themselves (e.g., by flow cytometry), proteins that they express, and/or nucleic acids that they express.

[0289] Thus, also described herein is a population of expanded and stimulated NK cells comprising residual feeder cells (live cells or dead cells) or residual feeder cell cellular impurities (e.g., residual feeder cell proteins or portions thereof, and/or genetic material such as a nucleic

acid or portion thereof). In some cases, the expanded and stimulated NK cells comprise more than 0% and, but 0.3% or less residual feeder cells, e.g., eHuT-78 feeder cells.

[0290] In some cases, the expanded and stimulated NK cells comprise residual feeder cell nucleic acids, e.g., encoding residual 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and/or mutant TNFalpha (SEQ ID NO: 3) or portion(s) thereof. In some cases, the membrane bound IL-21 comprises a CD8 transmembrane domain

[0291] In some cases, the expanded and stimulated NK cells comprise a % residual feeder cells of more than 0% and less than or equal to 0.2%, as measured, e.g., by the relative proportion of a feeder cell specific protein or nucleic acid sequence (that is, a protein or nucleic acid sequence not expressed by the natural killer cells) in the sample. For example, by qPCR, e.g., as described herein.

[0292] In some embodiments, the residual feeder cells are CD4(+) T cells. In some embodiments, the residual feeder cells are engineered CD4(+) T cells. In some embodiments, the residual feeder cell cells are engineered to express at least one gene selected from the group consisting of 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and mutant TNFalpha (SEQ ID NO: 3) (“eHuT-78 cells”), or variants thereof. Thus, in some cases, the feeder cell specific protein is 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and/or mutant TNFalpha (SEQ ID NO: 3). And, therefore, the feeder cell specific nucleic acid is a nucleic acid encoding 4-1BBL (UniProtKB P41273, SEQ ID NO: 1), membrane bound IL-21 (SEQ ID NO: 2), and/or mutant TNFalpha (SEQ ID NO: 3), or portion thereof. In some cases, the membrane bound IL-21 comprises a CD8 transmembrane domain.

[0293] A wide variety of different methods can be used to analyze and detect the presence of nucleic acids or protein gene products in a biological sample. As used herein, “detecting” can refer to a method used to discover, determine, or confirm the existence or presence of a compound and/or substance (e.g., a cell, a protein and/or a nucleic acid). In some embodiments, a detecting method can be used to detect a protein. In some embodiments, detecting can include chemiluminescence or fluorescence techniques. In some embodiments, detecting can include immunological-based methods (e.g., quantitative enzyme-linked immunosorbent assays (ELISA), Western blotting, or dot blotting) wherein antibodies are used to react specifically with entire proteins or specific epitopes of a protein. In some embodiments, detecting can include immunoprecipitation of the protein (Jungblut et al., *J Biotechnol.* 31; 41(2-3):111-20 (1995); Franco et al., *Eur J Morphol.* 39(1):3-25 (2001)). In some embodiments, a detecting method can be used to detect a nucleic acid (e.g., DNA and/or RNA). In some embodiments, detecting can include Northern blot analysis, nucleic acid protection assays (NPA), in situ hybridization, or reverse transcription-polymerase chain reaction (RT-PCR) (Raj et al., *Nat. Methods* 5, 877-879 (2008); Jin et al., *J Clin Lab Anal.* 11(1):2-9 (1997); Ahmed, *J Environ Sci Health C Environ Carcinog Ecotoxicol Rev.* 20(2):77-116 (2002)).

[0294] Thus, also described herein, are methods for detecting a population of expanded and stimulated NK cells, e.g., expanded and stimulated using the methods described herein, that have been co-cultured with engineered feeder cells, e.g., eHuT-78 feeder cells described herein.

II. NATURAL KILLER CELL ENGINEERING

[0295] In some embodiments, the natural killer cells are engineered, e.g., to produce CAR-NK(s) and/or IL-15 expressing NK(s).

[0296] In some embodiments, the natural killer cells are engineered, e.g., transduced, during expansion and stimulation, e.g., expansion and stimulation described herein. In some embodiments, the natural killer cells are engineered during expansion and stimulation, e.g., during production of a MCB, as described herein. In some embodiments, the natural killer cells are engineered during expansion and stimulation, e.g., during production of NK cells suitable for use in an injection-ready drug product and/or during production of a MCB, as described above. Thus, in some embodiments, the NK cell(s) are host cells and provided herein are NK host cell(s) expressing a heterogeneous protein, e.g., as described herein.

[0297] In some embodiments, the natural killer cells are engineered prior to expansion and stimulation. In some embodiments, the natural killer cells are engineered after expansion and stimulation.

[0298] In some embodiments, the NK cells are engineered by transducing with a vector. Suitable vectors are described herein, e.g., lentiviral vectors, e.g., a lentiviral vectors comprising a heterologous protein, e.g., as described herein. In some embodiments, the NK cells are transduced during production of a first MCB, as described herein.

[0299] In some embodiments, the NK cell(s) are transduced at a multiplicity of infection of from or from about 1 to or to about 40 viral particles per cell. In some embodiments, the NK cell(s) are transduced at a multiplicity of infection of or of about 1, of or of about 5, of or of about 10, of or of about 15, of or of about 20, of or of about 25, of or of about 30, of or of about 35, or of or of about 40 viral particles per cell.

A. Chimeric Antigen Receptors

[0300] In some embodiments, the heterologous protein is a fusion protein, e.g., a fusion protein comprising a chimeric antigen receptor (CAR) is introduced into the NK cell, e.g., during the expansion and stimulation process.

[0301] In some embodiments, the CAR comprises one or more of: a signal sequence, an extracellular domain, a hinge, a transmembrane domain, and one or more intracellular signaling domain sequences. In some embodiments, the CAR further comprises a spacer sequence.

[0302] In some embodiments, the CAR comprises (from N- to C-terminal): a signal sequence, an extracellular domain, a hinge, a spacer, a transmembrane domain, a first signaling domain sequence, a second signaling domain sequence, and a third signaling domain sequence.

[0303] In some embodiments, the CAR comprises (from N- to C-terminal): a signal sequence, an extracellular domain, a hinge, a transmembrane domain, a first signaling domain sequence, a second signaling domain sequence, and a third signaling domain sequence.

[0304] In some embodiments the extracellular domain comprises an antibody or antigen-binding portion thereof.

[0305] In some embodiments, one or more of the intracellular signaling domain sequence(s) is a CD28 intracellular signaling sequence. In some embodiments, the CD28 intracellular signaling sequence comprises or consists of SEQ ID NO: 5.

[0306] In some embodiments, one or more of the intracellular signaling domain sequence(s) is an OX40L signaling sequence. See, e.g., Matsumura et al., "Intracellular Signaling of gp34, the OX40 Ligand: Induction of c-jun and c-fos mRNA Expression Through gp34 upon Binding of Its Receptor, OX40," *J. Immunol* 163:3007-11 (1999), which is hereby incorporated by reference in its entirety. In some embodiments, the OX40L signaling sequence comprises or consists of SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.

[0307] In some embodiments, one or more of the intracellular signaling sequence(s) is a CD3 ζ intracellular signaling domain sequence. In some embodiments, the CD3 ζ intracellular signaling sequence comprises or consists of SEQ ID NO: 13.

[0308] In some embodiments, the CAR comprises a CD28 intracellular signaling sequence (SEQ ID NO: 5), an OX40L intracellular signaling sequence (SEQ ID NO: 8), and a CD3 ζ intracellular signaling sequence (SEQ ID NO: 13).

[0309] In some embodiments, the CAR comprises an intracellular signaling domain comprising or consisting of SEQ ID NO: 25.

[0310] In some embodiments, the CAR does not comprise an OX40L intracellular signaling domain sequence.

[0311] In some embodiments, the CAR comprises a CD28 intracellular signaling sequence (SEQ ID NO: 5), and a CD3 ζ intracellular signaling sequence (SEQ ID NO: 13), but not an OX40L intracellular signaling domain sequence.

B. IL-15

[0312] In some embodiments, the NK cell is engineered to express IL-15, e.g., human IL-15 (UniProtKB #P40933; NCBI Gene ID #3600), e.g., soluble human IL-15 or an ortholog thereof, or a variant of any of the foregoing. In some embodiments, the IL-15 is expressed as part of a fusion protein further comprising a cleavage site. In some embodiments, the IL-15 is expressed as part of a polypeptide comprising a self-cleaving peptide such as T2A ribosomal skip sequence site See, e.g., Radcliffe & Mitrophanous, "Multiple Gene Products from a Single Vector: 'Self-Cleaving' 2A Peptides," *Gene Therapy* 11:1673-4 (2004); see also Liu et al., "Systematic Comparison of 2A Peptides for Cloning Multi-Genes in a Polycistronic Vector," *Scientific Reports* 7(1):2193 (2017).

[0313] In some embodiments, the IL-15 comprises or consists of SEQ ID NO: 22.

[0314] In some embodiments, the self-cleaving peptide is a 2A self-cleaving peptide. In some embodiments, the self-cleaving peptide is a T2A, P2A, E2A, or F2A self-cleaving peptide. In some embodiments, the self-cleaving peptide comprises SEQ ID NO: 16. In some embodiments, the self-cleaving peptide comprises or consists of SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 20, or SEQ ID NO: 21.

[0315] In some embodiments, the T2A cleavage site comprises or consists of SEQ ID NO: 17.

[0316] In some embodiments, the IL-15 is expressed as part of a fusion protein comprising a CAR, e.g., a CAR described herein.

[0317] In some embodiments, the fusion protein comprises (oriented from N-terminally to C-terminally): a CAR comprising, a cleavage site, and IL-15.

[0318] In some embodiments, the fusion protein comprises SEQ ID NO: 26.

C. Inhibitory Receptors

[0319] In some embodiments, the NK cell is engineered to alter, e.g., reduce, expression of one or more inhibitor receptor genes.

[0320] In some embodiments, the inhibitory receptor gene is a HLA-specific inhibitory receptor. In some embodiments, the inhibitory receptor gene is a non-HLA-specific inhibitory receptor.

[0321] In some embodiments, the inhibitor receptor gene is selected from the group consisting of KIR, CD94/NKG2A, LILRB1, PD-1, IRp60, Siglec-7, LAIR-1, and combinations thereof.

D. Polynucleic Acids, Vectors, and Host Cells

[0322] Also provided herein are polynucleic acids encoding the fusion protein(s) or portions thereof, e.g., the polynucleotide sequences encoding the polypeptides described herein, as shown in the Table of sequences provided herein

[0323] Also provided herein are vector(s) comprising the polynucleic acids, and cells, e.g., NK cells, comprising the vector(s).

[0324] In some embodiments, the vector is a lentivirus vector. See, e.g., Milone et al., "Clinical Use of Lentiviral Vectors," *Leukemia* 32:1529-41 (2018). In some embodiments, the vector is a retrovirus vector. In some embodiments, the vector is a gamma retroviral vector. In some embodiments, the vector is a non-viral vector, e.g., a piggyback non-viral vector (PB transposon, see, e.g., Wu et al., "piggyback is a Flexible and Highly Active Transposon as Compared to Sleeping Beauty, Tol2, and Mos1 in Mammalian Cells," *PNAS* 103(41):15008-13 (2006)), a sleeping beauty non-viral vector (SB transposon, see, e.g., Hudecek et al., "Going Non-Viral: the Sleeping Beauty Transposon System Breaks on Through to the Clinical Side," *Critical Reviews in Biochemistry and Molecular Biology* 52(4):355-380 (2017)), or an mRNA vector.

III. PHARMACEUTICAL COMPOSITIONS

[0325] Provided herein are pharmaceutical compositions comprising the natural killer cells described herein and dosage units of the pharmaceutical compositions described herein.

[0326] In some cases, the dosage unit comprises between 100 million and 1.5 billion cells, e.g., 100 million, 200 million, 300 million, 400 million, 500 million, 600 million, 700 million, 800 million, 900 million, 1 billion, 1.1 billion, 1.2 billion, 1.3 billion, 1.4 billion, or 1.5 billion.

[0327] Pharmaceutical compositions typically include a pharmaceutically acceptable carrier. As used herein the language "pharmaceutically acceptable carrier" includes saline, solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration.

[0328] In some embodiments, the pharmaceutical composition comprises: a) natural killer cell(s) described herein; and b) a cryopreservation composition.

[0329] Suitable cryopreservation compositions are described herein.

[0330] In some embodiments, the composition is frozen. In some embodiments, the composition has been frozen for at least three months, e.g., at least six months, at least nine

months, at least 12 months, at least 15 months, at least 18 months, at least 24 months, or at least 36 months.

[0331] In some embodiments, at least 60%, e.g., at least 70%, at least 80%, at least 90% at least 95%, at least 99%, or 100% of the natural killer cells are viable after being thawed.

[0332] In some embodiments, the pharmaceutical composition comprises: a) a cryopreservation composition described herein; and b) therapeutic cell(s).

[0333] In some embodiments, the therapeutic cell(s) are animal cell(s). In some embodiments, the therapeutic cell(s) are human cell(s).

[0334] In some embodiments, the therapeutic cell(s) are immune cell(s). In some embodiments, the immune cell(s) are selected from basophils, eosinophils, neutrophils, mast cells, monocytes, macrophages, neutrophils, dendritic cells, natural killer cells, B cells, T cells, and combinations thereof.

[0335] In some embodiments, the immune cell(s) are natural killer (NK) cells. In some embodiments, the natural killer cell(s) are expanded and stimulated by a method described herein, e.g., the CAR-NKs described herein.

[0336] In some embodiments, the pharmaceutical composition further comprises: c) a buffer solution. Suitable buffer solutions are described herein, e.g., as for cryopreservation compositions.

[0337] In some embodiments, the pharmaceutical composition comprises from or from about 1×10^7 to or to about 1×10^9 cells/mL. In some embodiments, the pharmaceutical composition comprises 1×10^8 cells/mL. In some embodiments, the pharmaceutical composition comprises about 1×10^8 cells/mL.

[0338] In some embodiments, the pharmaceutical composition further comprises an antibody or antigen binding fragment thereof, e.g., an antibody described herein.

[0339] Pharmaceutical compositions are typically formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral, e.g., intravenous, intradermal, subcutaneous, oral (e.g., inhalation), transdermal (topical), transmucosal, and rectal administration.

[0340] Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., Remington: The Science and Practice of Pharmacy, 21st ed., 2005; and the books in the series Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs (Dekker, NY). For example, solutions or suspensions used for parenteral, intradermal, or subcutaneous application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

[0341] Pharmaceutical compositions suitable for injectable use can include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dis-

persion. For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). In all cases, the composition must be sterile and should be fluid to the extent that easy syringability exists. It should be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate and gelatin.

[0342] Sterile injectable solutions can be prepared by incorporating the active compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle, which contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying, which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

IV. METHODS OF TREATMENT

[0343] The NK cells described herein, e.g., the CAR-NK cells described herein, find use for treating cancer or other proliferative disorders.

[0344] Thus, also provided herein are methods of treating a patient suffering from a disorder, e.g., a disorder associated with a cancer, comprising administering the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein.

[0345] Also provided herein are methods of preventing, reducing and/or inhibiting the recurrence, growth, proliferation, migration and/or metastasis of a cancer cell or population of cancer cells in a subject in need thereof, comprising administering the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein.

[0346] Also provided herein are methods of enhancing, improving, and/or increasing the response to an anticancer therapy in a subject in need thereof, comprising administering the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein.

[0347] Also provided herein are methods for inducing the immune system in a subject in need thereof comprising administering the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein.

[0348] The methods described herein include methods for the treatment of disorders associated with abnormal apoptotic or differentiative processes, e.g., cellular proliferative disorders or cellular differentiative disorders, e.g., cancer, including both solid tumors and hematopoietic cancers. Generally, the methods include administering a therapeutically effective amount of a treatment as described herein, to a subject who is in need of, or who has been determined to be in need of, such treatment. In some embodiments, the methods include administering a therapeutically effective amount of a treatment comprising NK cells, e.g., CAR-NK cells described herein.

[0349] As used herein, the terms “treatment,” “treat,” and “treating” refer to reversing, alleviating, delaying the onset of, or inhibiting the progress of a disorder associated with abnormal apoptotic or differentiative processes. For example, a treatment can result in a reduction in tumor size or growth rate. Administration of a therapeutically effective amount of a compound described herein for the treatment of a condition associated with abnormal apoptotic or differentiative processes will result in a reduction in tumor size or decreased growth rate, a reduction in risk or frequency of reoccurrence, a delay in reoccurrence, a reduction in metastasis, increased survival, and/or decreased morbidity and mortality, among other things. In some embodiments, treatment may be administered after one or more symptoms have developed. In other embodiments, treatment may be administered in the absence of symptoms. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment may also be continued after symptoms have resolved, for example to prevent or delay their recurrence.

[0350] As used herein, the terms “inhibition”, as it relates to cancer and/or cancer cell proliferation, refer to the inhibition of the growth, division, maturation or viability of cancer cells, and/or causing the death of cancer cells, individually or in aggregate with other cancer cells, by cytotoxicity, nutrient depletion, or the induction of apoptosis.

[0351] As used herein, “delaying” development of a disease or disorder, or one or more symptoms thereof, means to defer, hinder, slow, retard, stabilize and/or postpone development of the disease, disorder, or symptom thereof. This delay can be of varying lengths of time, depending on the history of the disease and/or subject being treated. As is evident to one skilled in the art, a sufficient or significant delay can, in effect, encompass prevention, in that the subject does not develop the disease, disorder, or symptom thereof. For example, a method that “delays” development of cancer is a method that reduces the probability of disease development in a given time frame and/or reduces extent of the disease in a given time frame, when compared to not using the method. Such comparisons may be based on clinical studies, using a statistically significant number of subjects.

[0352] As used herein, “prevention” or “preventing” refers to a regimen that protects against the onset of the disease or disorder such that the clinical symptoms of the disease do not develop. Thus, “prevention” relates to administration of a therapy (e.g., administration of a therapeutic substance) to a subject before signs of the disease are detectable in the subject and/or before a certain stage of the disease (e.g., administration of a therapeutic substance to a subject with a cancer that has not yet metastasized). The subject may be an

individual at risk of developing the disease or disorder, or at risk of disease progression, e.g., cancer metastasis. Such as an individual who has one or more risk factors known to be associated with development or onset of the disease or disorder. For example, an individual may have mutations associated with the development or progression of a cancer. Further, it is understood that prevention may not result in complete protection against onset of the disease or disorder. In some instances, prevention includes reducing the risk of developing the disease or disorder. The reduction of the risk may not result in complete elimination of the risk of developing the disease or disorder.

[0353] An “increased” or “enhanced” amount (e.g., with respect to antitumor response, cancer cell metastasis) refers to an increase that is 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.5, 3, 3.5, 4, 4.5, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, or 50 or more times (e.g., 100, 500, 1000 times) (including all integers and decimal points in between and above 1, e.g., 2.1, 2.2, 2.3, 2.4, etc.) an amount or level described herein. It may also include an increase of at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 100%, at least 150%, at least 200%, at least 500%, or at least 1000% of an amount or level described herein.

[0354] A “decreased” or “reduced” or “lesser” amount (e.g., with respect to tumor size, cancer cell proliferation or growth) refers to a decrease that is about 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.5, 3, 3.5, 4, 4.5, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, or 50 or more times (e.g., 100, 500, 1000 times) (including all integers and decimal points in between and above 1, e.g., 1.5, 1.6, 1.7, 1.8, etc.) an amount or level described herein. It may also include a decrease of at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, at least 100%, at least 150%, at least 200%, at least 500%, or at least 1000% of an amount or level described herein.

A. Disorders

[0355] Methods and manufactured compositions disclosed herein find use in targeting a number of disorders, such as cellular proliferative disorders. A benefit of the approaches herein is that allogenic cells are used to target specific cells. Unlike previous therapies, such as chemo or radiotherapy, using the approaches and pharmaceutical compositions herein, one is able to specifically target cells exhibiting detrimental proliferative activity, potentially without administering a systemic drug or toxin that impacts proliferating cells indiscriminately.

[0356] Examples of cellular proliferative and/or differentiative disorders include cancer, e.g., carcinoma, sarcoma, metastatic disorders or hematopoietic neoplastic disorders, e.g., leukemias. A metastatic tumor can arise from a multitude of primary tumor types, including but not limited to those of prostate, colon, lung, breast and liver origin.

[0357] As used herein, the terms “cancer”, “hyperproliferative” and “neoplastic” refer to cells having the capacity for autonomous growth, i.e., an abnormal state or condition characterized by rapidly proliferating cell growth. Hyperproliferative and neoplastic disease states may be categorized as pathologic, i.e., characterizing or constituting a disease state, or may be categorized as non-pathologic, i.e., a deviation from normal but not associated with a disease state. The term is meant to include all types of cancerous growths or oncogenic processes, metastatic tissues or malign-

nantly transformed cells, tissues, or organs, irrespective of histopathologic type or stage of invasiveness. "Pathologic hyperproliferative" cells occur in disease states characterized by malignant tumor growth. Examples of non-pathologic hyperproliferative cells include proliferation of cells associated with wound repair.

[0358] The terms "cancer" or "neoplasms" include malignancies of the various organ systems, such as affecting lung, breast, thyroid, lymphoid, gastrointestinal, and genito-urinary tract, as well as adenocarcinomas which include malignancies such as most colon cancers, renal-cell carcinoma, prostate cancer and/or testicular tumors, non-small cell carcinoma of the lung, cancer of the small intestine and cancer of the esophagus.

[0359] The term "carcinoma" is art recognized and refers to malignancies of epithelial or endocrine tissues including respiratory system carcinomas, gastrointestinal system carcinomas, genitourinary system carcinomas, testicular carcinomas, breast carcinomas, prostatic carcinomas, endocrine system carcinomas, and melanomas. In some embodiments, the disease is renal carcinoma or melanoma. Exemplary carcinomas include those forming from tissue of the cervix, lung, prostate, breast, head and neck, colon and ovary. The term also includes carcinosarcomas, e.g., which include malignant tumors composed of carcinomatous and sarcomatous tissues. An "adenocarcinoma" refers to a carcinoma derived from glandular tissue or in which the tumor cells form recognizable glandular structures.

[0360] The term "sarcoma" is art recognized and refers to malignant tumors of mesenchymal derivation.

[0361] Additional examples of proliferative disorders include hematopoietic neoplastic disorders. As used herein, the term "hematopoietic neoplastic disorders" includes diseases involving hyperplastic/neoplastic cells of hematopoietic origin, e.g., arising from myeloid, lymphoid or erythroid lineages, or precursor cells thereof. Preferably, the diseases arise from poorly differentiated acute leukemias, e.g., erythroblastic leukemia and acute megakaryoblastic leukemia. Additional exemplary myeloid disorders include, but are not limited to, acute promyeloid leukemia (APML), acute myelogenous leukemia (AML) and chronic myelogenous leukemia (CML) (reviewed in Vaickus, L. (1991) Crit Rev. in Oncol./Hematol. 11:267-97); lymphoid malignancies include, but are not limited to acute lymphoblastic leukemia (ALL) which includes B-lineage ALL and T-lineage ALL, chronic lymphocytic leukemia (CLL), prolymphocytic leukemia (PLL), hairy cell leukemia (HLL) and Waldenstrom's macroglobulinemia (WM). Additional forms of malignant lymphomas include, but are not limited to non-Hodgkin lymphoma and variants thereof, peripheral T cell lymphomas, adult T cell leukemia/lymphoma (ATL), cutaneous T-cell lymphoma (CTCL), large granular lymphocytic leukemia (LGF), Hodgkin's disease and Reed-Sternberg disease.

[0362] In some embodiments, the cancer is selected from the group consisting of: acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), adrenocortical carcinoma, Kaposi sarcoma, AIDS-related lymphoma, primary CNS lymphoma, anal cancer, appendix cancer, astrocytoma, typical teratoid/rhabdoid tumor, basal cell carcinoma, bile duct cancer, bladder cancer, bone cancer, brain tumor, breast cancer, bronchial tumor, Burkitt lymphoma, carcinoid, cardiac tumors, medulloblastoma, germ cell tumor, primary CNS lymphoma, cervical cancer, cholangiocarcinoma, chor-

doma, chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), chronic myeloproliferative neoplasms, colorectal cancer, craniopharyngioma, cutaneous T-cell lymphoma, ductal carcinoma in situ, embryonal tumors, endometrial cancer, ependymoma, esophageal cancer, esthesioneuroblastoma, Ewing sarcoma, extracranial germ cell tumor, extragonadal germ cell tumor, eye cancer (e.g., intraocular melanoma or retinoblastoma), fallopian tube cancer, fibrous histiocytoma of bone, osteosarcoma, gallbladder cancer, gastric cancer, gastrointestinal carcinoid tumor, gastrointestinal stromal tumors (GIST), germ cell tumors, gestational trophoblastic disease, hairy cell leukemia, head and neck cancer, heart tumor, hepatocellular cancer, histiocytosis, Hodgkin lymphomas, hypopharyngeal cancer, intraocular melanoma, islet cell tumors, pancreatic neuroendocrine tumors, kidney (renal cell) carcinoma, Langerhans cell histiocytosis, laryngeal cancer, leukemia, lip and oral cavity cancer, liver cancer, lung cancer (e.g., non-small cell lung cancer, small cell lung cancer, pleuropulmonary blastoma, and tracheobronchial tumor), lymphoma, male breast cancer, malignant fibrous histiocytoma of bone, melanoma, Merkel cell carcinoma, mesothelioma, metastatic cancer, metastatic squamous neck cancer, midline tract carcinoma, mouth cancer, multiple endocrine neoplasia syndromes, multiple myeloma/plasma cell neoplasms, mycosis fungoides, myelodysplastic syndromes, myelodysplastic/myeloproliferative neoplasms, myeloproliferative neoplasms, nasal cavity and paranasal sinus cancer, nasopharyngeal cancer, neuroblastoma, non-Hodgkin lymphoma, oral cancer, lip and oral cavity cancer, oropharyngeal cancer, osteosarcoma, malignant fibrous histiocytoma, ovarian cancer, pancreatic cancer, pancreatic neuroendocrine tumors, papillomatosis, paraganglioma, paranasal sinus and nasal cavity cancer, parathyroid cancer, penile cancer, pharyngeal cancer, pheochromocytomas, pituitary tumor, plasma cell neoplasm, multiple myeloma, pleuropulmonary blastoma, pregnancy and breast cancer, primary central nervous system lymphoma, primary peritoneal cancer, prostate cancer, rectal cancer, recurrent cancer, renal cell cancer, retinoblastoma, rhabdomyosarcoma, salivary gland cancer, sarcoma (e.g., childhood rhabdomyosarcoma, childhood vascular tumors, Ewing sarcoma, Kaposi sarcoma, osteosarcoma, soft tissue sarcoma, uterine sarcoma), Sezary syndrome, skin cancer, small intestine cancer, soft tissue sarcoma, squamous cell carcinoma, squamous neck cancer, stomach cancer, T-cell lymphomas, testicular cancer, throat cancer, nasopharyngeal cancer, oropharyngeal cancer, hypopharyngeal cancer, thymoma and thymic carcinomas, thyroid cancer, tracheobronchial tumors, transitional cell cancer of the renal pelvis and ureter, urethral cancer, uterine cancer, uterine sarcoma, vaginal cancer, vascular tumors, vulvar cancer, and Wilms tumor.

[0363] In some embodiments, the cancer is a solid tumor.

[0364] In some embodiments, the cancer is metastatic.

B. Patients

[0365] Suitable patients for the compositions and methods herein include those who are suffering from, who have been diagnosed with, or who are suspected of having a cellular proliferative and/or differentiative disorder, e.g., a cancer. Patients subjected to technology of the disclosure herein generally respond better to the methods and compositions herein, in part because the pharmaceutical compositions are allogeneic and target cells identified by the antigen binding

domain, rather than targeting proliferating cells generally. As a result, there is less off-target impact and the patients are more likely to complete treatment regimens without substantial detrimental off-target effects.

[0366] In some embodiments, the methods of treatment provided herein may be used to treat a subject (e.g., human, monkey, dog, cat, mouse) who has been diagnosed with or is suspected of having a cellular proliferative and/or differentiative disorder, e.g., a cancer. In some embodiments, the subject is a mammal. In some embodiments, the subject is a human.

[0367] In some embodiments, the mammal is selected from the group consisting of an armadillo, an ass, a bat, a bear, a beaver, a cat, a chimpanzee, a cow, a coyote, a deer, a dog, a dolphin, an elephant, a fox, a panda, a gibbon, a giraffe, a goat, a gopher, a hedgehog, a hippopotamus, a horse, a humpback whale, a jaguar, a kangaroo, a koala, a leopard, a lion, a llama, a lynx, a mole, a monkey, a mouse, a narwhal, an orangutan, an orca, an otter, an ox, a pig, a polar bear, a porcupine, a puma, a rabbit, a raccoon, a rat, a rhinoceros, a sheep, a squirrel, a tiger, a walrus, a weasel, a wolf, a zebra, a goat, a horse, and combinations thereof.

[0368] In some embodiments, the mammal is a human.

[0369] As used herein, a subject refers to a mammal, including, for example, a human.

[0370] The subject, e.g., the human subject, can be a child, e.g., from or from about 0 to or to about 14 years in age. The subject can be a youth, e.g., from or from about 15 to or to about 24 years in age. The subject can be an adult, e.g., from or from about 25 to or to about 64 years in age. The subject can be a senior, e.g., 65+ years in age.

[0371] In some embodiments, the subject may be a human who exhibits one or more symptoms associated with a cellular proliferative and/or differentiative disorder, e.g., a cancer, e.g., a tumor. Any of the methods of treatment provided herein may be used to treat cancer at various stages. By way of example, the cancer stage includes but is not limited to early, advanced, locally advanced, remission, refractory, reoccurred after remission and progressive. In some embodiments, the subject is at an early stage of a cancer. In other embodiments, the subject is at an advanced stage of cancer. In various embodiments, the subject has a stage I, stage II, stage III or stage IV cancer. The methods of treatment described herein can promote reduction or retraction of a tumor, decrease or inhibit tumor growth or cancer cell proliferation, and/or induce, increase or promote tumor cell killing. In some embodiments, the subject is in cancer remission. The methods of treatment described herein can prevent or delay metastasis or recurrence of cancer.

[0372] In some embodiments, the subject is at risk, or genetically or otherwise predisposed (e.g., risk factor), to developing a cellular proliferative and/or differentiative disorder, e.g., a cancer, that has or has not been diagnosed.

[0373] As used herein, an “at risk” individual is an individual who is at risk of developing a condition to be treated, e.g., a cellular proliferative and/or differentiative disorder, e.g., a cancer. Generally, an “at risk” subject may or may not have detectable disease, and may or may not have displayed detectable disease prior to the treatment methods described herein. “At risk” denotes that an individual has one or more so-called risk factors, which are measurable parameters that correlate with development of a disease or condition and are known in the art. For example, an at risk subject may have one or more risk factors, which are measurable parameters

that correlate with development of cancer. A subject having one or more of these risk factors has a higher probability of developing cancer than an individual without these risk factor(s). In general, risk factors may include, for example, age, sex, race, diet, history of previous disease, presence of precursor disease, genetic (e.g., hereditary) considerations, and environmental exposure. In some embodiments, the subjects at risk for cancer include, for example, those having relatives who have experienced the disease, and those whose risk is determined by analysis of genetic or biochemical markers.

[0374] In addition, the subject may be undergoing one or more standard therapies, such as chemotherapy, radiotherapy, immunotherapy, surgery, or combination thereof. Accordingly, one or more kinase inhibitors may be administered before, during, or after administration of chemotherapy, radiotherapy, immunotherapy, surgery or combination thereof.

[0375] In certain embodiments, the subject may be a human who is (i) substantially refractory to at least one chemotherapy treatment, or (ii) is in relapse after treatment with chemotherapy, or both (i) and (ii). In some of embodiments, the subject is refractory to at least two, at least three, or at least four chemotherapy treatments (including standard or experimental chemotherapies).

C. Lymphodepletion

[0376] In some embodiments, the patient is lymphodepleted before treatment.

[0377] Illustrative lymphodepleting chemotherapy regimens, along with correlative beneficial biomarkers, are described in WO 2016/191756 and WO 2019/079564, hereby incorporated by reference in their entirety. In certain embodiments, the lymphodepleting chemotherapy regimen comprises administering to the patient doses of cyclophosphamide (between 200 mg/m²/day and 2000 mg/m²/day) and doses of fludarabine (between 20 mg/m²/day and 900 mg/m²/day).

[0378] In some embodiments, lymphodepletion comprises administration of or of about 250 to about 500 mg/m² of cyclophosphamide, e.g., from or from about 250 to or to about 500, 250, 400, 500, about 250, about 400, or about 500 mg/m² of cyclophosphamide.

[0379] In some embodiments, lymphodepletion comprises administration of or of about 20 mg/m²/day to or to about 40 mg/m²/day fludarabine, e.g., 30 or about 30 mg/m²/day.

[0380] In some embodiments, lymphodepletion comprises administration of both cyclophosphamide and fludarabine.

[0381] In some embodiments, the patient is lymphodepleted by intravenous administration of cyclophosphamide (250 mg/m²/day) and fludarabine (30 mg/m²/day).

[0382] In some embodiments, the patient is lymphodepleted by intravenous administration of cyclophosphamide (500 mg/m²/day) and fludarabine (30 mg/m²/day).

[0383] In some embodiments, the lymphodepletion occurs no more than 5 days prior to the first dose of NK cells. In some embodiments, the lymphodepletion occurs no more than 7 days prior to the first dose of NK cells.

[0384] In some embodiments, lymphodepletion occurs daily for 3 consecutive days, starting 5 days before the first dose of NK cells (i.e., from Day -5 through Day -3).

[0385] In some embodiments, the lymphodepletion occurs on day -5, day -4 and day -3.

D. Administration

1. NK Cells

[0386] In some embodiments, the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein are administered as part of a pharmaceutical composition, e.g., a pharmaceutical composition described herein. Cells are administered after thawing, in some cases without any further manipulation in cases where their cryoprotectant is compatible for immediate administration. For a given individual, a treatment regimen often comprises administration over time of multiple aliquots or doses of NK cells drawn from a common batch or donor.

[0387] In some embodiments, the NK cells, e.g., the NK cells described herein, e.g., the CAR-NK cells described herein, are administered at or at about 1×10^6 to or to about 1×10^9 NK cells per dose. In some embodiments, the NK cells are administered at or at about 1×10^6 , at or at about 1×10^7 , at or at about 3×10^7 , at or at about 1×10^8 , at or at about 3×10^8 , or at or at about 1×10^9 cells per dose.

[0388] In some embodiments, the NK cells are administered weekly. In some embodiments, the NK cells are administered for or for about 8 weeks.

[0389] In some embodiments, the NK cells are administered between one and four times over the course of nine months.

[0390] In some embodiments, the NK cells are cryopreserved in an infusion-ready media, e.g., a cryopreservation composition suitable for intravenous administration, e.g., as described herein.

[0391] In some embodiments, the NK cells are cryopreserved in vials containing from or from about 1×10^7 to or to about 1×10^9 cells per vial. In some embodiments, the NK cells are cryopreserved in vials containing a single dose.

[0392] In some embodiments, the cells are thawed, e.g., in a 37° C. water bath, prior to administration.

[0393] In some embodiments, the thawed vial(s) of NK cells are aseptically transferred to a single administration vessel, e.g., administration bag using, e.g., a vial adapter and a sterile syringe. The NK cells can be administered to the patient from the vessel through a Y-type blood/solution set filter as an IV infusion, by gravity.

[0394] In some embodiments, the NK cells are administered as soon as practical, preferably less than 90 minutes, e.g., less than 80, 70, 60, 50, 40, 30, 20, or 10 minutes after thawing. In some embodiments, the NK cells are administered within 30 minutes of thawing.

[0395] In some embodiments, the pharmaceutical composition is administered intravenously via syringe.

[0396] In some embodiments, 1 mL, 4 mL, or 10 mL of drug product is administered to the patient intravenously via syringe.

2. Cytokines

[0397] In some embodiments, a cytokine is administered to the patient.

[0398] In some embodiments, the cytokine is administered together with the NK cells as part of a pharmaceutical composition. In some embodiments, the cytokine is administered separately from the NK cells, e.g., as part of a separate pharmaceutical composition.

[0399] In some embodiments, the cytokine is IL-2.

[0400] In some embodiments, the IL-2 is administered subcutaneously.

[0401] In some embodiments, the IL-2 is administered from between 1 to 4 or about 1 to about 4 hours following the conclusion of NK cell administration. In some embodiments, the IL-2 is administered at least 1 hour following the conclusion of NK cell administration. In some embodiments, the IL-2 is administered no more than 4 hours following the conclusion of NK cell administration. In some embodiments, the IL-2 is administered at least 1 hour after and no more than 4 hours following the conclusion of NK cell administration.

[0402] In some embodiments, the IL-2 is administered at up to 10 million IU/M², e.g., up to 1 million, 2 million, 3 million, 4 million, 5 million, 6 million, 7 million, 8 million, 9 million, or 10 million IU/m².

[0403] In some embodiments, the IL-2 is administered at or at about 1 million, at or at about 2 million, at or at about 3 million, at or at about 4 million, at or at about 5 million, at or at about 6 million, at or at about 7 million, at or at about 8 million, at or at about 9 million, at or at about 10 million IU/M².

[0404] In some embodiments, the IL-2 is administered at or at about 1×10^6 IU/M². In some embodiments, the IL-2 is administered at or at about 2×10^6 IU/M².

[0405] In some embodiments, less than 1×10^6 IU/M² IL-2 is administered to the patient.

[0406] In some embodiments, a flat dose of IL-2 is administered to the patient. In some embodiments, a flat dose of 6 million IU or about 6 million IU is administered to the patient.

[0407] In some embodiments, IL-2 is not administered to the patient.

E. Dosing

[0408] An “effective amount” is an amount sufficient to effect beneficial or desired results. For example, a therapeutic amount is one that achieves the desired therapeutic effect. This amount can be the same or different from a prophylactically effective amount, which is an amount necessary to prevent onset of disease or disease symptoms. An effective amount can be administered in one or more administrations, applications or dosages. A therapeutically effective amount of a therapeutic compound (i.e., an effective dosage) depends on the therapeutic compounds selected. The compositions can be administered one from one or more times per day to one or more times per week; including once every other day. The skilled artisan will appreciate that certain factors may influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health and/or age of the subject, and other diseases present. Moreover, treatment of a subject with a therapeutically effective amount of the therapeutic compounds described herein can include a single treatment or a series of treatments.

[0409] Dosage, toxicity and therapeutic efficacy of the therapeutic compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD50 (the dose lethal to 50% of the population) and the ED50 (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD50/ED50. Compounds

which exhibit high therapeutic indices are preferred. While compounds that exhibit toxic side effects may be used, care should be taken to design a delivery system that targets such compounds to the site of affected tissue in order to minimize potential damage to uninfected cells and, thereby, reduce side effects.

[0410] The data obtained from cell culture assays and animal studies can be used in formulating a range of dosage for use in humans. The dosage of such compounds may be within a range of circulating concentrations that include the ED50 with little or no toxicity. The dosage may vary within this range depending upon the dosage form employed and the route of administration utilized. For any compound used in the method of the invention, the therapeutically effective dose can be estimated initially from cell culture assays. A dose may be formulated in animal models to achieve a circulating plasma concentration range that includes the IC50 (i.e., the concentration of the test compound which achieves a half-maximal inhibition of symptoms) as determined in cell culture. Such information can be used to more accurately determine useful doses in humans. Levels in plasma may be measured, for example, by high performance liquid chromatography.

F. Combination Therapies

[0411] FOR CAR APPS INCLUDE THIS PARA: In some embodiments, the method comprises administering the NK

cells described herein, e.g., the CAR-NK cells described herein, in combination with another therapy, e.g., an antibody, an NK cell engager, an antibody drug conjugate (ADC), a chemotherapy drug, e.g., a small molecule drug, an immune checkpoint inhibitor, and combinations thereof.

1. Antibodies

[0412] In some embodiments, the other therapy is an antibody.

[0413] In some embodiments, the antibody binds to a target selected from the group consisting of CD20, HER-2, EGFR, CD38, SLAMF7, GD2, ALK1, AMHR2, CCR2, CD137, CD19, CD26, CD32b, CD33, CD37, CD70, CD73, CD74, CD248, CLDN6, Clever-1, c-MET, CSF-1R, CXCR4, DKK1, DR5, EphA3, FGFR2b, FGFR3, FLT3, FOLR1, Globo-H, Glypican3, GM1, Grp78, HER-3, HGF, IGF-1R, IL1RAP, IL-8R, ILT4, Integrin alpha V, M-CSF, Mesothelin, MIF, MUC1, MUC16, MUC5AC, Myostatin, NKG2A, NOTCH, NOTCH2/3, PIGF, PRL3, PSMA, ROR1, SEMA4D, Sialyl Lewis A, Siglec15, TGF- β , TNFR3, TRAIL-R2, VEGF, VEGFR1, VEGFR2, Vimentin, and combinations thereof.

[0414] Suitable antibodies include, but are not limited to those shown in Table 3.

TABLE 3

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
CD20	Rituxan	Rituximab	DLBCL/FL, NHL, CLL, RA, GPA, MPA	Du et al., Auto Immun Highlights (2017) 8(1): 12
CD20	Gazyva	Obinutuzumab	CLL, FL	Gagez et al., Curr Opin Oncol. 2014 September; 26(5): 484-91
CD20	Arzerra	Ofatumumab	CLL	Robak, Curr Opin Mol Ther. 2008 June; 10(3): 294-309
CD20	Ocrevus	Ocrelizumab	RMS, PPMS	Genovese et al., Arthritis Rheum. 2008 September; 58(9): 2652-61
CD20	Zevalin	Ibritumomab	NHL	Wiseman et al., Eur J Nucl Med. 2000 July; 27(7): 766-77
CD20		Veltuzumab	NHL, CLL	Kalaycio et al. Leuk Lymphoma. 2016; 57(4): 803-11
CD20	Bexxar	Tositumomab and Iodine I 131 tositumomab	NHL	Vose et al., J Clin Oncol. 2000 March; 18(6): 1316-23
CD20		Ublituximab	NHL, CLL, RMS	Sawas et al., Br J Haematol. 2017 April; 177(2): 243-253
HER-2	Herceptin	Trastuzumab	Breast, Gastric	Goldenberg, Clin Ther. 1999 February; 21(2): 309-18
HER-2	Perjeta	Pertuzumab	Breast	Agus et al., J Clin Oncol. 2005 Apr. 10; 23(11): 2534-43
HER-2	Margenza	Margetuximab	Breast	Bang et al., Ann Oncol. 2017 Apr. 1; 28(4): 855-861
EGFR	Erbix	Cetuximab	CRC, HNC	Jonker et al., N Engl J Med 2007; 357: 2040-2048
EGFR	Vectibix	Panitumumab	CRC	Gibson et al., Clin Colorectal Cancer. 2006 May; 6(1): 29-31
EGFR	Portrazza	Necitumumab	NSCLC	Kuenen et al., Clin Cancer Res. 2010 Mar. 15; 16(6): 1915-23
CD38	Darzalex	Daratumumab	MM	de Weers et al., J Immunol. 2011 Feb. 1; 186(3): 1840-8
CD38	Sarcclisa	Isatuximab	MM	Martin et al., Blood Cancer J. 2019 Mar. 29; 9(4): 41
SLAMF7	Empliciti	Elotuzumab	MM	Lonial et al., N Engl J Med 2015; 373: 621-631
GD2	Unituxin	Dinutuximab	NB	Hoy, Target Oncol. 2016 April; 11(2):247-53

TABLE 3-continued

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
GD2	Danyelza	Naxitamab	NB	Markham, Drugs. 2021 February; 81(2): 291-296
ALK1	PF-03446962	Ascricinivacumab	Liver cancer	Simonelli et al., Ann Oncol. 2016 September; 27(9): 1782-7
AMHR2	GM-102	Murlentamab	Ovarian Cancer	Leary et al., J Clin Oncol. 2019 37: 15_suppl, 2521-2521
CCR2	TAK-202	Plozalizumab	Atherosclerosis, Melanoma	Gilbert et al., Am J Cardiol. 2011 Mar. 15; 107(6): 906-11
CD137	BMS-663513	Urelumab	Melanoma, Myeloma, NSCLC	Segal et al., Clin Cancer Res. 2017 Apr. 15; 23(8): 1929-1936
CD137	PF-05082566	Utomilumab	Ovarian Cancer	Segal et al., Clin Cancer Res. 2018 Apr. 15; 24(8): 1816-1823
CD19	AMG103	Blinatumomab	ALL, NHL	Nadafi et al., Int J Mol Cell Med (2015) 4(3): 143-151
CD19	SAR3419	Coltuximab Ravtansine	ALL, NHL	Nadafi et al.
CD19	XmAb 5574	MOR208	ALL, NHL, CLL	Nadafi et al.
CD19	MEDI-551	MEDI-551	B-cell malignancies, CLL, Multiple Myeloma, Scleroderma	Nadafi et al.
CD19	SGN-19A	Denintuzumab	NHL	Nadafi et al.
CD19		Mafodotin DI-B4	B-cell malignancies	Nadafi et al.
CD19	Taplitumomabpaptox	Taplitumomabpapt OX	B-cell malignancies	Nadafi et al.
CD19	XmAb 5871	XmAb 5871	Autoimmune Diseases	Nadafi et al.
CD19	MDX-1342	MDX-1342	CLL, Rheumatoid Arthritis	Nadafi et al.
CD19	AFM11	AFM11	NHL	Nadafi et al.
CD19	ADCT-402	Loncastuximab Tesirine	ALL, NHL	Yu et al., Journal of Hematology & Oncology (2019) 12(94)
CD19	Monjuvi	Tafasitamab	NHL (DLBCL)	Hoy, Drugs. 2020 November; 80(16): 1731-1737
CD26	Begedina	Begelomab	Graft versus host disease	Bacigalupo et al., Bone Marrow Transplant. 2020 August; 55(8): 1580-1587
CD32b	BI-1206	BI-1206	BCL, CLL	Trial ID: NCT04219254
CD33	Mylotarg	Gemtuzumab Ozogamicin	AML	Stasi, Expert Opin Biol Ther. 2008 April; 8(4): 527-40
CD33	SGN-33	Lintuzumab	AML	Trial ID: NCT02998047
CD37	BI 836826	BI 836826	DLBCL, CLL, NHL	Trial ID: NCT02538614
CD37	IMGN529	Naratuximab emtansine	DLBCL, NHL	Yu et al., Journal of Hematology & Oncology (2019) 12(94)
CD37	AGS67E	AGS67E	DLBCL, NHL	Yu et al.
CD70	BMS-936561	MDX-1203	DLBCL, MCL	Yu et al.
CD70	SGN-75	Vorsetuzumab mafodotin	NHL	Yu et al.
CD73	MEDI9447	Oleclumab	Pancreatic cancer	Geoghegan et al., Mabs. 2016; 8(3): 454-67
CD73	AK119	AK119	Covid-19, Solid Tumors	Trial ID: NCT04516564
CD74	hLL1-DOX	Milatumumab doxorubicin	MM	Yu et al.
CD74	STRO-001	STRO-001	MM, NHL	Trial ID: NCT03424603
CD248	Ontecizumab	Ontuxizumab	MM, Soft tissue sarcoma	D'Angelo et al., Invest New Drugs. 2018 February; 36(1): 103-113
CLDN6	IMAB027	ASP1650	Testicular cancer	Trial ID: NCT03760081
Clever-1 c-MET	Clevegen MetMAb	Bexmarilimab Onartuzumab	Solid tumors NSCLC	Trial ID: NCT03733990 Hughes et al., Trends Cancer (2018) 4(2): 94-97

TABLE 3-continued

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
c-MET	AMG-102	Rilotumumab	Gastric cancer	Waddell et al., Immunotherapy. 2014; 6(12): 1243-53
CSF-1R	FPA-008	Cabiralizumab	MM, NSCLC	Trial ID: NCT04050462
CSF-1R	RG-7155	Emactuzumab	Ovarian cancer	Trial ID: NCT03708224
CSF-1R	IMC CS4	LY3022855	MM	Trial ID: NCT03153410
CSF-1R	AMB 051	AMG 820	Solid tumors	Trial ID: NCT04731675
CSF-1R	SNDX-6352	Axatilimab	Graft versus host disease	Trial ID: NCT04710576
CXCR4	BMS-936564	Ulocuplumab	Leukemia	Bobkov et al., Mol Pharmacol (2019) 96: 753-764
CXCR4	LY2624587	LY2624587	Metastatic Cancer	Bobkov et al.
CXCR4	PF-06747143	PF-06747143	AML	Bobkov et al.
CXCR4	F50067	hz515H7	MM	Bobkov et al.
CXCR4	MEDI3185	MEDI3185	Hematologic malignancies	Bobkov et al.
DKK1	DKN-01	DKN-01	Gastric cancer	Wall et al., Expert Opin Investig Drugs. 2020 July; 29(7): 639-644
DKK1	BHQ880	BHQ880	MM	Fulciniti et al., Blood. 2009 Jul. 9; 114(2):371-9
DR5	AD5-10	Zaptuzumab	Solid tumors	Zhang et al., Theranostics. 2019 Jul. 13; 9(18): 5412-5423
DR5	AMG655	Conatumumab	Colon, pancreatic cancer	Rosevear et al., Curr Opin Investig Drugs. 2010 June; 11(6): 688-98
DR5	PRO955780	Drozitumab	NHL, NSCLC	Kang et al., Clin Cancer Res. 2011 May 15; 17(10): 3181-92
DR5	ETR2-ST01	Lexatumumab	Solid tumors	Plummer et al., Clin Cancer Res. 2007 Oct. 15; 13(20): 6187-94
DR5	CS-1008	Tigatuzumab	Solid tumors	Reck et al., Lung Cancer. 2013 December; 82(3): 441-8
DR5		DS-8273a	Solid tumors	Forero et al., Invest New Drugs. 2017 June; 35(3): 298-306
Epha3	KB004	KB004	Glioblastoma	Swords et al., Leuk Res. 2016 November; 50: 123-131
FGFR2b	FPA-144	Bemarituzumab	Gastric cancer	Catenacci et al., J Clin Oncol. 2020 Jul. 20; 38(21): 2418-2426
FGFR2b	BAY 1187982	Aprutumab	Solid tumors	Kim et al., Target Oncol. 2019 October; 14(5): 591-601
FGFR2b	BAY-1179470	Aprutumab	Solid tumors	Trial ID: NCT01881217
FGFR3	LY3076226	LY3076226	Solid tumors	Trial ID: NCT02529553
FLT3	IMC-EB10	IMC-EB10	AML	Piloto et al., Cancer Res. 2006 May 1; 66(9): 4843-51
FOLRI	AGS 62P1	ASP1235	AML	Trial ID: NCT02864290
	MORAb-003	Farletuzumab	Ovarian cancer	Sato et al., Onco Targets Ther. 2016 Mar. 7; 9: 1181-8
Globo-H	OBI-833	OBI-833	Solid tumors	Trial ID: NCT02310464
Globo-H	OBI-888	OBI-888	Solid tumors	Trial ID: NCT03573544
Globo-H	OBI-999	OBI-999	Solid tumors	Trial ID: NCT04084366
Glypican3	GC33	Codrituzumab	Liver cancer	Abou-Alfa et al., J Hepatol. 2016 August; 65(2): 289-95
Glypican3		ERY974	Solid tumors	Ishiguro et al., Sci Transl Med. 2017 Oct. 4; 9(410)
GM1	BMS986012	BMS-986012	Lung cancer	Ponath et al., Clin Cancer Res. 2018 Oct. 15; 24(20): 5178-5189
Grp78	PAT-SM6	PAT-SM6	Multiple myeloma	Hensel et al., Melanoma Res. 2013 August; 23(4): 264-75
HER-3	U3-1402	Patritumab	NSCLC, Solid tumors	Hashimoto et al., Clin Cancer Res. 2019 Dec. 1; 25(23): 7151-7161
HGF	AMG-102	Rilotumumab	Solid tumors	Waddell et al., Immunotherapy. 2014; 6(12): 1243-53
HGF	AV-299	Ficlatuzumab	AML, NSCLC	Bauman et al., Cancers (Basel). 2020 June 11; 12(6): 1537
HGF	L2G7	TAK-701	Solid tumors	Okamoto et al., Mol Cancer Ther. 2010 October; 9(10): 2785-92
IGF-1R	IMC-A12	Cixutumumab	EWS, HCC	Chen et al., Chin J Cancer (2013) 32(5): 242-252
IGF-1R	CP-751	Figitumumab	EWS, ACC	Chen et al.
IGF-1R	MK-0646	Dalotuzumab	Colorectal cancer	Chen et al.

TABLE 3-continued

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
IGF-1R	AMG 479	Ganitumab	EWS, DRCT	Chen et al.
IGF-1R		R1507	EWS	Chen et al.
IGF-1R	AVE-1642	VRDN 001	MM, Breast cancer	Trial ID: NCT01233895
IL1RAP	CAN04	Nidanilimab	NSCLC	Awada et al., J Clin Oncol. 2019 May; 37: 2504-2504
IL-8R	BMS-986253	HuMax-IL8	Covid-19, NSCLC	Bihusic et al., J Immunother Cancer. 2019 Sep. 5; 7(1): 240
ILT4	JTX-8064	JTX-8064	Solid tumors	Trial ID: NCT04669899
Integrin alpha V	IMGN388	IMGN388	Solid tumors	Trial ID: NCT00721669
Integrin alpha V	CNTO-95	Intetumumab	MM	O'Day et al., Br J Cancer. 2011 Jul. 26; 105(3): 346-52
Integrin alpha V	EMD52579	Abituzumab	Colorectal cancer	Jiang et al., Mol Cancer Res. 2017 July; 15(7): 875-883
Integrin alpha V	MEDI-522	Etaracizumab	MM, Colorectal cancer	Hersey et al., Cancer. 2010 Mar. 15; 116(6): 1526-34
Integrin alpha V	VPI-2690B	VPI-2690B	Diabetic nephropathies	Trial ID: NCT02251067
M-CSF	MCS-110	Lacnotuzumab	Breast cancer, Gastric cancer	Pognan et al., J Pharmacol Exp Ther. 2019 June; 369(3): 428-442
Mesothelin	MORAb-009	amatuximab	Mesothelioma	Baldo et al., Onco Targets Ther. 2017 Nov. 8; 10: 5337-5353
Mesothelin	SS1(dsFv)-PE38	SS1P	Neoplasms	Hassan et al., J Clin Oncol. 2016 December; 34(34): 4171-4179
Mesothelin	BAY 94-9343	Anetumab ravtansine	Mesothelioma	Hassan et al., J Clin Oncol. 2020 Jun. 1; 38(16): 1824-1835
Mesothelin	RG7600	DMOT4039A	Pancreatic cancer, ovarian cancer	Hassan et al., J Clin Oncol. 2016 December; 34(34): 4171-4179
Mesothelin	BMS-986148	BMS-986148	Solid Tumors	Hassan et al., J Clin Oncol. 2016 December; 34(34): 4171-4179
MIF	BAX69	Imalumab	Colorectal cancer	Mahalingham et al., Br J Clin Pharmacol. 2020 September; 86(9): 1836-1848
MUC1	huC242-DM1	Cantuzumab mertansine	Pancreatic cancer	Tolcher et al., J Clin Oncol. 2003 Jan. 15; 21(2): 211-22
MUC1	hPAM4	Clivatuzumab	Pancreatic cancer	Liu et al., Oncotarget. 2015 Feb. 28; 6(6): 4274-85
MUC1	GT-MAB 2.5-GEXTM	Gatipotuzumab	Ovarian cancer	Heublin et al., Int J Mol Sci. 2019 Jan. 12; 20(2): 295
MUC1	mAb-AR20.5	AR20.5	Pancreatic cancer	de Bono et al., Ann Oncol. 2004 December; 15(12): 1825-33
MUC16	ACA 125	Abagovomab	Ovarian cancer	Sabbatini et al., J Clin Oncol. 2013 Apr. 20; 31(12): 1554-61
MUC16	DMUC5754 A	Softuzumab vedotin	Ovarian cancer	Liu et al., Ann Oncol. 2016 November; 27(11): 2124-2130
MUC16	DMUC4064 A	THIOMABTM	Ovarian cancer	Trial ID: NCT02146313
MUC5AC	PAM4	Clivatuzumab	PDAC	Gold et al., Molecular Cancer (2013) 12: 143
MUC5AC	NPC-1C	Ensituximab	Pancreatic cancer	Kim et al., Clin Cancer Res. 2020 Jul. 15; 26(14): 3557-3564
Myostatin	MYO-029	Stamulumab	Muscular atrophy, Muscular dystrophies	Trial ID: NCT00563810
Myostatin	PF-06252616	Domagrozumab	Duchenne muscular dystrophy	Wagner et al., Neuromuscul Disord. 2020 June; 30(6): 492-502
Myostatin	LY-2495655	Landogrozumab	Muscular atrophy, Pancreatic cancer	Golan et al., J Cachexia Sarcopenia Muscle. 2018 October; 9(5): 871-879
Myostatin	REGN-1033	Trevogrumab	Muscular atrophy	Trial ID: NCT01720576
Myostatin	SRK-015	Apitegromab	Spinal muscular atrophy	Trial ID: NCT03921528

TABLE 3-continued

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
NKG2A	IPH2201	Monalizumab	Breast cancer; NSCLC	Andre et al., Cell. 2018 Dec. 13; 175(7): 1731-1743
NOTCH	OMP-21M18	Demcizumab	NSCLC	Takebe et al., Pharmacol Ther (2014) 141(2): 140-149
NOTCH	REGN421/S AR153192	Enoticumab	NSCLC, Ovarian cancer	Takebe et al.
NOTCH	OPM-52M51	Brontictuzumab	Solid tumors	Takebe et al.
NOTCH 2/3	OMP-59R5	Tarextumab	Sarcomas, Rectal cancer	Takebe et al.
PIGF	RO5323441	TB-403	Solid tumors	Martinsson-Niskanen et al., Clin Ther. 2011 September; 33(9): 1142-9
PRL3	PRL3-ZUMAB	PRL3-zumab	Solid tumors	Trial ID: NCT04452955
PSMA	Capromab	Capromab pendetide	Prostate cancer	Trial ID: NCT00992745
PSMA	MT112	Pasotuxizumab	Prostate cancer	Hummel et al., Immunotherapy. 2021 February; 13(2): 125-141
PSMA		MDX1201-A488	Prostate cancer	Trial ID: NCT02048150
PSMA	APVO 414	MOR209/ES414	Prostate cancer	Hernandez-Hoyos et al., Mol Cancer Ther. 2016 September; 15(9): 2155-65
PSMA	ARX-517	ARX517	Prostate cancer	Trial ID: NCT04662580
PSMA	ADCT 401	MEDI3726	Prostate cancer	Cho et al., Mol Cancer Ther. 2018 October; 17(10): 2176-2186
PSMA		JNJ-63898081	Prostate cancer	Trial ID: NCT03926013
PSMA	PSMA TTC	BAY 2315497	Prostate cancer	Hammer et al., Clin Cancer Res. 2020 Apr. 15; 26(8): 1985-1996
PSMA		TLX592	Prostate cancer	Trial ID: NCT04726033
PSMA	DOTA-HUJ-591	Rosopatamab tetraxetan	Prostate cancer	Vallabhajosula et al., Curr Radiopharm. 2016; 9(1): 44-53
PSMA		PSMA ADC	Prostate cancer	Petrylak et al., Prostate. 2020 January; 80(1): 99-108
ROR1	UC-961	Cirmtuzumab	CLL, MCL	Choi et al., Cell Stem Cell. 2018 Jun. 1; 22(6): 951-959
SEMA4D	VX15/2503	Pepinemab	NSCLC, MM	Gupta et al., J Gastrointest Oncol. 2020 April; 11(2): 231-235
Sialyl Lewis A	MVT-5873	MVT-5873	Colorectal cancer	
Sialyl Lewis A	AbGn-7	AbGn-7	Gastric cancer	Trial ID: NCT01466569
Siglec 15	NC318	NC318	Solid tumors	Trial ID: NCT03665285
TGF-b		SRK-181	Solid tumors	Trial ID: NCT04291079
TGF-b	M-7824	Bintrafusp alfa	NSCLC, Solid tumors	Yoo et al., J Immunother Cancer. 2020 May; 8(1): e000564
TGF-b	GC-1008	Fresolimumab	MM	Rice et al., J Clin Invest. 2015 Jul. 1; 125(7): 2795-807
TGF-b		LY2382770	Diabetic nephropathie S	Trial ID: NCT01113801
TGF-b	NIS-793	NIS793	Pancreatic cancer	Trial ID: NCT04390763:
TGF-b		SAR439459	Solid tumors	Trial ID: NCT03192345
TGF-b		Metelimumab	Cancer, Scleroderma	Lord et al., MAbs. 2018 April; 10(3): 444-452
TGF-b	IMC TR1	LY3022859	Solid tumors	Tolcher et al., Cancer Chemother Pharmacol. 2017 April; 79(4): 673-680
TNFR3	Bamnercept	BG9924	Rheumatoid arthritis	Trial ID: NCT00664716
TRAIL-R2	CS-1008	Tigatuzumab	Breast cancer, NSCLC	Cheng et al., J Hepatol. 2015 October; 63(4): 896-904
TRAIL-R2	AMG-655	Conatumumab	Solid tumors	Bajaj et al., Expert Opin Biol Ther. 2011 November; 11(11): 1519-24
TRAIL-R2	PRO-95780	Drozitumab	NHL, NSCLC	Lima et al., Cancer Invest. 2012 December; 30(10): 727-31
TRAIL-R2	HGS-ETR2	Lexatumumab	Solid tumors	Plummer et al., Clin Cancer Res. 2007 Oct. 15; 13(20): 6187-94

TABLE 3-continued

Antibodies for Combination Therapy				
Target	Drug Name	Brand Name	Indication(s)	Reference
TRAIL-R2	TAS-266	TAS266	Solid tumors	Trial ID: NCT01529307
TRAIL-R2	GEN1029	Benufutamab	Solid tumors	Overdijk et al., Mol Cancer Ther. 2020 October; 19(10): 2126-2138
TRAIL-R2	RO-6874813	RG7386	Solid tumors	Brunker et al., Mol Cancer Ther. 2016 May; 15(5): 946-57
TRAIL-R2	JCT-205	INBRX-109	Solid tumors	Trial ID: NCT03715933
VEGF	Avastin	Bevacizumab	NSCLC, MM	Garcia et al., Cancer Treat Rev. 2020 June; 86: 102017
VEGF	Lucentis	Ranibizumab	Macular degeneration	Gross et al., JAMA Ophthalmol. 2018 Oct. 1; 136(10): 1138-1148
VEGFR1	IMC-18F1	Icrucumab	Breast cancer	LoRusso et al., Invest New Drugs. 2014 April; 32(2): 303-11
VEGFR2	Cyramza	Ramucirumab	NSCLC, Colorectal cancer	Khan et al., Expert Opin Biol Ther. 2019 November; 19(11): 1135-1141
VEGFR2	Tanibirumab	Olinvacimab	Glioblastoma	Lee et al., Drug Des Devel Ther. 2018 Mar. 8; 12: 495-504
VEGFR2		Gentuximab	Solid tumors	Chamie et al., JAMA Oncol. 2017 Jul. 1; 3(7): 913-920
VEGFR2	CDP-791	Alacizumab pegol	NSCLC	Trial ID: NCT00152477
VEGFR2	HLX-06	Vulinacimab	Solid tumors	Trial ID: NCT03494231
VEGFR2		MSB0254	Solid tumors	Trial ID: NCT04381325
VEGFR2		AK109	Solid tumors	Trial ID: NCT04547205
Vimentin	CLNH11	Pritumumab	Glioma	Babic et al., Hum Antibodies. 2018 Feb. 5; 26(2): 95-101
Vimentin		86C	Glioblastoma	Stouhalova et al., Cancers (2020) 12(1): 184

2. Small Molecule/Chemotherapy Drugs

[0415] In some embodiments, the additional therapy is a small molecule drug. In some embodiments, the additional therapy is a chemotherapy drug. In some embodiments, the additional therapy is a small molecule chemotherapy drug. Such small molecule drugs can include existing standard-of-care treatment regimens to which adoptive NK cell therapy is added. In some cases, the use of the NK cells described herein can enhance the effects of small molecule drugs, including by enhancing the efficacy, reducing the amount of small molecule drug necessary to achieve a desired effect, or reducing the toxicity of the small molecule drug.

[0416] In some embodiments, the drug is selected from the group consisting of

[0417] In some embodiments, the drug is [(1S,2S,3R,4S,7R,9S,10S,12R,15S)-4-acetyloxy-1,9,12-trihydroxy-15-[(2R,3S)-2-hydroxy-3-[(2-methylpropan-2-yl)oxycarbonylamino]-3-phenylpropanoyl]oxy-10,14,17,17-tetramethyl-11-oxo-6-oxatetracyclo[11.3.1.0^{3,10}.0^{4,7}]heptadec-13-en-2-yl]benzoate (docetaxel) or a pharmaceutically acceptable salt thereof.

[0418] In some embodiments, the drug is [(1S,2S,3R,4S,7R,9S,10S,12R,15S)-4,12-diacetyloxy-15-[(2R,3S)-3-benzamido-2-hydroxy-3-phenylpropanoyl]oxy-1,9-dihydroxy-10,14,17,17-tetramethyl-11-oxo-6-oxatetracyclo[11.3.1.0^{3,10}.0^{4,7}]heptadec-13-en-2-yl]benzoate (paclitaxel) or a pharmaceutically acceptable salt thereof.

[0419] In some embodiments, the drug is 6-N-(4,4-dimethyl-5H-1,3-oxazol-2-yl)-4-N-[3-methyl-4-([1,2,4]triazolo[1,5-a]pyridin-7-yloxy)phenyl]quinazoline-4,6-diamine (tucatinib) or a pharmaceutically acceptable salt thereof.

[0420] In some embodiments, the drug is pentyl N-[1-[(2R,3R,4S,5R)-3,4-dihydroxy-5-methyloxolan-2-yl]-5-fluoro-2-oxopyrimidin-4-yl]carbamate (capecitabine) or a pharmaceutically acceptable salt thereof.

[0421] In some embodiments, the drug is azanide; cyclobutane-1,1-dicarboxylic acid; platinum (2+) (carboplatin) or a pharmaceutically acceptable salt thereof.

[0422] In some embodiments, the drug is methyl (1R,9R,10S,11R,12R,19R)-11-acetyloxy-12-ethyl-4-[(12S,14R)-16-ethyl-12-methoxycarbonyl-1,10-diazatetracyclo[12.3.1.0^{3,11}.0^{4,9}]octadeca-3(11),4,6,8,15-pentaen-12-yl]-10-hydroxy-5-methoxy-8-methyl-8,16-diazapentacyclo[10.6.1.0^{1,9}.0^{2,7}.0^{16,19}]nonadeca-2,4,6,13-tetraene-10-carboxylate (vinorelbine) or a pharmaceutically acceptable salt thereof.

[0423] In some embodiments, the drug is N-[3-chloro-4-[(3-fluorophenyl) methoxy]phenyl]-6-[5-[(2-methylsulfonyl)ethylamino)methyl]furan-2-yl]quinazolin-4-amine (lapatinib) or a pharmaceutically acceptable salt thereof.

[0424] In some embodiments, the drug is (E)-N-[4-[3-chloro-4-(pyridin-2-ylmethoxy)anilino]-3-cyano-7-ethoxyquinolin-6-yl]-4-(dimethylamino)but-2-enamide (neratinib) or a pharmaceutically acceptable salt thereof.

[0425] In some embodiments, the drug is 6-acetyl-8-cyclopentyl-5-methyl-2-[(5-piperazin-1-ylpyridin-2-yl)amino]pyrido[2,3-d]pyrimidin-7-one (palbociclib) or a pharmaceutically acceptable salt thereof.

[0426] In some embodiments, the drug is 7-cyclopentyl-N,N-dimethyl-2-[(5-piperazin-1-ylpyridin-2-yl)amino]pyrrolo[2,3-d]pyrimidine-6-carboxamide (ribociclib) or a pharmaceutically acceptable salt thereof.

[0427] In some embodiments, the drug is N-[5-[(4-ethylpiperazin-1-yl)methyl]pyridin-2-yl]-5-fluoro-4-(7-fluoro-

2-methyl-3-propan-2-ylbenzimidazol-5-yl)pyrimidin-2-amine (abemaciclib) or a pharmaceutically acceptable salt thereof.

[0428] In some embodiments, the drug is (1R,9S,12S,15R,16E,18R,19R,21R,23S,24E,26E,28E,30S,32S,35R)-1,18-dihydroxy-12-[(2R)-1-[(1S,3R,4R)-4-(2-hydroxyethoxy)-3-methoxycyclohexyl]propan-2-yl]-19,30-dimethoxy-15,17,21,23,29,35-hexamethyl-11,36-dioxo-4-azatricyclo[30.3.1.0^{4,9}]hexatriaconta-16,24,26,28-tetraene-2,3,10,14,20-pentone (everolimus) or a pharmaceutically acceptable salt thereof.

[0429] In some embodiments, the drug is (2S)-1-N-[4-methyl-5-[2-(1,1,1-trifluoro-2-methylpropan-2-yl)pyridin-4-yl]-1,3-thiazol-2-yl]pyrrolidine-1,2-dicarboxamide (alpelisib) or a pharmaceutically acceptable salt thereof.

[0430] In some embodiments, the drug is 4-[[3-[4-(cyclopropanecarbonyl)piperazine-1-carbonyl]-4-fluorophenyl]methyl]-2H-phthalazin-1-one (olaparib) or a pharmaceutically acceptable salt thereof.

[0431] In some embodiments, the drug is (11S,12R)-7-fluoro-11-(4-fluorophenyl)-12-(2-methyl-1,2,4-triazol-3-yl)-2,3,10-triazatricyclo[7.3.1.0^{5,13}]trideca-1,5(13),6,8-tetraen-4-one (talazoparib) or a pharmaceutically acceptable salt thereof.

[0432] In some embodiments, the drug is N-[2-[2-(dimethylamino)ethyl-methylamino]-4-methoxy-5-[[4-(1-methylindol-3-yl)pyrimidin-2-yl]amino]phenyl]prop-2-enamid (osimertinib) or a pharmaceutically acceptable salt thereof.

[0433] In some embodiments, the drug is N-(3-chloro-4-fluorophenyl)-7-methoxy-6-(3-morpholin-4-ylpropoxy)quinazolin-4-amine (gefitinib) or a pharmaceutically acceptable salt thereof.

[0434] In some embodiments, the drug is N-(3-ethynylphenyl)-6,7-bis(2-methoxyethoxy)quinazolin-4-amine (erlotinib) or a pharmaceutically acceptable salt thereof.

[0435] In some embodiments, the drug is (E)-N-[4-(3-chloro-4-fluoroanilino)-7-[(3S)-oxolan-3-yl]oxyquinazolin-6-yl]-4-(dimethylamino)but-2-enamide (afatinib) or a pharmaceutically acceptable salt thereof.

[0436] In some embodiments, the drug is azane; dichloroplatinum (cisplatin, platino) or a pharmaceutically acceptable salt thereof.

[0437] In some embodiments, the drug is azanide; cyclobutane-1,1-dicarboxylic acid; platinum (2+) (carboplatin) or a pharmaceutically acceptable salt thereof.

[0438] In some embodiments, the drug is 4-amino-1-[(2R,4R,5R)-3,3-difluoro-4-hydroxy-5-(hydroxymethyl)oxolan-2-yl]pyrimidin-2-one (gemcitabine) or a pharmaceutically acceptable salt thereof.

[0439] In some embodiments, the drug is (2S)-2-[[4-[2-(2-amino-4-oxo-3,7-dihydropyrrolo[2,3-d]pyrimidin-5-yl)ethyl]benzoyl]amino]pentanedioic acid (pemetrexed) or a pharmaceutically acceptable salt thereof.

[0440] In some embodiments, the drug is N,N-bis(2-chloroethyl)-2-oxo-1,3,2λ⁵-oxazaphosphinan-2-amine (cyclophosphamide) or a pharmaceutically acceptable salt thereof.

[0441] In some embodiments, the drug is (2R,3S,4S,5R)-2-(6-amino-2-fluoropurin-9-yl)-5-(hydroxymethyl)oxolane-3,4-diol (fludarabine) or a pharmaceutically acceptable salt thereof.

[0442] In some embodiments, the drug is (7S,9S)-7-[(2R,4S,5S,6S)-4-amino-5-hydroxy-6-methyloxan-2-yl]oxy-6,9,11-trihydroxy-9-(2-hydroxyacetyl)-4-methoxy-8,10-di-

hydro-7H-tetracene-5,12-dione (doxorubicin) or a pharmaceutically acceptable salt thereof.

[0443] In some embodiments, the drug is methyl (1R,9R,10S,11R,12R,19R)-11-acetyloxy-12-ethyl-4-[(13S,15S,17S)-17-ethyl-17-hydroxy-13-methoxycarbonyl-1,11-diazatetracyclo[13.3.1.0^{4,12}.0^{5,10}]nonadeca-4(12),5,7,9-tetraen-13-yl]-8-formyl-10-hydroxy-5-methoxy-8,16-diazapentacyclo[10.6.1.0^{1,9}.0^{2,7}.0^{16,19}]nonadeca-2,4,6,13-tetraene-10-carboxylate (vincristine) or a pharmaceutically acceptable salt thereof.

[0444] In some embodiments, the drug is (8S,9S,10R,13S,14S,17R)-17-hydroxy-17-(2-hydroxyacetyl)-10,13-dimethyl-6,7,8,9,12,14,15,16-octahydrocyclopenta[a]phenanthrene-3,11-dione (prednisone) or a pharmaceutically acceptable salt thereof.

[0445] In some embodiments, the drug is N,3-bis(2-chloroethyl)-2-oxo-1,3,2λ⁵-oxazaphosphinan-2-amine (ifosfamide) or a pharmaceutically acceptable salt thereof.

[0446] In some embodiments, the drug is (5S,5aR,8aR,9R)-5-[[[(2R,4aR,6R,7R,8R,8aS)-7,8-dihydroxy-2-methyl-4,4a,6,7,8,8a-hexahydropyrano[3,2-d][1,3]dioxin-6-yl]oxy]-9-(4-hydroxy-3,5-dimethoxyphenyl)-5a,6,8a,9-tetrahydro-5H-[2]benzofuro[6,5-f][1,3]benzodioxol-8-one (etoposide) or a pharmaceutically acceptable salt thereof.

[0447] In some embodiments, the drug is (8S,9R,10S,11S,13S,14S,16R,17R)-9-fluoro-11,17-dihydroxy-17-(2-hydroxyacetyl)-10,13,16-trimethyl-6,7,8,11,12,14,15,16-octahydrocyclopenta[a]phenanthren-3-one (dexamethasone) or a pharmaceutically acceptable salt thereof.

[0448] In some embodiments, the drug is (8S,9R,10S,11S,13S,14S,16R,17R)-9-fluoro-11,17-dihydroxy-17-(2-hydroxyacetyl)-10,13,16-trimethyl-6,7,8,11,12,14,15,16-octahydrocyclopenta[a]phenanthren-3-one (cytarabine) or a pharmaceutically acceptable salt thereof.

3. NK Cell Engagers

[0449] In some embodiments, the additional therapy is an NK cell engager, e.g., a bispecific or trispecific antibody.

[0450] In some embodiments, the NK cell engager is a bispecific antibody against CD16 and a disease-associated antigen, e.g., cancer-associated antigen, e.g., an antigen of cancers described herein. In some embodiments, the NK cell engager is a trispecific antibody against CD16 and two disease-associated antigens, e.g., cancer-associated antigens, e.g., antigens of cancers described herein.

4. Checkpoint Inhibitors

[0451] In some embodiments, the additional therapy is an immune checkpoint inhibitor.

[0452] In some embodiments, the immune checkpoint inhibitor is selected from the group consisting of a PD-1 inhibitor, a PD-L1 inhibitor, a CTLA-4 inhibitor, and combinations thereof.

[0453] In some embodiments, the immune checkpoint inhibitor is selected from the group consisting of a PD-1 inhibitor, a PD-L1 inhibitor, a CTLA-4 inhibitor, a VISTA inhibitor, a BTLA inhibitor, a TIM-3 inhibitor, a KIR inhibitor, a LAG-3 inhibitor, a TIGIT inhibitor, a CD-96 inhibitor, a SIRPα inhibitor, and combinations thereof.

[0454] In some embodiments, the immune checkpoint inhibitor is selected from the group consisting of a PD-1 inhibitor, a PD-L1 inhibitor, a CTLA-4 inhibitor, a LAG-3 (CD223) inhibitor, a TIM-3 inhibitor, a B7-H3 inhibitor, a

B7-H4 inhibitor, an A2aR inhibitor, a CD73 inhibitor, a NKG2A inhibitor, a PVRIG/PVRL2 inhibitor, a CEACAM1 inhibitor, a CEACAM 5 inhibitor, a CEACAM 6 inhibitor, a FAK inhibitor, a CCL2 inhibitor, a CCR2 inhibitor, a LIF inhibitor, a CD47 inhibitor, a SIRP α inhibitor, a CSF-1 inhibitor, an M-CSF inhibitor, a CSF-1R inhibitor, an IL-1 inhibitor, an IL-1R3 inhibitor, an IL-RAP inhibitor, an IL-8

inhibitor, a SEMA4D inhibitor, an Ang-2 inhibitor, a CELVER-1 inhibitor, an Axl inhibitor, a phosphatidylserine inhibitor, and combinations thereof.

[0455] In some embodiments, the immune checkpoint inhibitor is selected from those shown in Table 4, or combinations thereof.

TABLE 4

Exemplary Immune Checkpoint Inhibitors	
Target	Inhibitor
LAG-3 (CD223)	LAG525 (IMP701), REGN3767 (R3767), BI 754,091, tebotelimab (MGD013), eftilagimod alpha (IMP321), FS118
TIM-3	MBG453, Sym023, TSR-022
B7-H3, B7-H4	MGC018, FPA150
A2aR	EOS100850, AB928
CD73	CPI-006
NKG2A	Monalizumab
PVRIG/PVRL2	COM701
CEACAM1	CM24
CEACAM 5/6	NEO-201
FAK	Defactinib
CCL2/CCR2	PF-04136309
LIF	MSC-1
CD47/SIRP	Hu5F9-G4 (SF9), ALX148, TTI-662, RRx-001
CSF-1	Lacnotuzumab (MCS110), LY3022855, SNDX-6352, emactuzumab
(M-CSF)/CSF-1R	(RG7155), pexidartinib (PLX3397)
IL-1 and IL-1R3	CAN04, Canakinumab (ACZ885)
(IL-IRAP)	
IL-8	BMS-986253
SEMA4D	Pepinemab (VX15/2503)
Ang-2	Trebananib
CLEVER-1	FP-1305
Axl	Enapotamab vedotin (EnaV)
Phosphatidylserine	Bavituximab

[0456] In some embodiments, the immune checkpoint inhibitor is an antibody.

[0457] In some embodiments, the PD-1 inhibitor is selected from the group consisting of pembrolizumab, nivolumab, toripalimab, cemiplimab-rwlc, sintilimab, and combinations thereof.

[0458] In some embodiments, the PD-L1 inhibitor is selected from the group consisting of atezolizumab, durvalumab, avelumab, and combinations thereof.

[0459] In some embodiments, the CTLA-4 inhibitor is ipilimumab.

[0460] In some embodiments, the PD-1 inhibitor is selected from the group of inhibitors shown in Table 5.

TABLE 5

Exemplary PD-1 Inhibitor Antibodies			
Name	Internal Name	Antigen	Company
nivolumab	Opdivo, ONO-4538, MDX-1106, BMS-936558, 5C4	PD-1	BMS, Medarex, Ono
pembrolizumab	Keytruda, MK-3475, SCH 900475, lambrolizumab	PD-1	Merck (MSD), Schering-Plough
toripalimab	JS001, JS-001, TAB001, Triprizumab	PD-1	Junmeng Biosciences, Shanghai Junshi, TopAlliance Bio
cemiplimab-rwlc	Libtayo, cemiplimab, REGN2810	PD-1	Regeneron, Sanofi

TABLE 5-continued

Exemplary PD-1 Inhibitor Antibodies			
Name	Internal Name	Antigen	Company
sintilimab	Tyvyt, IBI308	PD-1	Adimab, Innovent, Lilly
MEDI0680	AMP-514	PD-1	Amplimmune, Medimmune
LZM009		PD-1	Livzon
vudalimab	XmAb20717	CTLA4, PD-1	Xencor
SI-B003		CTLA4, PD-1	Sichuan Baili Pharma, Systimmune
Sym021	Symphogen patent anti-PD-1	PD-1	Symphogen
LVGN3616		PD-1	Lyvgen Biopharma
MGD019		CTLA4, PD-1	MacroGenics
MEDI5752		CTLA4, PD-1	Medimmune
CS1003		PD-1	CStone Pharma
IBI319	IBI-319	PD-1, Undisclosed	Innovent, Lilly
IBI315	IBI-315	HER2/neu, PD-1	Beijing Hanmi, Innovent
budigalimab	ABBV-181, PR-1648817	PD-1	Abbvie
Sunshine Guojian patent anti-PD-1 F520	609A	PD-1	Sunshine Guojian Pharma
RO7247669		PD-1	Shandong New Time Pharma
izuralimab	XmAb23104	LAG-3, PD-1	Roche
LY3434172		ICOS, PD-1	Xencor
SG001		PD-1, PD-L1	Lilly, Zymeworks
QL1706	PSB205	PD-1	CSPC Pharma
AMG 404	AMG404	CTLA4, PD-1	Sound Biologics
MW11		PD-1	Amgen
GNR-051		PD-1	Mabwell
Ningbo Cancer Hosp. anti-PD-1 CAR	HerinCAR-PD1	PD-1	IBC Generium
Chinese PLA Gen.Hosp. anti-PD-1		PD-1	Ningbo Cancer Hosp.
cetrelimab	JNJ-63723283	PD-1	Chinese PLA Gen.Hosp.
TY101		PD-1	Janssen Biotech
AK112		PD-1, VEGF	Tayu Huaxia
EMB-02		LAG-3, PD-1	Akeso
pidilizumab	CT-011, hBat-1, MDV9300	PD-1	EpimAb
sasanlimab	PF-06801591, RN-888	PD-1	CureTech, Medivation, Teva
balstilimab	AGEN2034, AGEN-2034	PD-1	Pfizer
geptanolimab	CBT-501, GB226, GB 226, Genolimzumab, Genormab	PD-1	Agenus, Ludwig Inst., Sloan-Kettering
RO7121661		PD-1, TIM-3	CBT Pharma, Genor
AK104		CTLA4, PD-1	Roche
pimivalimab	JTX-4014	PD-1	Akeso
IBI318	IBI-318	PD-1, PD-L1	Jounce
BAT1306		PD-1	Innovent, Lilly
ezabenlimab	BI754091, BI 754091	PD-1	Bio-Thera Solutions
Henan Cancer Hospital anti-PD-1	Teripalimab	PD-1	Boehringer
tebotelimab		LAG-3, PD-1	Henan Cancer Hospital
sindelizumab		PD-1	MacroGenics
dostarlimab	ANB011, TSR-042, ABT1	PD-1	Nanjing Medical U.
tislelizumab	BGB-A317	PD-1	AnaplysBio, Tesaro
spartalizumab	PDR001, BAP049	PD-1	BeiGene, Celgene
retifanlimab	MGA012, INCMGA00012	PD-1	Dana-Farber, Novartis
camrelizumab	SHR-1210	PD-1	Incyte, MacroGenics
zimberelimab	WBP3055, GLS-010, AB122	PD-1	Incyte, Jiangsu Hengrui, Shanghai Hengrui
penpulimab	AK105	PD-1	Arcus, Guangzhou Gloria Bio, Harbin Gloria
prolgolimab	BCD-100	PD-1	Pharma, WuXi Biologics
HX008		PD-1	Akeso, HanX Bio, Taizhou Hanzhong Bio
			Biocad
			Taizhou Hanzhong Bio, Taizhou HoudeAoke Bio

TABLE 5-continued

Exemplary PD-1 Inhibitor Antibodies			
Name	Internal Name	Antigen	Company
SCT-I10A serplulimab	HLX10	PD-1 PD-1	Sinocelltech Henlix

[0461] In some embodiments, the PD-L1 inhibitor is selected from the group of inhibitors shown in Table 6.

TABLE 6

Exemplary PD-L1 Inhibitor Antibodies			
Name	Internal Name	Antigen	Company
durvalumab	Imfinzi, MEDI-4736, MEDI4736	PD-L1	AstraZeneca, Celgene, Medimmune
atezolizumab	Tecentriq, MPDL3280A, RG7446, YW243.55.S70, RO5541267	PD-L1	Genentech
avelumab	Bavencio, MSB0010718C, A09-246-2	PD-L1	Merck Serono, Pfizer
AMP-224	CK-301, TG-1501	PD-L1	Amplimmune, GSK, Medimmune
cosibelimab		PD-L1	Checkpoint Therapeutics, Dana-Farber, Novartis, TG Therapeutics
Iodapolimab	LY3300054	PD-L1	Lilly
MCLA-145	ES101	4-1BB, PD-L1	Merus
FS118		LAG-3, PD-L1	f-star, Merck Serono
INBRX-105	ES101	4-1BB, PD-L1	Elpiscience, Inhibrx
Suzhou Nanomab patent anti-PD-L1		PD-L1	Suzhou Nanomab
MSB2311	HLX20, HLX09	PD-L1	Mabspace
BCD-13		PD-L1	Biocad
opucolimab	HLX20, HLX09	PD-L1	Henlix
IBI322		CD47, PD-L1	Innovent
LY3415244	IBI-322	PD-L1, TIM-3	Lilly, Zymeworks
GR1405	BMS-936559, 12A4	PD-L1	Genrix Biopharma
LY3434172		PD-1, PD-L1	Lilly, Zymeworks
CDX-527	BMS-936559, 12A4	CD27, PD-L1	Celldex
FS222		4-1BB, PD-L1	f-star
LDP	BMS-936559, 12A4	PD-L1	Dragonboat Biopharma
ABL503		4-1BB, PD-L1	ABL Bio
HB0025	BMS-936559, 12A4	PD-L1, VEGF	Huabo Biopharm
MDX-1105		PD-L1	Medarex
garivulimab	BGB-A333	PD-L1	BeiGene
GEN1046		4-1BB, PD-L1	BioNTech, Genmab
NM21-1480	M7824, MSB0011359C	4-1BB, PD-L1, Serum Albumin	Numab
bintrafusp alfa		PD-L1, TGFBR1	Merck Serono, NCI
pacmilimab	CX-072	PD-L1	CytomX
A167		PD-L1	Harbour Biomed Ltd., Sichuan Kelun Pharma
IBI318	IBI-318	PD-1, PD-L1	Innovent, Lilly
KN046		CTLA4, PD-L1	Alphamab
STI-3031	IMC-001	PD-L1	Sorrento
SHR-1701		PD-L1	Jiangsu Hengrui
LP002	ZKAB001	PD-L1	Taizhou HoudeAoke Bio
STI-1014		PD-L1	Lee's Pharm, Sorrento
envafolimab	KN035	PD-L1	Alphamab
adebreliumab	SHR-1316	PD-L1	Jiangsu Hengrui, Shanghai Hengrui
CS1001	CBT-502	PD-L1	CStone Pharma
TQB2450		PD-L1	CBT Pharma, Chia Tai Tianqing Pharma

[0462] In some embodiments, the CTLA-4 inhibitor is selected from the group of inhibitors shown in

TABLE 7

CTLA4 Inhibitor			
Name	Internal Name	Antigen	Company
ipilimumab	Yervoy, MDX-010, MDX101, 10D1, BMS-734016	CTLA4	Medarex
ATOR-1015	ADC-1015	CTLA4, OX40	Alligator
vudalimab	XmAb20717	CTLA4, PD-1	Xencor
SI-B003		CTLA4, PD-1	Sichuan Baili Pharma, Systimmune
MGD019		CTLA4, PD-1	MacroGenics
MEDI5752		CTLA4, PD-1	Medimmune
ADU-1604		CTLA4	Aduro
BCD-145	Q3W	CTLA4	Biocad
CS1002		CTLA4	CStone Pharma
REGN4659		CTLA4	Regeneron
pavunalimab	XmAb22841	CTLA4, LAG-3	Xencor
AGEN1181		CTLA4	Agenus
QL1706	PSB205	CTLA4, PD-1	Sound Biologics
ADG126		CTLA4	Adagene
KN044		CTLA4	Changchun Intelli-Crown
ONC-392		CTLA4	OncoImmune, Pfizer
BMS-986218		CTLA4	BMS
BMS-986249		CTLA4	BMS
BT-001	TG6030	CTLA4	BioInvent
quavonlimab	MK-1308	CTLA4	Merck (MSD)
zalifrelimab	AGEN1884	CTLA4	Agenus, Ludwig Inst., Sloan-Kettering
AK104		CTLA4, PD-1	Akeso
IBI310	IBI-310	CTLA4	Innovent
KN046		CTLA4, PD-L1	Alphamab
tremelimumab	ticilimumab, CP-675206, clone 11.2.1	CTLA4	Amgen, Medimmune, Pfizer

[0463] In some embodiments, the immune checkpoint inhibitor is a small molecule drug. Small molecule checkpoint inhibitors are described, e.g., in WO2015/034820A1, WO2015/160641A2, WO2018/009505 A1, WO2017/066227 A1, WO2018/044963 A1, WO2018/026971 A1, WO2018/045142 A1, WO2018/005374 A1, WO2017/202275 A1, WO2017/202273 A1, WO2017/202276 A1, WO2018/006795 A1, WO2016/142852 A1, WO2016/142894 A1, WO2015/033301 A1, WO2015/033299 A1, WO2016/142886 A2, WO2016/142833 A1, WO2018/051255 A1, WO2018/051254 A1, WO2017/205464 A1, US2017/0107216 A1, WO2017/070089A1, WO2017/106634A1, US2017/0174679 A1, US2018/0057486 A1, WO2018/013789 A1, US2017/0362253 A1, WO2017/192961 A1, WO2017/118762 A1, US2014/199334 A1, WO2015/036927 A1, US2014/0294898 A1, US2016/0340391 A1, WO2016/039749 A1, WO2017/176608 A1, WO2016/077518 A1, WO2016/100608 A1, US2017/0252432 A1, WO2016/126646 A1, WO2015/044900 A1, US2015/0125491 A1, WO2015/033303 A1, WO2016/142835 A1, WO2019/008154 A1, WO2019/008152 A1, and WO2019023575A1.

[0464] In some embodiments, the PD-1 inhibitor is 2-[[4-amino-1-[5-(1-amino-2-hydroxypropyl)-1,3,4-oxadiazol-2-yl]-4-oxobutyl]carbamoylamino]-3-hydroxypropanoic acid (CA-170).

[0465] In some embodiments, the immune checkpoint inhibitor is (S)-1-(3-Bromo-4-((2-bromo-[1,1'-biphenyl]-3-yl)methoxy)benzyl)piperidine-2-carboxylic Acid.

[0466] In some embodiments, the immune checkpoint inhibitor is a peptide. See, e.g., Sasikumar et al., "Peptide

and Peptide-Inspired Checkpoint Inhibitors: Protein Fragments to Cancer Immunotherapy," *Medicine in Drug Discovery* 8:100073 (2020).

V. VARIANTS

[0467] In some embodiments, the fusion protein(s) or components thereof described herein, or the NK cell genotypes described herein, are at least 80%, e.g., at least 85%, 90%, 95%, 98%, or 100% identical to the amino acid sequence of an exemplary sequence (e.g., as provided herein), e.g., have differences at up to 1%, 2%, 5%, 10%, 15%, or 20% of the residues of the exemplary sequence replaced, e.g., with conservative mutations, e.g., including or in addition to the mutations described herein. In preferred embodiments, the variant retains desired activity of the parent.

[0468] To determine the percent identity of two nucleic acid sequences, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second amino acid or nucleic acid sequence for optimal alignment and non-homologous sequences can be disregarded for comparison purposes). The length of a reference sequence aligned for comparison purposes is at least 80% of the length of the reference sequence, and in some embodiments is at least 90% or 100%. The nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position (as used herein nucleic acid "identity" is equivalent to nucleic acid "homology").

The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences.

[0469] Percent identity between a subject polypeptide or nucleic acid sequence (i.e. a query) and a second polypeptide or nucleic acid sequence (i.e. target) is determined in various ways that are within the skill in the art, for instance, using publicly available computer software such as Smith Waterman Alignment (Smith, T. F. and M. S. Waterman (1981) *J Mol Biol* 147:195-7); “BestFit” (Smith and Waterman, *Advances in Applied Mathematics*, 482-489 (1981)) as incorporated into GeneMatcher Plus™, Schwarz and Dayhof (1979) *Atlas of Protein Sequence and Structure*, Dayhof, M.O., Ed, pp 353-358; BLAST program (Basic Local Alignment Search Tool; (Altschul, S. F., W. Gish, et al. (1990) *J Mol Biol* 215:403-10), BLAST-2, BLAST-P, BLAST-N, BLAST-X, WU-BLAST-2, ALIGN, ALIGN-2, CLUSTAL, or Megalign (DNASTAR) software. In addition, those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the length of the sequences being compared. In general, for target proteins or nucleic acids, the length of comparison can be any length, up to and including full length of the target (e.g., 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 100%). For the purposes of the present disclosure, percent identity is relative to the full length of the query sequence.

[0470] For purposes of the present disclosure, the comparison of sequences and determination of percent identity between two sequences can be accomplished using a Blossum 62 scoring matrix with a gap penalty of 12, a gap extend penalty of 4, and a frameshift gap penalty of 5.

[0471] Conservative substitutions typically include substitutions within the following groups: glycine, alanine; valine, isoleucine, leucine; aspartic acid, glutamic acid, asparagine, glutamine; serine, threonine; lysine, arginine; and phenyl-alanine, tyrosine.

VI. DEFINITIONS

[0472] Unless defined otherwise, all terms of art, notations and other technical and scientific terms or terminology used herein are intended to have the same meaning as is commonly understood by one of ordinary skill in the art to which the claimed subject matter pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art.

[0473] Throughout this application, various embodiments may be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosure. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within

that range, for example, 1, 2, 3, 4, 5, and 6. This applies regardless of the breadth of the range.

[0474] As used in the specification and claims, the singular forms “a”, “an” and “the” include plural references unless the context clearly dictates otherwise. For example, the term “a sample” includes a plurality of samples, including mixtures thereof.

[0475] The terms “determining,” “measuring,” “evaluating,” “assessing,” “assaying,” and “analyzing” are often used interchangeably herein to refer to forms of measurement. The terms include determining if an element is present or not (for example, detection). These terms can include quantitative, qualitative or quantitative and qualitative determinations. Assessing can be relative or absolute. “Detecting the presence of” can include determining the amount of something present in addition to determining whether it is present or absent depending on the context.

[0476] The terms “subject,” “individual,” or “patient” are often used interchangeably herein.

[0477] The term “in vivo” is used to describe an event that takes place in a subject’s body.

[0478] The term “ex vivo” is used to describe an event that takes place outside of a subject’s body. An ex vivo assay is not performed on a subject. Rather, it is performed upon a sample separate from a subject. An example of an ex vivo assay performed on a sample is an “in vitro” assay.

[0479] The term “in vitro” is used to describe an event that takes place contained in a container for holding laboratory reagent such that it is separated from the biological source from which the material is obtained. In vitro assays can encompass cell-based assays in which living or dead cells are employed. In vitro assays can also encompass a cell-free assay in which no intact cells are employed.

[0480] As used herein, the term “about” a number refers to that number plus or minus 10% of that number. The term “about” a range refers to that range minus 10% of its lowest value and plus 10% of its greatest value.

[0481] As used herein, the term “buffer solution” refers to an aqueous solution consisting of a mixture of a weak acid and its conjugate base, or vice versa.

[0482] As used herein, the term “cell culture medium” refers to a mixture for growth and proliferation of cells in vitro, which contains essential elements for growth and proliferation of cells such as sugars, amino acids, various nutrients, inorganic substances, etc.

[0483] A buffer solution, as used herein, is not a cell culture medium.

[0484] As used herein, the term “bioreactor” refers to a culture apparatus capable of continuously controlling a series of conditions that affect cell culture, such as dissolved oxygen concentration, dissolved carbon dioxide concentration, pH, and temperature.

[0485] The term “vector,” as used herein, refers to a nucleic acid molecule capable of propagating another nucleic acid to which it is linked. The term includes the vector as a self-replicating nucleic acid structure as well as the vector incorporated into the genome of a host cell into which it has been introduced. Some vectors are suitable for delivering the nucleic acid molecule(s) or polynucleotide(s) of the present application. Certain vectors are capable of directing the expression of nucleic acids to which they are operatively linked. Such vectors are referred to herein as expression vectors.

[0486] The term “operably linked” refers to two or more nucleic acid sequence or polypeptide elements that are usually physically linked and are in a functional relationship with each other. For instance, a promoter is operably linked to a coding sequence if the promoter is able to initiate or regulate the transcription or expression of a coding sequence, in which case, the coding sequence should be understood as being “under the control of” the promoter.

[0487] The terms “host cell,” “host cell line,” and “host cell culture” are used interchangeably and refer to cells into which exogenous nucleic acid has been introduced, including the progeny of such cells. Host cells include “engineered cells,” “transformants,” and “transformed cells,” which include the primary engineered (e.g., transformed) cell and progeny derived therefrom without regard to the number of passages. Progeny may not be completely identical in nucleic acid content to a parent cell, but may contain mutations. Mutant progeny that have the same function or biological activity as screened or selected for in the originally transformed cell are included herein.

[0488] As appropriate, the host cells can be stably or transiently transfected with a polynucleotide encoding a fusion protein, as described herein.

[0489] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

VII. EXAMPLES

[0490] The following examples are included for illustrative purposes only and are not intended to limit the scope of the invention.

Example 1: CAR Costimulatory Structure Comprising OX40L

[0491] In some embodiments, the NK cells are CAR-NK cells. As shown in Error! Reference source not found., CAR-NKs comprising a co-stimulatory domain comprising OX40L exhibited greater cytotoxic potential than those without OX40L. In this example, the CAR-NK cells comprise an anti-HER2 scFv as described in US20200399397A1, which is hereby incorporated by reference in its entirety.

[0492] In vitro efficacy, proliferation, CAR expression, and in vitro efficacy was compared for NK-CARs comprising the CARs with anti-HER2 scFv with (SEQ ID NO: 64) and without OX40L (SEQ ID NO: 66) (Error! Reference source not found.). As shown in Error! Reference source not found. and Error! Reference source not found., both CAR-NK structures proliferated in tumor negative control cells and expressed the CAR. In vitro efficacy (CD107a expression, cytokine production, and percent lysis) is shown for various cell lines (HER2 positive and trastuzumab sensitive target cells (SKBR3, NCI-N87, and SKOV-3), HER2 positive and trastuzumab resistant target cells (HCC1954), and HER2 negative target cells (MDA-MB-468) in Error! Reference source not found., Error! Reference source not found., and Error! Reference source not found.). The OX40L containing CAR showed greater cytotoxic potential than that the CAR without OX40L against HER2 positive cell lines.

Example 2: AB-201

[0493] A CAR-NK expressing the fusion protein having SEQ ID NO: 27 was produced by transducing NK cells with a vector comprising SEQ ID NO: 28.

Example 3: CAR Constructs Expressing IL-15 have Increased Cytotoxicity

[0494] To investigate whether a CAR-NK which expresses both CAR and IL-15 has a synergistic effect on cytotoxicity, CAR-NK structures were generated as shown in Error! Reference source not found.

[0495] NK cells including NK, mock-NK, CAR-NK, CAR (t)-IL-15-NK, and CAR-IL-15-NK (AB-201) group were generated from cord blood of a healthy donor. The CD3 negative cells in cord blood unit were purified by using CD3+ cells positive isolation kit, and then they were used as seed cells.

[0496] The seed cells included CD56+ NK cells were stimulated with irradiated eHuT-78P cells and OKT3 and recombinant IL-2 (Proleukin) in complete serum-free medium (CellGro) on day 0.

[0497] The cultured NK cells were transduced by lentiviral vector on day 6 or 8 and were stimulated again with the irradiated eHuT-78P cells and OKT3 and IL-2 on day 14. At day 22, the cell groups were divided two groups again and cultured in the presence or absence of IL-2, respectively. Both transduced and non-transduced NK cells were cultured for 35 days in the presence of IL-2. As shown in Error! Reference source not found., IL-15 secreting transduced expressed the CAR stably until day 35. As shown in Error! Reference source not found., only IL-15 secreting transduced NK cells survived and expressed the CAR in the absence of IL-2. Moreover, the results show that expression of IL-15 increases the proportion of cells that are CAR+ in the absence of IL-2 (compare CAR-NK (43%) and CAR-IL-15-NK (91.3%) at day 29). As shown in Error! Reference source not found., NK cells not secreting IL-15 did not proliferate after day 22 and, as shown in Error! Reference source not found., their viability decreased rapidly after day 22. The results show that recombinant expression of IL-15 extends survival of NK cells even in the absence of IL-2.

[0498] To measure cytotoxicity, NK cells were cultured in the presence of IL-2 until day 22, and then cultured four more days in the absence of IL-2. At day 26, the NK cells were co-cultured with HCC1954 or SKOV3 at the E:T=0. 3:1 ratio for long-term killing assay (Error! Reference source not found.) or E:T=1:1 ratio for IFNg ELISA (Error! Reference source not found.) for 6 days in the absence of IL-2. CAR-IL-15-NK cells had a higher cytotoxicity than that of other NK cells. These results show that the CAR comprising an OX40L costimulatory domain and IL-15 expression exhibited better and more sustained killing activity. The CAR-NK cells lacking IL-15 expression showed significantly reduced killing activity compared to the cells expressing IL-15 under these conditions. On day 32, the amount of IFNg in the culture supernatant was measured. The CAR-IL-15-NK cells produced the highest amount of IFNg, and it was correlated to the cytolytic activity results in Error! Reference source not found.

[0499] To measure IL-15 production, the non-transduced or transduced NK cells were cultured in the presence of IL-2 until day 28, and then cultured four more days in the absence of IL-2. At day 32, the indicated NK cells were co-cultured

with HCC1954 or SKOV3 for 72 hours in the absence of IL-2, and the IL-15 levels in the cultured supernatant were determined by ELISA, as shown in Error! Reference source not found. These results show that co-culturing CAR-IL-15-NK cells in the presence of HER2+ target cells increased the amount of IL-15 produced by the NK cells. In contrast, the CAR (t)-IL-15-NK cells without costimulatory domains generated relatively constant amounts of basal IL-15 expression in both the absence of and presence of target cells. NK cells that lacked recombinant nucleic acids encoding IL-15 did not generate significant levels of IL-15 expression.

Example 4: Secretion of IL-15 Maintains the Survival of Bystander NK Cells

[0500] NK cells and CAR-IL-15-NK (AB-201) cells were generated from two different donors. Cells were transduced to produce CAR-IL-15-NK. At day 14, the NK cells were re-stimulated and CAR-IL-15-NK cells were re-stimulated and sorted. At day 19, the NK cells were CFSE labeled and co-cultures were created by mixing CFSE NK cells and CAR-IL-15-NK cells at a 1:1 ratio. Cocultures either with or without IL-2 were carried out for 5 days. Fixable viability dye (Invitrogen #65-0865) was used to detect viable cells. As shown in Error! Reference source not found., despite the absence of IL-2, the frequency of living NK cells that co-cultured with CAR-IL-15-NK cells was not decreased in the experiments using two different donors.

Example 5: Long Term Stability and Survival of CAR-NKs Expressing IL-15

[0501] NK cells expressing CARs with and without IL-15 (Error! Reference source not found.) were cultured as described in Example 4 to day 19. At day 19, they were cultured without IL-2. As shown in Error! Reference source not found., the CAR-expressing cells lacking IL-15 (3rd CAR) had significantly reduced CAR expression levels (e.g., only 55.2% of max at day 30) as compared to cells expressing the CAR with IL-15 (4th CAR) (e.g., 97.1% of max at day 56). The CAR-expressing cells lacking IL-15 (3rd CAR) also failed to persist as long, as none survived until day 44. As shown in Error! Reference source not found. and Error! Reference source not found., the cells expressing the CAR lacking IL-15 did not survive past day 37, whereas

the cells expressing the CAR with IL-15 survived at least up to day 62, and also maintained viability. As shown in Error! Reference source not found., cells expressing IL-15 persisted better than cells lacking heterologous expression of IL-15 in the presence target cells.

Example 6: AB-202 in Vitro Cytotoxicity

[0502] AB-202 displays significant cytotoxic activity against CD19+ target cells (Ramos). As shown in Error! Reference source not found., IFN-gamma secretion from AB-202 cells was increased when exposed to CD19+ Ramos tumor cells. As shown in Error! Reference source not found., AB-202 demonstrated greater activation and cytotoxic activity against CD19+ Ramos cell than non-CAR NK cells.

[0503] As shown in Error! Reference source not found., AB-202 displays significantly greater killing of CD19+ target cells than the CD-19 CAR-NKs based on Liu et al., “Use of CAR-Transduced Natural Killer Cells in CD19-Positive Lymphoid Tumors,” *N Engl J Med* 382:545-53 (2020). The AB-202 CAR (SEQ ID NO: 38) comprises an anti-CD19 scFv based on the FMC63 clone (SEQ ID NO: 37), an IgG1 hinge, an IgG1 spacer, a CD28 transmembrane domain, a CD28 costimulatory domain, an OX40L costimulatory domain, and a CD3ζ costimulatory domain. The NK cells expressing the AB-202 CAR were compared to NK cells expressing CD19-CAR CD28-CD3z-IL15 (SEQ ID NO: 39), which includes the same FMC63 clone (SEQ ID NO: 37), IgG1 hinge, IgG1 spacer, CD28 transmembrane domain, CD28 costimulatory domain, and CD3ζ costimulatory domain. Notably, the CD19-CAR CD28-CD3z-IL15 lacks the OX40L costimulatory domain.

[0504] The NK cells were incubated with Raji B-cell lymphoma cells at an E:T ratio of 0.3:1 for more than four days. The growth of Raji cells was monitored by measuring phase area confluence using Incucyte® live cell imaging of the cultures. Both AB-202 and the NK cells expressing CD19-CAR CD28-CD3z-IL15 inhibited Raji cell growth compared to Raji cells grown in the absence of NK cells (Error! Reference source not found.). AB-202 cells, however, inhibited significantly more growth and was more cytotoxic than NK cells expressing CD19-CAR CD28-CD3z-IL15. The results show that the use of a CAR costimulatory structure that includes the intracellular signaling portion of OX40L improves the cell killing activity of NK cells.

SEQUENCES	
SEQ ID NO: and DESCRIPTION	SEQUENCE
SEQ ID NO: 1 Sequence of 4-1BBL that can be expressed by feeder cells	MEYASDASLDPEAPWPPAPRARACRVLPWALVAGLLLLLLAAACAVFLACPWAVSGA RASPGSAASPRLREGPELSPDDPAGLLDLRQGMFAQLVAQNVLIDGPLSWYSDPGLA GVSLTGGLSYKEDTKELVVAKAGVYVFFQLELRRVVAGEGSGSVSLALHLQPLRSAA GAAALALTVDLPPASSEARNSAFGFQGRLLHLSAGQRLGVHLHTEARARHAWQLTQGA TVLGLFRVTPEIPAGLPSRSE
SEQ ID NO: 2 Sequence of a membrane bound IL-21 (mbIL-21) that can be expressed by feeder cells	MALPVTALLLPLALLHAARPQDRHMIRMRLIDIVDQLKNYVNDLVPEFLPAPEDVE TNCEWSAFSCFQKAQLKSANTGNNERIINVSIKLKRKPPSTNAGRQKHRLTCPSCD SYEKKPPKEFLERFKSLLQKMIHQHLSRTHGSEDSAKPTTTPAPRPPTPAPTIASQP LSLRPEACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLY

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SEQUENCES	
SEQ ID NO: and DESCRIPTION	SEQUENCE
SEQ ID NO: 3 Sequence of a mutated TNF alpha (mTNF-a) that can be expressed by feeder cells	MSTESMIRDVELAEEALPKKTGGPQGSRRCLFLSLFSFLIVAGATTLFCLLHFGVIGP QREEFPRDLISPLAQPVRRSSRTSPDKPVAHVVANPQAEGLQWLNRRANALLANG VELRDNLVVPSEGLYLIYSQVLFKGQCPSTHVLLTHTISRIVSYQTKVNLLSAIK SPCQRETPEGAEAKPWYEPIYLGGVFQLEKGDRLSAEINRPDYLDFAESGQVYFGIILA
SEQ ID NO: 4 Sequence of OX40L that can be expressed by feeder cells	MERVQPLEENVGNAARPRFERNKLLVASVIQGLLLLCFTYICLHFSALQVSHRYPR IQSIKVQFTEYKKEKGFILTSQKEDEIMKVQNNSVIINCDGFYLISLKGYSQEVNIS LHYQKDEEPLFQLKKVRSVNSLMVASLTYKDKVYLNVTDDNTSLDDFHVNGGELILIH QNPGEFCVL
SEQ ID NO: 5 CD28 intracellular signaling domain	RSKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFAAYRS
SEQ ID NO: 6 CD28 intracellular signaling domain	AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTACATGAACATGACTCCCCGCCGCC CCGGGCCACCCGCAAGCATTACCAGCCCTATGCCACCACGCGACTTCGCAGCCTA TCGCTCC
SEQ ID NO: 7 Codon Optimized CD28 intracellular signaling domain	CGGAGCAAGAGGTCCCGCCTGCTGCACAGCGACTATATGAACATGACCCACGGAGAC CCGGCCCTACACGGAACATTACCAGCCCTATGCTCCACCCCGGGACTTCGCAGCTTA CAGAAGT
SEQ ID NO: 8 OX40L intracellular signaling domain	ERVQPLEENVGNAARPRFERNK
SEQ ID NO: 9 OX40L intracellular signaling domain functional domain	LEENVGNAARPRFERNK
SEQ ID NO: 10 OX40L intracellular signaling domain functional domain	RPRFERNK
SEQ ID NO: 11 OX40L intracellular signaling domain	GAAAGGTTCAACCCCTGGAAGAGAATGTGGGAAATGCAGCCAGGCCAAGATTCGAGA GGAACAAG
SEQ ID NO: 12 Codon optimized OX40L intracellular signaling domain	GAAAGAGTGCAGCCCTGGAAGAGAATGTGGGAATGCCGCTCGCCCAAGATTGAAA GGAACAAA
SEQ ID NO: 13 CD3ζ signaling sequence	RVKFSRSADAPAYQQGNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGL YNELQKDKMAEAYSEIGMKGERRRGKHGDLGYQLSTATKDYDALHMQALPPR
SEQ ID NO: 14 CD3ζ signaling sequence	AGAGTGAAGTTCAGCAGGAGCGCAGACGCCCCCGCTACCAGCAGGGCCAGAACCAGC TCTATAACGAGCTCAATCTAGGACGAAGAGAGGAGTACGATGTTTTGGACAAGAGACG TGCCCGGACCCCTGAGATGGGGGGAAGCCGAGAAGGAAGAACCCCTCAGGAAGGCCTG TACAATGAAGTGCAGAAAGATAAGATGGCGGAGGCCTACAGTGAGATTGGGATGAAAG GCGAGCGCCGAGGGGCAAGGGGCACGATGGCCTTTACCAGGGTCTCAGTACAGCCAC CAAGGACACCTACGACGCCCTTACATGCAGGCCCTGCCCCCTCGC
SEQ ID NO: 15 Codon optimized CD3ζ signaling sequence	CGAGTGAAGTTCAGCAGGTCCGCCGACGCTCCTGCATACCAGCAGGGACAGAACCAGC TGTATAACGAGCTGAATCTGGGCCGGAGAGAGGAATACGACGTGCTGGACAAAAGGCG GGGCCGGACCCCGAAATGGGAGGGAAGCCACGACGGAAAAACCCCAAGGAGGCCTG TACAATGAGCTGCAAAAGGACAAAAATGGCCGAGGCTTATTCTGAAATCGGGATGAAGG GAGAGAGAAGGCGCGAAAAGGCCACGATGGCCTGTACCAGGGGCTGAGCACCGCTAC AAAGGACACCTATGATGCACCTGCACATGCAGGCCCTGCCCCCTCGC
SEQ ID NO: 16 2A cleavage motif	GDVEXNPGP
SEQ ID NO: 17 T2A cleavage site	GSGEGRGSLLTCDGVEENPGP

-continued

SEQUENCES	
SEQ ID NO: and DESCRIPTION	SEQUENCE
SEQ ID NO: 18 T2A cleavage site	GGCTCAGGTGAGGGGCGCGGGAGCCTGCTGACTTGTGGGGATGTAGAGGAAATCCTG GTCCT
SEQ ID NO: 19 P2A cleavage site	GSGATNFSLLKQAGDVEENPGP
SEQ ID NO: 20 E2A cleavage site	GSGQCTNYALLKLAGDVESNPGP
SEQ ID NO: 21 F2A cleavage site	GSGVKQTLNFDLLKLAGDVESNPGP
SEQ ID NO: 22 IL-15	MRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFILGCFsAGLPKTEANWVNVISDLK KIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLQLQVISLES GDASIHDTVENLI I LANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMPINTS-
SEQ ID NO: 23 IL-15	ATGAGAATCAGCAAACACACCTCCGGAGCATATCAATCCAGTGTTACTTGTGCGCTTC TTTTGAACTCCCATTTCTCACCAGGAGGAGGCATTATGTGTTTCATATTGGGGTGCTT TAGTGCTGGGCTTCCGAAACGGAAGCTAACTGGGTAAACGTCATCAGTGACCTAAA AAAATTGAGGATCTTATCCAATCAATGCACATCGACGCGACTCTCTACACAGAATCTG ACGTACACCCGTCATGCAAAGTCACGGCAATGAAGTGTTCCTTCGAGCTCCAAGT AATTTCCCTGGAGTCTGGCGATGCCTCCATCCACGATACGGTTGAAAATCTGATTATA TTGGCCAACAATAGCCTCAGTTCTAACGGTAACGTGACTGAAAGTGGCTGCAAAGAGT GCGAAGAGCTCGAAGAAAAGAAATCAAGGAGTTCTCCAATCATTTGTTTACATTTGT GCAAATGTTTATCAACACCTCTTGA
SEQ ID NO: 24 IL-15	ATGCGCATAAAGTAAGCCTCATCTGCGGTCCATTCTATACAATGTTATCTGTGCTTGC TTTTGAACTCCCATTTCTTACGGAAGCAGGCATTATGTGTTTCATTTCTGGGTGTTT TTCTGCCGGGCTGCCAAAACCGAGGCCAACTGGGTCAACGTGATCAGCGACCTCAAG AAGATCGAGGATTGATTCAAAGTATGCATATAGACGCCACACTCTATACTGAGTCCG ACGTTCAACCCGAGTTGTAAAGTTACGGCTATGAAGTGCTTTTGTGGAACCTCAGGT GATTTCCCTTGAATCCGGCGATGCGAGCATCCACGATACGGTAGAGAATCTTATTATT CTGGCGAATAATTCTCTGTCTTCAAATGGGAATGTAACGTGAGAGCGGTGTAAAGAAT GCGAAGAACTTGAAGAAAAGAAATCAAGGAATTTCTTCAGAGTTTCGTGCATATTGT TCAAATGTTTCATCAACACATCCTGA
SEQ ID NO: 25 CD28/OX40L/CD ζ	RSKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFAAYRSERVQPLEENVGNAARPR FERNKRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKN PQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGHGLYQGLSTATKDTYDALHMQALPP R
SEQ ID NO: 26 CD28/OX40L/CD ζ /T2 A/IL1-5	RSKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFAAYRSERVQPLEENVGNAARPR FERNKRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKN PQEGLYNELQKDKMAEAYSEIGMKGERRRGKGGHGLYQGLSTATKDTYDALHMQALPP RSGGEGRGSLLTCGDVEENPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFIL GCFsAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCELLE LQVISLES GDASIHDTVENLI I LANNSLSSNGNVTESGCKECELEEKNIKEFLQSFV HIVQMPINTS-
SEQ ID NO: 27 HER2 CAR with T2A and IL-15	MALPVTALLLPLALLLHAARPD IQMTQSPSSLSASVGDVRTITCKASQDINSYLSWFQ QKPGKAPKTLIYRANRLVDGVPsRFSGSGSQDYTLTISLQPEDFATYYCLQYDEFP WTFGQGTKEI KGGGSGGGGSGGGGSQVLVQSGSELKKPGASVKVSCKASGYTFN YGVNwVRQAPGQGLEWMGWINTHTGEPTYAEEFKGRFVFSLDTSVSTAYLQISSLKAE DTAVYYCARDYYVRVDYWGQGTTVTVSSAKPTTTPAPRPPTPAPTIASQPLSLRPEA CRPAAGGAVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLHSDY MNMTPRRPGPTRKHYPYAPPRDFAAYRSERVQPLEENVGNAARPRFERNKRVKESRS ADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGGHGLYQGLSTATKDTYDALHMQALPPRSGGEGRGSLLT CGDVEENPGPMRISKPHLRSISIQCYLCLLLNSHFLTEAGIHVFILGCFsAGLPKTEA NwVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLQLQVISLES GDAS IHDTVENLI I LANNSLSSNGNVTESGCKECELEEKNIKEFLQSFVHIVQMPINTS-
SEQ ID NO: 28 Codon Optimized HER2 CAR with T2A and IL-15	ATGGCACTTCTGTACAGCCCTCCTGCTCCCACTGGCTTGTGCTGCTGCATGCTGCAC GACCGGACATCCAGATGACCCAGAGCCCGAGCAGCTGAGCGCCAGCGTGGGCGACAG AGTGACCATCACCTGCAAGGCCAGCCAGGACATCAACAGCTACCTGAGCTGGTTCCAG CAGAAGCCCCGGAAGCCCCCAAGACCTGATCTACAGAGCCAACAGACTGGTGGACG GCGTGCCAGCAGATTGACGGGAGCGGAGCGGCGAGGACTACACCTGACCATCAG CAGCCTGCAGCCGAGGACTTCGCCACCTACTACTGCTGCAGTACGACGAGTTCCCC TGGACCTTCGGCCAGGGCACCAGGTGGAGATCAAGGGTGGCGGTGGATCGGGCGGTG GTGGATCTGGAGGAGTGGCTCCAGGTGCAGCTGGTGCAGAGCGGCAGCGAGCTGAA

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SEQUENCES	
SEQ ID NO: and DESCRIPTION	SEQUENCE
	GAAGCCCGCGCCAGCGTGAAGGTGAGCTGCAAGGCCAGCGGTACACCTTCACCAAC TACGGCTGAACTGGGTGAGACAGGCCCGCGCCAGGGCTGGAGTGGATGGGCTGGA TCAACACCCACACCGGCGAGCCACCTACGCGGAGGAGTTCAAGGGCAGATTCTGTGT CAGCCTGGACACCAGCGTGAGCACCGCCTACCTGCAGATCAGCAGCCTGAAGGCCGAG GACACCGCGCTGTACTACTGCGCCAGAGACGACTACTACGTGAGAGTGGACTACTGGG GCCAGGGCACCACCGTGACCGTGAGCAGCGCAAAACCTACCACAACCTCTGCACCACG CCCCCCTACTCCAGCACCTACCATCGCATCTCAGCCACTGAGTCTGCGACCAGAGGCC TGCCGCGCCGCGCCGCGGGGCGCTCCATACCAGAGGGCTGGACTTTGCCTGCGATT TTTGGGTCTGTGGTCTGTGGGAGGGGTGCTGGCATGTTACTCACTGCTGGTCACCGT GGCTTTCATCATCTTCTGGGTGCGGAGCAAGAGGTCCCGCCTGCTGCACAGCGACTAT ATGAACATGACCCACGGAGACCCGCGCTACACGGAAACATTACCAGCCCTATGCTC CACCCCGGGACTTCGCAGCTTACAGAAGTGAAGAGTGCAGCCCTGGAAGAGAATGT CGGGAATGCGCTCGCCCCAAGATTTGAAAGGAACAAACGAGTGAAGTTGAGCAGGTCC CCGCAGCTCCTGCATACCAGCAGGACAGAACACGCTGTATAACGAGCTGAATCTGG GCCGGAGAGAGGAATACGACGTGCTGGACAAAAGCGGGGCCGGGACCCGAAATGGG AGGGAAGCCACGACGGAAAAACCCCGAGGAGGGCTGTACAATGAGCTGCAAAAGGAC AAATGGCCGAGGCTTATTCTGAAATCGGGATGAAGGGAGAGAGAGGCGCGGAAAAAG GCCACGATGGCCTGTACCAGGGGCTGAGCACCGCTACAAAGGACACCTATGATGCACT GCACATGCAGGCCCTGCCCTCGGGGCTCAGGTGAGGGGCGCGGGAGCCTGCTGACT TGTGGGATGTAGAGGAAATCCTGGTCTATGAGAATCAGCAACACACCTCCGGA GCATATCAATCCAGTGTACTTGTGCCTTCTTTGAACTCCCATTTCTCACCGAGGC AGGCATTATGATGTTTATATTGGGGTGTCTTAGTGCTGGGCTTCCGAAACCGAAGCT AACTGGGTAAACGTCATCAGTGACCTTAAAAAATTGAGGATCTTATCCAATCAATGC ACATCGACGCGACTCTCTACACAGAATCTGACGTACACCCGTCATGCAAAGTCACGGC ATGAAGTGTCTTCTCGAGCTCCAAGTAATTTCCCTGGAGTCTGGCGATGCCTCC ATCCAGATACGGTTGAAATCTGATTATATTGGCCCAACAATAGCCTCAGTTCTAACG GTAACGTGACTGAAAGTGGCTGCAAAGAGTGCGAAGAGCTCGAAGAAAAGAATATCAA GGAGTTCCTCCAATCATTGTTCACATTGTGCAAAATGTTTATCAACACCTCT
SEQ ID NO: 29 CD8α hinge	AKPTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACD
SEQ ID NO: 30 CD8α hinge	GCGAAGCCACCACGACGCCAGCGCCGCGACCAACACCGGCGCCACCATCGCGT CGCAGCCCCGTGCTCCCTGCGCCAGAGGCGTGCCGGCCAGCGCGGGGGCGCAGTGCA CACGAGGGGCTGGACTTCGCCTGTGAT
SEQ ID NO: 31 Codon Optimized CD8α hinge	GCAAAACCTACCACAACTCCTGCACCACGCCCCCTACTCCAGCACCTACCATCGCAT CTCAGCCACTGAGTCTGCGACAGAGGCTGCCGGCCCGCCCGCGGGGCGGTCCA TACCAGAGGGCTGGACTTTGCCTGCGAT
SEQ ID NO: 32 CD28 transmembrane domain	FWVLVVVGGVLACYSLLVTVAFIIFWV
SEQ ID NO: 33 CD28 transmembrane domain	TTTTGGGTGCTGGTGGTGGTGGTGGAGTCTGGCTTGCTATAGCTTGCTAGTAACAG TGGCCTTTATTATTTCTGGGTG
SEQ ID NO: 34 Codon Optimized CD28 transmembrane domain	TTTTGGGTCTGGTGGTGGTGGGAGGGGTGCTGGCATGTTACTCACTGCTGGTCACCG TGGCCTTCATCATCTTCTGGGTG
SEQ ID NO: 35 CD19 CAR-MU-28	DIQMTQTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDGTVKLLIYHTRSRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPYTFGGGTKLEITGGGGSGGGG SGGGGSEVKLQESGPGLVAPSQLSVTCTVSGVSLPDYGVSWIRQPPRKGLEWLGVIW GSETTYNSALKSRLLTIIKDNSKQVFLKMNSLQTDDBTAIYYCAKHYYYGGSYAMDYW GQGTSTVTSSEPKCDKHTCPPCPAPPVAGPSVFLFPPKPKDITLMISRTPEVTCVVV DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYQSTYRVVSVLTVLHQDWLNGKEYKCK VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVE WESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQVMHEALHNHYTQ KSLSLSPGKFWVLVVVGGVLACYSLLVTVAFIIFWVRSKRSLRLHSDYMNMTPRRPGP TRKHYPYAPPRDFAYRSERVQPLEENVGNAARPRFERNKRVKFSRSADAPAYQQGQ NQLYNELNLGRREYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQDKMAEAYSEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
SEQ ID NO: 36 CD19 CAR-MU-28 with T2A and IL-15	MALPVTALLLPLALLLHAARPDQMTQTSSLSASLGDRVTISCRASQDISKYLNWYQ QKPDGTVKLLIYHTRSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLP YTFGGGTKLEITGGGGSGGGGSGGGSEVKLQESGPGLVAPSQLSVTCTVSGVSLPD YGVSWIRQPPRKGLEWLGVIWGGSETTYNSALKSRLLTIIKDNSKQVFLKMNSLQTDDB TAIYYCAKHYYYGGSYAMDYWGQGTSTVTSSEPKCDKHTCPPCPAPPVAGPSVLELF

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SEQUENCES	
SEQ ID NO: and DESCRIPTION	SEQUENCE
	PPPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYQSTYR VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDEL KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRW QQGNVFSCSVMHEALHNNHTQKSLSPGKFWLVVVGVLACYSLLTVAFIIFWVR SKRSRLHSDYMNMTPRRPGPTRKHYPYAPPRDFAAYRSERVQPLEENVGNAARPRF ERNKRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDRRGRDPGEMGGKPRRKNP QEGLYNELQDKMAEAYSIEIGMKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR GSGEGRGSLTTCGDVEENPGPMRISKPHLRSISIQCYLCLLLNSHELTEAGIHVFILG CFSAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCCKVTAMKCFLEEL QVISLESGLDASIHDTVENLIILANNSLSNGNVTESGCKECEELEEKNIKEFLQSFVH IVQMFINTS-
SEQ ID NO: 37 anti-CD19 scFv	DIQMTQTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDGTVKLLIYHSTRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPYTFGGGTKEITGGGGSGGGG SGGGGSEVKLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLEWLGVIW GSETTYNSALKSRLTIIKDNSKSKQVFLKMNLSLQTDDTAIYYCAKHYYGGSYAMDYW GQGTSTVTSS
SEQ ID NO: 38 CD19 CAR-MU-28	DIQMTQTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDGTVKLLIYHSTRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPYTFGGGTKEITGGGGSGGGG SGGGGSEVKLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLEWLGVIW GSETTYNSALKSRLTIIKDNSKSKQVFLKMNLSLQTDDTAIYYCAKHYYGGSYAMDYW GQGTSTVTSSSEPKSCDKTHTCPPCPAPPVAGPSVFLPPKPKDITLMISRTPEVTCVVV DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYQSTYRVVSVLTVLHQDWLNGKEYKCK VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVE WESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNNHTQ KSLSLSPGKFWLVVVGVLACYSLLTVAFIIFWVRSKRSRLHSDYMNMTPRRPGP TRKHYPYAPPRDFAAYRSERVQPLEENVGNAARPRFRERNKRVKFSRSADAPAYQQGQ NQLYNELNLGRREEYDVLDRRGRDPGEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIG MKGERRRGKGDGLYQGLSTATKDTYDALHMQALPPR
SEQ ID NO: 39 Anti-CD19 CAR (CD28-CD3z-IL15)	DIQMTQTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDGTVKLLIYHSTRLHSGV PSRFSGSGSGTDYSLTISNLEQEDIATYFCQQGNTLPYTFGGGTKEITGGGGSGGGG SGGGGSEVKLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLEWLGVIW GSETTYNSALKSRLTIIKDNSKSKQVFLKMNLSLQTDDTAIYYCAKHYYGGSYAMDYW GQGTSTVTSSSEPKSCDKTHTCPPCPAPPVAGPSVFLPPKPKDITLMISRTPEVTCVVV DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYQSTYRVVSVLTVLHQDWLNGKEYKCK VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVE WESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNNHTQ KSLSLSPGKFWLVVVGVLACYSLLTVAFIIFWVRSKRSRLHSDYMNMTPRRPGP TRKHYPYAPPRDFAAYRSRVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDRR GRDPGEMGGKPRRKNPQEGLYNELQDKMAEAYSIEIGMKGERRRGKGDGLYQGLSTAT KDTYDALHMQALPPR

Other Embodiments

[0505] It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate

and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 39

<210> SEQ ID NO 1

<211> LENGTH: 254

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: 4-1BBL expressed by feeder cells

<400> SEQUENCE: 1

Met Glu Tyr Ala Ser Asp Ala Ser Leu Asp Pro Glu Ala Pro Trp Pro

1 5 10 15

Pro Ala Pro Arg Ala Arg Ala Cys Arg Val Leu Pro Trp Ala Leu Val

-continued

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Ala	Gly	Leu	Leu	Leu	Leu	Leu	Leu	Ala	Ala	Ala	Cys	Ala	Val	Phe	
	35						40				45				
Leu	Ala	Cys	Pro	Trp	Ala	Val	Ser	Gly	Ala	Arg	Ala	Ser	Pro	Gly	Ser
	50					55					60				
Ala	Ala	Ser	Pro	Arg	Leu	Arg	Glu	Gly	Pro	Glu	Leu	Ser	Pro	Asp	Asp
	65				70					75				80	
Pro	Ala	Gly	Leu	Leu	Asp	Leu	Arg	Gln	Gly	Met	Phe	Ala	Gln	Leu	Val
			85						90					95	
Ala	Gln	Asn	Val	Leu	Leu	Ile	Asp	Gly	Pro	Leu	Ser	Trp	Tyr	Ser	Asp
			100					105						110	
Pro	Gly	Leu	Ala	Gly	Val	Ser	Leu	Thr	Gly	Gly	Leu	Ser	Tyr	Lys	Glu
		115					120					125			
Asp	Thr	Lys	Glu	Leu	Val	Val	Ala	Lys	Ala	Gly	Val	Tyr	Tyr	Val	Phe
	130					135					140				
Phe	Gln	Leu	Glu	Leu	Arg	Arg	Val	Val	Ala	Gly	Glu	Gly	Ser	Gly	Ser
	145				150					155					160
Val	Ser	Leu	Ala	Leu	His	Leu	Gln	Pro	Leu	Arg	Ser	Ala	Ala	Gly	Ala
			165						170					175	
Ala	Ala	Leu	Ala	Leu	Thr	Val	Asp	Leu	Pro	Pro	Ala	Ser	Ser	Glu	Ala
		180						185						190	
Arg	Asn	Ser	Ala	Phe	Gly	Phe	Gln	Gly	Arg	Leu	Leu	His	Leu	Ser	Ala
	195						200					205			
Gly	Gln	Arg	Leu	Gly	Val	His	Leu	His	Thr	Glu	Ala	Arg	Ala	Arg	His
	210					215					220				
Ala	Trp	Gln	Leu	Thr	Gln	Gly	Ala	Thr	Val	Leu	Gly	Leu	Phe	Arg	Val
	225				230					235					240
Thr	Pro	Glu	Ile	Pro	Ala	Gly	Leu	Pro	Ser	Pro	Arg	Ser	Glu		
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<210> SEQ ID NO 2

<211> LENGTH: 223

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mbIL-21 expressed by feeder cells

<400> SEQUENCE: 2

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1				5						10				15	
His	Ala	Ala	Arg	Pro	Gln	Asp	Arg	His	Met	Ile	Arg	Met	Arg	Gln	Leu
			20					25					30		
Ile	Asp	Ile	Val	Asp	Gln	Leu	Lys	Asn	Tyr	Val	Asn	Asp	Leu	Val	Pro
	35						40					45			
Glu	Phe	Leu	Pro	Ala	Pro	Glu	Asp	Val	Glu	Thr	Asn	Cys	Glu	Trp	Ser
	50					55					60				
Ala	Phe	Ser	Cys	Phe	Gln	Lys	Ala	Gln	Leu	Lys	Ser	Ala	Asn	Thr	Gly
	65				70					75				80	
Asn	Asn	Glu	Arg	Ile	Ile	Asn	Val	Ser	Ile	Lys	Lys	Leu	Lys	Arg	Lys
			85					90						95	
Pro	Pro	Ser	Thr	Asn	Ala	Gly	Arg	Arg	Gln	Lys	His	Arg	Leu	Thr	Cys
			100					105					110		
Pro	Ser	Cys	Asp	Ser	Tyr	Glu	Lys	Lys	Pro	Pro	Lys	Glu	Phe	Leu	Glu

-continued

115	120	125
Arg Phe Lys Ser Leu Leu Gln Lys Met Ile His Gln His Leu Ser Ser		
130	135	140
Arg Thr His Gly Ser Glu Asp Ser Ala Lys Pro Thr Thr Thr Pro Ala		
145	150	155
Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser		
165	170	175
Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr		
180	185	190
Arg Gly Leu Asp Phe Ala Cys Asp Ile Tyr Ile Trp Ala Pro Leu Ala		
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Gly Thr Cys Gly Val Leu Leu Leu Ser Leu Val Ile Thr Leu Tyr		
210	215	220

<210> SEQ ID NO 3
 <211> LENGTH: 233
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: TNF alpha expressed by feeder cells

<400> SEQUENCE: 3

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20 25 30
Leu Ser Leu Phe Ser Phe Leu Ile Val Ala Gly Ala Thr Thr Leu Phe
35 40 45
Cys Leu Leu His Phe Gly Val Ile Gly Pro Gln Arg Glu Glu Phe Pro
50 55 60
Arg Asp Leu Ser Leu Ile Ser Pro Leu Ala Gln Pro Val Arg Ser Ser
65 70 75 80
Ser Arg Thr Pro Ser Asp Lys Pro Val Ala His Val Val Ala Asn Pro
85 90 95
Gln Ala Glu Gly Gln Leu Gln Trp Leu Asn Arg Arg Ala Asn Ala Leu
100 105 110
Leu Ala Asn Gly Val Glu Leu Arg Asp Asn Gln Leu Val Val Pro Ser
115 120 125
Glu Gly Leu Tyr Leu Ile Tyr Ser Gln Val Leu Phe Lys Gly Gln Gly
130 135 140
Cys Pro Ser Thr His Val Leu Leu Thr His Thr Ile Ser Arg Ile Ala
145 150 155 160
Val Ser Tyr Gln Thr Lys Val Asn Leu Leu Ser Ala Ile Lys Ser Pro
165 170 175
Cys Gln Arg Glu Thr Pro Glu Gly Ala Glu Ala Lys Pro Trp Tyr Glu
180 185 190
Pro Ile Tyr Leu Gly Gly Val Phe Gln Leu Glu Lys Gly Asp Arg Leu
195 200 205
Ser Ala Glu Ile Asn Arg Pro Asp Tyr Leu Asp Phe Ala Glu Ser Gly
210 215 220
Gln Val Tyr Phe Gly Ile Ile Ala Leu
225 230

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<210> SEQ ID NO 4
<211> LENGTH: 183
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: OX40L expressed by feeder cells

<400> SEQUENCE: 4

Met Glu Arg Val Gln Pro Leu Glu Glu Asn Val Gly Asn Ala Ala Arg
 1             5             10            15

Pro Arg Phe Glu Arg Asn Lys Leu Leu Leu Val Ala Ser Val Ile Gln
 20            25            30

Gly Leu Gly Leu Leu Leu Cys Phe Thr Tyr Ile Cys Leu His Phe Ser
 35            40            45

Ala Leu Gln Val Ser His Arg Tyr Pro Arg Ile Gln Ser Ile Lys Val
 50            55            60

Gln Phe Thr Glu Tyr Lys Lys Glu Lys Gly Phe Ile Leu Thr Ser Gln
 65            70            75            80

Lys Glu Asp Glu Ile Met Lys Val Gln Asn Asn Ser Val Ile Ile Asn
 85            90            95

Cys Asp Gly Phe Tyr Leu Ile Ser Leu Lys Gly Tyr Phe Ser Gln Glu
100           105           110

Val Asn Ile Ser Leu His Tyr Gln Lys Asp Glu Glu Pro Leu Phe Gln
115           120           125

Leu Lys Lys Val Arg Ser Val Asn Ser Leu Met Val Ala Ser Leu Thr
130           135           140

Tyr Lys Asp Lys Val Tyr Leu Asn Val Thr Thr Asp Asn Thr Ser Leu
145           150           155           160

Asp Asp Phe His Val Asn Gly Gly Glu Leu Ile Leu Ile His Gln Asn
165           170           175

Pro Gly Glu Phe Cys Val Leu
180

<210> SEQ ID NO 5
<211> LENGTH: 41
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD28 intracellular signaling domain

<400> SEQUENCE: 5

Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr
 1             5             10            15

Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
 20            25            30

Pro Arg Asp Phe Ala Ala Tyr Arg Ser
 35            40

<210> SEQ ID NO 6
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD28 intracellular signaling domain

<400> SEQUENCE: 6

Ala Gly Gly Ala Gly Thr Ala Ala Gly Ala Gly Gly Ala Gly Cys Ala
 1             5             10            15

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Gly Gly Cys Thr Cys Cys Thr Gly Cys Ala Cys Ala Gly Thr Gly Ala
 20 25 30
 Cys Thr Ala Cys Ala Thr Gly Ala Ala Cys Ala Thr Gly Ala Cys Thr
 35 40 45
 Cys Cys Cys Cys Gly Cys Cys Gly Cys Cys Cys Cys Gly Gly Gly Cys
 50 55 60
 Cys Cys Ala Cys Cys Cys Gly Cys Ala Ala Gly Cys Ala Thr Thr Ala
 65 70 75 80
 Cys Cys Ala Gly Cys Cys Cys Thr Ala Thr Gly Cys Cys Cys Cys Ala
 85 90 95
 Cys Cys Ala Cys Gly Cys Gly Ala Cys Thr Thr Cys Gly Cys Ala Gly
 100 105 110
 Cys Cys Thr Ala Thr Cys Gly Cys Thr Cys Cys
 115 120

<210> SEQ ID NO 7
 <211> LENGTH: 123
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Codon Optimized CD28 intracellular signaling domain

<400> SEQUENCE: 7

Cys Gly Gly Ala Gly Cys Ala Ala Gly Ala Gly Gly Thr Cys Cys Cys
 1 5 10 15
 Gly Cys Cys Thr Gly Cys Thr Gly Cys Ala Cys Ala Gly Cys Gly Ala
 20 25 30
 Cys Thr Ala Thr Ala Thr Gly Ala Ala Cys Ala Thr Gly Ala Cys Cys
 35 40 45
 Cys Cys Ala Cys Gly Gly Ala Gly Ala Cys Cys Cys Gly Gly Cys Cys
 50 55 60
 Cys Thr Ala Cys Ala Cys Gly Gly Ala Ala Ala Cys Ala Thr Thr Ala
 65 70 75 80
 Cys Cys Ala Gly Cys Cys Cys Thr Ala Thr Gly Cys Thr Cys Cys Ala
 85 90 95
 Cys Cys Cys Cys Gly Gly Gly Ala Cys Thr Thr Cys Gly Cys Ala Gly
 100 105 110
 Cys Thr Thr Ala Cys Ala Gly Ala Ala Gly Thr
 115 120

<210> SEQ ID NO 8
 <211> LENGTH: 22
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: OX40L intracellular signaling domain

<400> SEQUENCE: 8

Glu Arg Val Gln Pro Leu Glu Glu Asn Val Gly Asn Ala Ala Arg Pro
 1 5 10 15
 Arg Phe Glu Arg Asn Lys
 20

<210> SEQ ID NO 9
 <211> LENGTH: 17

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: OX40L intracellular signaling domain functional domain

<400> SEQUENCE: 9

Leu Glu Glu Asn Val Gly Asn Ala Ala Arg Pro Arg Phe Glu Arg Asn
1 5 10 15

Lys

<210> SEQ ID NO 10
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: OX40L intracellular signaling domain functional domain

<400> SEQUENCE: 10

Arg Pro Arg Phe Glu Arg Asn Lys
1 5

<210> SEQ ID NO 11
<211> LENGTH: 66
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: OX40L intracellular signaling domain

<400> SEQUENCE: 11

Gly Ala Ala Ala Gly Gly Gly Thr Cys Cys Ala Ala Cys Cys Cys Cys
1 5 10 15

Thr Gly Gly Ala Ala Gly Ala Gly Ala Thr Gly Thr Gly Gly Gly
20 25 30

Ala Ala Ala Thr Gly Cys Ala Gly Cys Cys Ala Gly Gly Cys Cys Ala
35 40 45

Ala Gly Ala Thr Thr Cys Gly Ala Gly Ala Gly Gly Ala Ala Cys Ala
50 55 60

Ala Gly
65

<210> SEQ ID NO 12
<211> LENGTH: 66
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Codon optimized OX40L intracellular signaling domain

<400> SEQUENCE: 12

Gly Ala Ala Ala Gly Ala Gly Thr Gly Cys Ala Gly Cys Cys Cys Cys
1 5 10 15

Thr Gly Gly Ala Ala Gly Ala Gly Ala Thr Gly Thr Cys Gly Gly
20 25 30

Gly Ala Ala Thr Gly Cys Cys Gly Cys Thr Cys Gly Cys Cys Ala
35 40 45

Ala Gly Ala Thr Thr Thr Gly Ala Ala Ala Gly Gly Ala Ala Cys Ala
50 55 60

Ala Ala

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65

<210> SEQ ID NO 13
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD3 zeta signaling sequence

<400> SEQUENCE: 13

Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly
1 5 10 15
Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr
20 25 30
Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys
35 40 45
Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys
50 55 60
Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg
65 70 75 80
Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala
85 90 95
Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
100 105 110

<210> SEQ ID NO 14
<211> LENGTH: 336
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD3 zeta signaling sequence

<400> SEQUENCE: 14

Ala Gly Ala Gly Thr Gly Ala Ala Gly Thr Thr Cys Ala Gly Cys Ala
1 5 10 15
Gly Gly Ala Gly Cys Gly Cys Ala Gly Ala Cys Gly Cys Cys Cys Cys
20 25 30
Cys Gly Cys Gly Thr Ala Cys Cys Ala Gly Cys Ala Gly Gly Gly Cys
35 40 45
Cys Ala Gly Ala Ala Cys Cys Ala Gly Cys Thr Cys Thr Ala Thr Ala
50 55 60
Ala Cys Gly Ala Gly Cys Thr Cys Ala Ala Thr Cys Thr Ala Gly Gly
65 70 75 80
Ala Cys Gly Ala Ala Gly Ala Gly Ala Gly Gly Ala Gly Thr Ala Cys
85 90 95
Gly Ala Thr Gly Thr Thr Thr Thr Gly Gly Ala Cys Ala Ala Gly Ala
100 105 110
Gly Ala Cys Gly Thr Gly Gly Cys Cys Gly Gly Gly Ala Cys Cys Cys
115 120 125
Thr Gly Ala Gly Ala Thr Gly Gly Gly Gly Gly Gly Ala Ala Ala Gly
130 135 140
Cys Cys Gly Ala Gly Ala Ala Gly Gly Ala Ala Gly Ala Ala Cys Cys
145 150 155 160
Cys Thr Cys Ala Gly Gly Ala Ala Gly Gly Cys Cys Thr Gly Thr Ala
165 170 175

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Cys Ala Ala Thr Gly Ala Ala Cys Thr Gly Cys Ala Gly Ala Ala Ala
 180 185 190
 Gly Ala Thr Ala Ala Gly Ala Thr Gly Gly Cys Gly Gly Ala Gly Gly
 195 200 205
 Cys Cys Thr Ala Cys Ala Gly Thr Gly Ala Gly Ala Thr Thr Gly Gly
 210 215 220
 Gly Ala Thr Gly Ala Ala Ala Gly Gly Cys Gly Ala Gly Cys Gly Cys
 225 230 235 240
 Cys Gly Gly Ala Gly Gly Gly Gly Cys Ala Ala Gly Gly Gly Gly Cys
 245 250 255
 Ala Cys Gly Ala Thr Gly Gly Cys Cys Thr Thr Thr Ala Cys Cys Ala
 260 265 270
 Gly Gly Gly Thr Cys Thr Cys Ala Gly Thr Ala Cys Ala Gly Cys Cys
 275 280 285
 Ala Cys Cys Ala Ala Gly Gly Ala Cys Ala Cys Cys Thr Ala Cys Gly
 290 295 300
 Ala Cys Gly Cys Cys Cys Thr Thr Cys Ala Cys Ala Thr Gly Cys Ala
 305 310 315 320
 Gly Gly Cys Cys Cys Thr Gly Cys Cys Cys Cys Thr Cys Gly Cys
 325 330 335

 <210> SEQ ID NO 15
 <211> LENGTH: 336
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Codon optimized CD3 zeta signaling sequence

 <400> SEQUENCE: 15

 Cys Gly Ala Gly Thr Gly Ala Ala Gly Thr Thr Cys Ala Gly Cys Ala
 1 5 10 15
 Gly Gly Thr Cys Cys Gly Cys Cys Gly Ala Cys Gly Cys Thr Cys Cys
 20 25 30
 Thr Gly Cys Ala Thr Ala Cys Cys Ala Gly Cys Ala Gly Gly Gly Ala
 35 40 45
 Cys Ala Gly Ala Ala Cys Cys Ala Gly Cys Thr Gly Thr Ala Thr Ala
 50 55 60
 Ala Cys Gly Ala Gly Cys Thr Gly Ala Ala Thr Cys Thr Gly Gly Gly
 65 70 75 80
 Cys Cys Gly Gly Ala Gly Ala Gly Ala Gly Gly Ala Ala Thr Ala Cys
 85 90 95
 Gly Ala Cys Gly Thr Gly Cys Thr Gly Gly Ala Cys Ala Ala Ala Ala
 100 105 110
 Gly Gly Cys Gly Gly Gly Gly Cys Cys Gly Gly Gly Ala Cys Cys Cys
 115 120 125
 Cys Gly Ala Ala Ala Thr Gly Gly Gly Ala Gly Gly Gly Ala Ala Gly
 130 135 140
 Cys Cys Ala Cys Gly Ala Cys Gly Gly Ala Ala Ala Ala Cys Cys
 145 150 155 160
 Cys Cys Cys Ala Gly Gly Ala Gly Gly Gly Cys Cys Thr Gly Thr Ala
 165 170 175
 Cys Ala Ala Thr Gly Ala Gly Cys Thr Gly Cys Ala Ala Ala Ala Gly
 180 185 190

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Gly Ala Cys Ala Ala Ala Ala Thr Gly Gly Cys Cys Gly Ala Gly Gly
 195 200 205
 Cys Thr Thr Ala Thr Thr Cys Thr Gly Ala Ala Ala Thr Cys Gly Gly
 210 215 220
 Gly Ala Thr Gly Ala Ala Gly Gly Gly Ala Gly Ala Gly Ala Gly Ala
 225 230 235 240
 Ala Gly Gly Cys Gly Cys Gly Gly Ala Ala Ala Ala Gly Gly Cys Cys
 245 250 255
 Ala Cys Gly Ala Thr Gly Gly Cys Cys Thr Gly Thr Ala Cys Cys Ala
 260 265 270
 Gly Gly Gly Gly Cys Thr Gly Ala Gly Cys Ala Cys Cys Gly Cys Thr
 275 280 285
 Ala Cys Ala Ala Ala Gly Gly Ala Cys Ala Cys Cys Thr Ala Thr Gly
 290 295 300
 Ala Thr Gly Cys Ala Cys Thr Gly Cys Ala Cys Ala Thr Gly Cys Ala
 305 310 315 320
 Gly Gly Cys Cys Cys Thr Gly Cys Cys Cys Cys Cys Thr Cys Gly Gly
 325 330 335

<210> SEQ ID NO 16
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: 2A cleavage motif
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (5)..(5)
 <223> OTHER INFORMATION: x=any amino acid

<400> SEQUENCE: 16

Gly Asp Val Glu Xaa Asn Pro Gly Pro
 1 5

<210> SEQ ID NO 17
 <211> LENGTH: 21
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: T2A cleavage site

<400> SEQUENCE: 17

Gly Ser Gly Glu Gly Arg Gly Ser Leu Leu Thr Cys Gly Asp Val Glu
 1 5 10 15
 Glu Asn Pro Gly Pro
 20

<210> SEQ ID NO 18
 <211> LENGTH: 63
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: T2A cleavage site

<400> SEQUENCE: 18

Gly Gly Cys Thr Cys Ala Gly Gly Thr Gly Ala Gly Gly Gly Cys
 1 5 10 15
 Gly Cys Gly Gly Gly Ala Gly Cys Cys Thr Gly Cys Thr Gly Ala Cys
 20 25 30

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Thr	Thr	Gly	Thr	Gly	Gly	Gly	Gly	Ala	Thr	Gly	Thr	Ala	Gly	Ala	Gly
		35					40					45			
Gly	Ala	Ala	Ala	Ala	Thr	Cys	Cys	Thr	Gly	Gly	Thr	Cys	Cys	Thr	
	50					55					60				

<210> SEQ ID NO 19
 <211> LENGTH: 22
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: P2A cleavage site

<400> SEQUENCE: 19

Gly	Ser	Gly	Ala	Thr	Asn	Phe	Ser	Leu	Leu	Lys	Gln	Ala	Gly	Asp	Val
1				5				10					15		

Glu	Glu	Asn	Pro	Gly	Pro
				20	

<210> SEQ ID NO 20
 <211> LENGTH: 23
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: E2A cleavage site

<400> SEQUENCE: 20

Gly	Ser	Gly	Gln	Cys	Thr	Asn	Tyr	Ala	Leu	Leu	Lys	Leu	Ala	Gly	Asp
1			5					10					15		

Val	Glu	Ser	Asn	Pro	Gly	Pro
				20		

<210> SEQ ID NO 21
 <211> LENGTH: 25
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: F2A cleavage site

<400> SEQUENCE: 21

Gly	Ser	Gly	Val	Lys	Gln	Thr	Leu	Asn	Phe	Asp	Leu	Leu	Lys	Leu	Ala
1			5					10					15		

Gly	Asp	Val	Glu	Ser	Asn	Pro	Gly	Pro
		20					25	

<210> SEQ ID NO 22
 <211> LENGTH: 162
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IL-15

<400> SEQUENCE: 22

Met	Arg	Ile	Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr
1			5					10					15		

Leu	Cys	Leu	Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His
		20						25					30		

Val	Phe	Ile	Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala
		35						40				45			

Asn	Trp	Val	Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile
	50					55					60				

-continued

Gln	Ser	Met	His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His
65					70					75					80
Pro	Ser	Cys	Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln
			85					90						95	
Val	Ile	Ser	Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu
			100					105					110		
Asn	Leu	Ile	Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val
		115					120					125			
Thr	Glu	Ser	Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile
	130					135					140				
Lys	Glu	Phe	Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met	Phe	Ile	Asn
145					150					155					160

Thr Ser

<210> SEQ ID NO 23
 <211> LENGTH: 489
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IL-15

<400> SEQUENCE: 23

Ala	Thr	Gly	Ala	Gly	Ala	Ala	Thr	Cys	Ala	Gly	Cys	Ala	Ala	Ala	Cys
1			5					10						15	
Cys	Ala	Cys	Ala	Cys	Cys	Thr	Cys	Cys	Gly	Gly	Ala	Gly	Cys	Ala	Thr
		20					25					30			
Ala	Thr	Cys	Ala	Ala	Thr	Cys	Cys	Ala	Gly	Thr	Gly	Thr	Thr	Ala	Cys
		35				40					45				
Thr	Thr	Gly	Thr	Gly	Cys	Cys	Thr	Thr	Cys	Thr	Thr	Thr	Thr	Gly	Ala
	50					55				60					
Ala	Cys	Thr	Cys	Cys	Cys	Ala	Thr	Thr	Thr	Cys	Cys	Thr	Cys	Ala	Cys
65					70					75					80
Cys	Gly	Ala	Gly	Gly	Cys	Ala	Gly	Gly	Cys	Ala	Thr	Thr	Cys	Ala	Thr
			85					90						95	
Gly	Thr	Gly	Thr	Thr	Cys	Ala	Thr	Ala	Thr	Thr	Gly	Gly	Gly	Gly	Thr
		100						105					110		
Gly	Cys	Thr	Thr	Thr	Ala	Gly	Thr	Gly	Cys	Thr	Gly	Gly	Gly	Cys	Thr
		115					120				125				
Thr	Cys	Cys	Gly	Ala	Ala	Ala	Ala	Cys	Gly	Gly	Ala	Ala	Gly	Cys	Thr
	130				135					140					
Ala	Ala	Cys	Thr	Gly	Gly	Gly	Thr	Ala	Ala	Ala	Cys	Gly	Thr	Cys	Ala
145					150					155					160
Thr	Cys	Ala	Gly	Thr	Gly	Ala	Cys	Cys	Thr	Thr	Ala	Ala	Ala	Ala	Ala
			165					170						175	
Ala	Ala	Thr	Thr	Gly	Ala	Gly	Gly	Ala	Thr	Cys	Thr	Thr	Ala	Thr	Cys
		180						185					190		
Cys	Ala	Ala	Thr	Cys	Ala	Ala	Thr	Gly	Cys	Ala	Cys	Ala	Thr	Cys	Gly
		195					200				205				
Ala	Cys	Gly	Cys	Gly	Ala	Cys	Thr	Cys	Thr	Cys	Thr	Ala	Cys	Ala	Cys
	210					215					220				
Ala	Gly	Ala	Ala	Thr	Cys	Thr	Gly	Ala	Cys	Gly	Thr	Ala	Cys	Ala	Cys
225					230					235					240

Cys Cys Gly Thr Cys Ala Thr Gly Cys Ala Ala Ala Gly Thr Cys Ala

-continued

245				250				255							
Cys	Gly	Gly	Cys	Ala	Ala	Thr	Gly	Ala	Ala	Gly	Thr	Gly	Thr	Thr	Thr
			260							265				270	
Thr	Cys	Thr	Thr	Cys	Thr	Cys	Gly	Ala	Gly	Cys	Thr	Cys	Cys	Ala	Ala
			275				280							285	
Gly	Thr	Ala	Ala	Thr	Thr	Thr	Cys	Cys	Cys	Thr	Gly	Gly	Ala	Gly	Thr
			290				295				300				
Cys	Thr	Gly	Gly	Cys	Gly	Ala	Thr	Gly	Cys	Cys	Thr	Cys	Cys	Ala	Thr
			305				310				315				320
Cys	Cys	Ala	Cys	Gly	Ala	Thr	Ala	Cys	Gly	Gly	Thr	Thr	Gly	Ala	Ala
			325								330				335
Ala	Ala	Thr	Cys	Thr	Gly	Ala	Thr	Thr	Ala	Thr	Ala	Thr	Thr	Gly	Gly
			340								345				350
Cys	Cys	Ala	Ala	Cys	Ala	Ala	Thr	Ala	Gly	Cys	Cys	Thr	Cys	Ala	Gly
			355				360								365
Thr	Thr	Cys	Thr	Ala	Ala	Cys	Gly	Gly	Thr	Ala	Ala	Cys	Gly	Thr	Gly
			370				375				380				
Ala	Cys	Thr	Gly	Ala	Ala	Ala	Gly	Thr	Gly	Gly	Cys	Thr	Gly	Cys	Ala
			385				390				395				400
Ala	Ala	Gly	Ala	Gly	Thr	Gly	Cys	Gly	Ala	Ala	Gly	Ala	Gly	Cys	Thr
			405								410				415
Cys	Gly	Ala	Ala	Gly	Ala	Ala	Ala	Ala	Gly	Ala	Ala	Thr	Ala	Thr	Cys
			420								425				430
Ala	Ala	Gly	Gly	Ala	Gly	Thr	Thr	Cys	Cys	Thr	Cys	Cys	Ala	Ala	Thr
			435				440								445
Cys	Ala	Thr	Thr	Thr	Gly	Thr	Thr	Cys	Ala	Cys	Ala	Thr	Thr	Gly	Thr
			450				455				460				
Gly	Cys	Ala	Ala	Ala	Thr	Gly	Thr	Thr	Thr	Ala	Thr	Cys	Ala	Ala	Cys
			465				470				475				480
Ala	Cys	Cys	Thr	Cys	Thr	Thr	Gly	Ala							
			485												

<210> SEQ ID NO 24

<211> LENGTH: 489

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: IL-15

<400> SEQUENCE: 24

Ala	Thr	Gly	Cys	Gly	Cys	Ala	Thr	Ala	Ala	Gly	Thr	Ala	Ala	Gly	Cys
1				5					10					15	
Cys	Thr	Cys	Ala	Thr	Cys	Thr	Gly	Cys	Gly	Gly	Thr	Cys	Cys	Ala	Thr
			20					25					30		
Thr	Thr	Cys	Thr	Ala	Thr	Ala	Cys	Ala	Ala	Thr	Gly	Thr	Thr	Ala	Thr
			35				40					45			
Cys	Thr	Gly	Thr	Gly	Cys	Thr	Thr	Gly	Cys	Thr	Thr	Thr	Thr	Gly	Ala
			50				55				60				
Ala	Cys	Thr	Cys	Cys	Cys	Ala	Cys	Thr	Thr	Thr	Cys	Thr	Thr	Ala	Cys
			65			70				75				80	
Gly	Gly	Ala	Ala	Gly	Cys	Ala	Gly	Gly	Cys	Ala	Thr	Thr	Cys	Ala	Thr
			85					90						95	
Gly	Thr	Gly	Thr	Thr	Cys	Ala	Thr	Thr	Cys	Thr	Gly	Gly	Gly	Thr	Thr

-continued

100							105					110				
Gly	Thr	Thr	Thr	Thr	Thr	Cys	Thr	Gly	Cys	Cys	Gly	Gly	Gly	Cys	Thr	
115						120						125				
Gly	Cys	Cys	Cys	Ala	Ala	Ala	Ala	Cys	Cys	Gly	Ala	Gly	Gly	Cys	Cys	
130						135						140				
Ala	Ala	Cys	Thr	Gly	Gly	Gly	Thr	Cys	Ala	Ala	Cys	Gly	Thr	Gly	Ala	
145				150						155				160		
Thr	Cys	Ala	Gly	Cys	Gly	Ala	Cys	Cys	Thr	Cys	Ala	Ala	Gly	Ala	Ala	
				165				170				175				
Gly	Ala	Thr	Cys	Gly	Ala	Gly	Gly	Ala	Thr	Thr	Thr	Gly	Ala	Thr	Thr	
		180						185				190				
Cys	Ala	Ala	Ala	Gly	Thr	Ala	Thr	Gly	Cys	Ala	Thr	Ala	Thr	Ala	Gly	
195						200						205				
Ala	Cys	Gly	Cys	Cys	Ala	Cys	Ala	Cys	Thr	Cys	Thr	Ala	Thr	Ala	Cys	
210						215						220				
Thr	Gly	Ala	Gly	Thr	Cys	Cys	Gly	Ala	Cys	Gly	Thr	Thr	Cys	Ala	Cys	
225				230						235				240		
Cys	Cys	Gly	Ala	Gly	Thr	Thr	Gly	Thr	Ala	Ala	Ala	Gly	Thr	Thr	Ala	
				245				250						255		
Cys	Gly	Gly	Cys	Thr	Ala	Thr	Gly	Ala	Ala	Gly	Thr	Gly	Cys	Thr	Thr	
		260						265				270				
Thr	Thr	Thr	Gly	Thr	Thr	Gly	Gly	Ala	Ala	Cys	Thr	Cys	Cys	Ala	Gly	
		275				280						285				
Gly	Thr	Gly	Ala	Thr	Thr	Thr	Cys	Cys	Cys	Thr	Thr	Gly	Ala	Ala	Thr	
290						295						300				
Cys	Cys	Gly	Gly	Cys	Gly	Ala	Thr	Gly	Cys	Gly	Ala	Gly	Cys	Ala	Thr	
305				310						315				320		
Cys	Cys	Ala	Cys	Gly	Ala	Thr	Ala	Cys	Gly	Gly	Thr	Ala	Gly	Ala	Gly	
				325				330						335		
Ala	Ala	Thr	Cys	Thr	Thr	Ala	Thr	Thr	Ala	Thr	Thr	Cys	Thr	Gly	Gly	
		340						345				350				
Cys	Gly	Ala	Ala	Thr	Ala	Ala	Thr	Thr	Cys	Thr	Cys	Thr	Gly	Thr	Cys	
355						360						365				
Thr	Thr	Cys	Ala	Ala	Ala	Thr	Gly	Gly	Gly	Ala	Ala	Thr	Gly	Thr	Ala	
370						375						380				
Ala	Cys	Thr	Gly	Ala	Gly	Ala	Gly	Cys	Gly	Gly	Thr	Thr	Gly	Thr	Ala	
385				390						395				400		
Ala	Ala	Gly	Ala	Ala	Thr	Gly	Cys	Gly	Ala	Ala	Gly	Ala	Ala	Cys	Thr	
				405				410						415		
Thr	Gly	Ala	Ala	Gly	Ala	Ala	Ala	Ala	Gly	Ala	Ala	Thr	Ala	Thr	Cys	
		420						425				430				
Ala	Ala	Gly	Gly	Ala	Ala	Thr	Thr	Thr	Cys	Thr	Thr	Cys	Ala	Gly	Ala	
435						440						445				
Gly	Thr	Thr	Thr	Cys	Gly	Thr	Gly	Cys	Ala	Thr	Ala	Thr	Thr	Gly	Thr	
450						455						460				
Thr	Cys	Ala	Ala	Ala	Thr	Gly	Thr	Thr	Cys	Ala	Thr	Cys	Ala	Ala	Cys	
465				470						475				480		
Ala	Cys	Ala	Thr	Cys	Cys	Thr	Gly	Ala								
				485												

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<211> LENGTH: 175
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD28/OX40L/CDzeta

<400> SEQUENCE: 25
Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr
1          5          10          15
Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
20          25          30
Pro Arg Asp Phe Ala Ala Tyr Arg Ser Glu Arg Val Gln Pro Leu Glu
35          40          45
Glu Asn Val Gly Asn Ala Ala Arg Pro Arg Phe Glu Arg Asn Lys Arg
50          55          60
Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln
65          70          75          80
Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp
85          90          95
Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro
100         105         110
Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp
115         120         125
Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg
130         135         140
Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr
145         150         155         160
Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg
165         170         175

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<210> SEQ ID NO 26
<211> LENGTH: 358
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CD28/OX40L/CDzeta/T2A/IL1-5

<400> SEQUENCE: 26
Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr
1          5          10          15
Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln Pro Tyr Ala Pro
20          25          30
Pro Arg Asp Phe Ala Ala Tyr Arg Ser Glu Arg Val Gln Pro Leu Glu
35          40          45
Glu Asn Val Gly Asn Ala Ala Arg Pro Arg Phe Glu Arg Asn Lys Arg
50          55          60
Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln Gln Gly Gln
65          70          75          80
Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu Glu Tyr Asp
85          90          95
Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly Gly Lys Pro
100         105         110
Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp
115         120         125
Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly Glu Arg Arg

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130	135	140
Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr		
145	150	155 160
Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro Pro Arg Gly		
	165	170 175
Ser Gly Glu Gly Arg Gly Ser Leu Leu Thr Cys Gly Asp Val Glu Glu		
	180	185 190
Asn Pro Gly Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser Ile Ser		
	195	200 205
Ile Gln Cys Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu Thr Glu		
	210	215 220
Ala Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly Leu Pro		
	225	230 235 240
Lys Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile		
	245	250 255
Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu		
	260	265 270
Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu		
	275	280 285
Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His		
	290	295 300
Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser		
	305	310 315 320
Asn Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu		
	325	330 335
Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln		
	340	345 350
Met Phe Ile Asn Thr Ser		
	355	

<210> SEQ ID NO 27

<211> LENGTH: 694

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: HER2 CAR with T2A and IL-15

<400> SEQUENCE: 27

Met Ala Leu Pro Val Thr Ala Leu Leu Leu Pro Leu Ala Leu Leu		
1	5	10 15
His Ala Ala Arg Pro Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu		
	20	25 30
Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln		
	35	40 45
Asp Ile Asn Ser Tyr Leu Ser Trp Phe Gln Gln Lys Pro Gly Lys Ala		
	50	55 60
Pro Lys Thr Leu Ile Tyr Arg Ala Asn Arg Leu Val Asp Gly Val Pro		
	65	70 75 80
Ser Arg Phe Ser Gly Ser Gly Gln Asp Tyr Thr Leu Thr Ile		
	85	90 95
Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Tyr		
	100	105 110
Asp Glu Phe Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys		

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115					120					125				
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Ser	Gln
130						135					140			
Val	Gln	Leu	Val	Gln	Ser	Gly	Ser	Glu	Leu	Lys	Lys	Pro	Gly	Ala
145					150					155				160
Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr
				165					170					175
Val	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	Glu	Trp	Met
				180					185				190	Gly
Trp	Ile	Asn	Thr	His	Thr	Gly	Glu	Pro	Thr	Tyr	Ala	Glu	Glu	Phe
		195					200					205		Lys
Gly	Arg	Phe	Val	Phe	Ser	Leu	Asp	Thr	Ser	Val	Ser	Thr	Ala	Tyr
	210					215					220			Leu
Gln	Ile	Ser	Ser	Leu	Lys	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
225					230					235				240
Arg	Asp	Asp	Tyr	Tyr	Val	Arg	Val	Asp	Tyr	Trp	Gly	Gln	Gly	Thr
				245					250					255
Val	Thr	Val	Ser	Ser	Ala	Lys	Pro	Thr	Thr	Thr	Pro	Ala	Pro	Arg
			260						265				270	Pro
Pro	Thr	Pro	Ala	Pro	Thr	Ile	Ala	Ser	Gln	Pro	Leu	Ser	Leu	Arg
		275					280					285		Pro
Glu	Ala	Cys	Arg	Pro	Ala	Ala	Gly	Gly	Ala	Val	His	Thr	Arg	Gly
	290					295					300			Leu
Asp	Phe	Ala	Cys	Asp	Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val
305					310					315				320
Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp
				325					330					335
Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu	His	Ser	Asp	Tyr	Met	Asn	Met
			340					345					350	Thr
Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg	Lys	His	Tyr	Gln	Pro	Tyr	Ala
		355					360					365		Pro
Pro	Arg	Asp	Phe	Ala	Ala	Tyr	Arg	Ser	Glu	Arg	Val	Gln	Pro	Leu
	370					375					380			Glu
Glu	Asn	Val	Gly	Asn	Ala	Ala	Arg	Pro	Arg	Phe	Glu	Arg	Asn	Lys
385				390						395				400
Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly
				405					410					415
Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr
		420						425					430	Asp
Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys
	435					440						445		Pro
Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys
	450					455					460			Asp
Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg
465					470					475				480
Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala
				485					490					495
Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg
			500					505					510	Gly
Ser	Gly	Glu	Gly	Arg	Gly	Ser	Leu	Leu	Thr	Cys	Gly	Asp	Val	Glu
		515					520					525		Glu

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Asn Pro Gly Pro Met Arg Ile Ser Lys Pro His Leu Arg Ser Ile Ser
 530                               535                               540

Ile Gln Cys Tyr Leu Cys Leu Leu Leu Asn Ser His Phe Leu Thr Glu
545                               550                               555                               560

Ala Gly Ile His Val Phe Ile Leu Gly Cys Phe Ser Ala Gly Leu Pro
                               565                               570                               575

Lys Thr Glu Ala Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile
                               580                               585                               590

Glu Asp Leu Ile Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu
                               595                               600                               605

Ser Asp Val His Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu
610                               615                               620

Leu Glu Leu Gln Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His
625                               630                               635                               640

Asp Thr Val Glu Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser
                               645                               650                               655

Asn Gly Asn Val Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu
660                               665                               670

Glu Lys Asn Ile Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln
675                               680                               685

Met Phe Ile Asn Thr Ser
690

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<210> SEQ ID NO 28
<211> LENGTH: 2082
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Codon Optimized HER2 CAR with T2A and IL-15

<400> SEQUENCE: 28

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Ala Thr Gly Gly Cys Ala Cys Thr Thr Cys Cys Thr Gly Thr Thr Ala
 1      5      10      15

Cys Ala Gly Cys Cys Cys Thr Cys Cys Thr Gly Cys Thr Cys Cys Cys
20      25      30

Ala Cys Thr Gly Gly Cys Thr Thr Thr Gly Cys Thr Gly Cys Thr Gly
35      40      45

Cys Ala Thr Gly Cys Thr Gly Cys Ala Cys Gly Ala Cys Cys Gly Gly
50      55      60

Ala Cys Ala Thr Cys Cys Ala Gly Ala Thr Gly Ala Cys Cys Cys Ala
65      70      75      80

Gly Ala Gly Cys Cys Cys Ala Gly Cys Ala Gly Cys Cys Thr Gly
85      90      95

Ala Gly Cys Gly Cys Cys Ala Gly Cys Gly Thr Gly Gly Gly Cys Gly
100     105     110

Ala Cys Ala Gly Ala Gly Thr Gly Ala Cys Cys Ala Thr Cys Ala Cys
115     120     125

Cys Thr Gly Cys Ala Ala Gly Gly Cys Cys Ala Gly Cys Cys Ala Gly
130     135     140

Gly Ala Cys Ala Thr Cys Ala Ala Cys Ala Gly Cys Thr Ala Cys Cys
145     150     155     160

Thr Gly Ala Gly Cys Thr Gly Gly Thr Thr Cys Cys Ala Gly Cys Ala
165     170     175

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Gly	Ala	Ala	Gly	Cys	Cys	Cys	Gly	Gly	Cys	Ala	Ala	Gly	Gly	Cys	Cys		180	185	190
Cys	Cys	Cys	Ala	Ala	Gly	Ala	Cys	Cys	Cys	Thr	Gly	Ala	Thr	Cys	Thr		195	200	205
Ala	Cys	Ala	Gly	Ala	Gly	Cys	Cys	Ala	Ala	Cys	Ala	Gly	Ala	Cys	Thr		210	215	220
Gly	Gly	Thr	Gly	Gly	Ala	Cys	Gly	Gly	Cys	Gly	Thr	Gly	Cys	Cys	Cys		225	230	235
Ala	Gly	Cys	Ala	Gly	Ala	Thr	Thr	Cys	Ala	Gly	Cys	Gly	Gly	Cys	Ala		245	250	255
Gly	Cys	Gly	Gly	Cys	Ala	Gly	Cys	Gly	Gly	Cys	Cys	Ala	Gly	Gly	Ala		260	265	270
Cys	Thr	Ala	Cys	Ala	Cys	Cys	Cys	Thr	Gly	Ala	Cys	Cys	Ala	Thr	Cys		275	280	285
Ala	Gly	Cys	Ala	Gly	Cys	Cys	Thr	Gly	Cys	Ala	Gly	Cys	Cys	Cys	Gly		290	295	300
Ala	Gly	Gly	Ala	Cys	Thr	Thr	Cys	Gly	Cys	Cys	Ala	Cys	Cys	Thr	Ala		305	310	315
Cys	Thr	Ala	Cys	Thr	Gly	Cys	Cys	Thr	Gly	Cys	Ala	Gly	Thr	Ala	Cys		325	330	335
Gly	Ala	Cys	Gly	Ala	Gly	Thr	Thr	Cys	Cys	Cys	Cys	Thr	Gly	Gly	Ala		340	345	350
Cys	Cys	Thr	Thr	Cys	Gly	Gly	Cys	Cys	Ala	Gly	Gly	Gly	Cys	Ala	Cys		355	360	365
Cys	Ala	Ala	Gly	Gly	Thr	Gly	Gly	Ala	Gly	Ala	Thr	Cys	Ala	Ala	Gly		370	375	380
Gly	Gly	Thr	Gly	Gly	Cys	Gly	Gly	Thr	Gly	Gly	Ala	Thr	Cys	Gly	Gly		385	390	395
Gly	Cys	Gly	Gly	Thr	Gly	Gly	Thr	Gly	Gly	Ala	Thr	Cys	Thr	Gly	Gly		405	410	415
Ala	Gly	Gly	Ala	Gly	Gly	Thr	Gly	Gly	Cys	Thr	Cys	Cys	Cys	Ala	Gly		420	425	430
Gly	Thr	Gly	Cys	Ala	Gly	Cys	Thr	Gly	Gly	Thr	Gly	Cys	Ala	Gly	Ala		435	440	445
Gly	Cys	Gly	Gly	Cys	Ala	Gly	Cys	Gly	Ala	Gly	Cys	Thr	Gly	Ala	Ala		450	455	460
Gly	Ala	Ala	Gly	Cys	Cys	Cys	Gly	Gly	Cys	Gly	Cys	Cys	Ala	Gly	Cys		465	470	475
Gly	Thr	Gly	Ala	Ala	Gly	Gly	Thr	Gly	Ala	Gly	Cys	Thr	Gly	Cys	Ala		485	490	495
Ala	Gly	Gly	Cys	Cys	Ala	Gly	Cys	Gly	Gly	Cys	Thr	Ala	Cys	Ala	Cys		500	505	510
Cys	Thr	Thr	Cys	Ala	Cys	Cys	Ala	Ala	Cys	Thr	Ala	Cys	Gly	Gly	Cys		515	520	525
Gly	Thr	Gly	Ala	Ala	Cys	Thr	Gly	Gly	Gly	Thr	Gly	Ala	Gly	Ala	Cys		530	535	540
Ala	Gly	Gly	Cys	Cys	Cys	Cys	Cys	Gly	Gly	Cys	Cys	Ala	Gly	Gly	Gly		545	550	555
Cys	Cys	Thr	Gly	Gly	Ala	Gly	Thr	Gly	Gly	Ala	Thr	Gly	Gly	Gly	Cys		565	570	575

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Thr	Gly	Gly	Ala	Thr	Cys	Ala	Ala	Cys	Ala	Cys	Cys	Cys	Ala	Cys	Ala
			580					585					590		
Cys	Cys	Gly	Gly	Cys	Gly	Ala	Gly	Cys	Cys	Cys	Ala	Cys	Cys	Thr	Ala
		595					600					605			
Cys	Gly	Cys	Cys	Gly	Ala	Gly	Gly	Ala	Gly	Thr	Thr	Cys	Ala	Ala	Gly
	610					615					620				
Gly	Gly	Cys	Ala	Gly	Ala	Thr	Thr	Cys	Gly	Thr	Gly	Thr	Thr	Cys	Ala
625					630					635					640
Gly	Cys	Cys	Thr	Gly	Gly	Ala	Cys	Ala	Cys	Cys	Ala	Gly	Cys	Gly	Thr
			645						650					655	
Gly	Ala	Gly	Cys	Ala	Cys	Cys	Gly	Cys	Cys	Thr	Ala	Cys	Cys	Thr	Gly
			660					665					670		
Cys	Ala	Gly	Ala	Thr	Cys	Ala	Gly	Cys	Ala	Gly	Cys	Cys	Thr	Gly	Ala
		675					680					685			
Ala	Gly	Gly	Cys	Cys	Gly	Ala	Gly	Gly	Ala	Cys	Ala	Cys	Cys	Gly	Cys
	690					695					700				
Cys	Gly	Thr	Gly	Thr	Ala	Cys	Thr	Ala	Cys	Thr	Gly	Cys	Gly	Cys	Cys
705					710					715					720
Ala	Gly	Ala	Gly	Ala	Cys	Gly	Ala	Cys	Thr	Ala	Cys	Thr	Ala	Cys	Gly
			725						730					735	
Thr	Gly	Ala	Gly	Ala	Gly	Thr	Gly	Gly	Ala	Cys	Thr	Ala	Cys	Thr	Gly
			740					745					750		
Gly	Gly	Gly	Cys	Cys	Ala	Gly	Gly	Gly	Cys	Ala	Cys	Cys	Ala	Cys	Cys
		755					760					765			
Gly	Thr	Gly	Ala	Cys	Cys	Gly	Thr	Gly	Ala	Gly	Cys	Ala	Gly	Cys	Gly
	770					775					780				
Cys	Ala	Ala	Ala	Ala	Cys	Cys	Thr	Ala	Cys	Cys	Ala	Cys	Ala	Ala	Cys
785					790					795					800
Thr	Cys	Cys	Thr	Gly	Cys	Ala	Cys	Cys	Ala	Cys	Gly	Cys	Cys	Cys	Cys
			805						810					815	
Cys	Cys	Thr	Ala	Cys	Thr	Cys	Cys	Ala	Gly	Cys	Ala	Cys	Cys	Thr	Ala
			820					825					830		
Cys	Cys	Ala	Thr	Cys	Gly	Cys	Ala	Thr	Cys	Thr	Cys	Ala	Gly	Cys	Cys
		835					840					845			
Ala	Cys	Thr	Gly	Ala	Gly	Thr	Cys	Thr	Gly	Cys	Gly	Ala	Cys	Cys	Ala
	850					855					860				
Gly	Ala	Gly	Gly	Cys	Cys	Thr	Gly	Cys	Cys	Gly	Gly	Cys	Cys	Cys	Gly
865					870					875					880
Cys	Cys	Gly	Cys	Cys	Gly	Gly	Cys	Gly	Gly	Gly	Gly	Cys	Cys	Gly	Thr
			885						890					895	
Cys	Cys	Ala	Thr	Ala	Cys	Cys	Ala	Gly	Ala	Gly	Gly	Gly	Cys	Thr	Gly
			900					905					910		
Gly	Ala	Cys	Thr	Thr	Thr	Gly	Cys	Cys	Thr	Gly	Cys	Gly	Ala	Thr	Thr
			915				920					925			
Thr	Thr	Thr	Gly	Gly	Gly	Thr	Cys	Cys	Thr	Gly	Gly	Thr	Gly	Gly	Thr
	930					935					940				
Cys	Gly	Thr	Gly	Gly	Gly	Ala	Gly	Gly	Gly	Gly	Thr	Gly	Cys	Thr	Gly
945					950					955					960
Gly	Cys	Ala	Thr	Gly	Thr	Thr	Ala	Cys	Thr	Cys	Ala	Cys	Thr	Gly	Cys
			965						970					975	
Thr	Gly	Gly	Thr	Cys	Ala	Cys	Cys	Gly	Thr	Gly	Gly	Cys	Cys	Thr	Thr

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980				985				990			
Cys	Ala	Thr	Cys	Ala	Thr	Cys	Thr	Thr	Cys	Thr	Gly
	995						1000				Gly
											Gly
											Gly
											Thr
											Cys
											Cys
Cys	Gly	Gly	Ala	Gly	Cys	Ala	Ala	Gly	Ala	Gly	Gly
	1010					1015					1020
Cys	Gly	Cys	Cys	Thr	Gly	Cys	Thr	Gly	Cys	Ala	Cys
	1025					1030					1035
Gly	Ala	Cys	Thr	Ala	Thr	Ala	Thr	Gly	Ala	Ala	Cys
	1040					1045					1050
Ala	Cys	Cys	Cys	Cys	Ala	Cys	Gly	Gly	Ala	Gly	Ala
	1055					1060					1065
Gly	Gly	Cys	Cys	Cys	Thr	Ala	Cys	Ala	Cys	Gly	Gly
	1070					1075					1080
Cys	Ala	Thr	Thr	Ala	Cys	Cys	Ala	Gly	Cys	Cys	Cys
	1085					1090					1095
Gly	Cys	Thr	Cys	Cys	Ala	Cys	Cys	Cys	Cys	Gly	Gly
	1100					1105					1110
Thr	Thr	Cys	Gly	Cys	Ala	Gly	Cys	Thr	Thr	Ala	Cys
	1115					1120					1125
Ala	Gly	Thr	Gly	Ala	Ala	Ala	Gly	Ala	Gly	Thr	Gly
	1130					1135					1140
Cys	Cys	Cys	Cys	Thr	Gly	Gly	Ala	Ala	Gly	Ala	Gly
	1145					1150					1155
Gly	Thr	Cys	Gly	Gly	Gly	Ala	Ala	Thr	Gly	Cys	Cys
	1160					1165					1170
Cys	Gly	Cys	Cys	Cys	Ala	Ala	Gly	Ala	Thr	Thr	Thr
	1175					1180					1185
Ala	Gly	Gly	Ala	Ala	Cys	Ala	Ala	Ala	Cys	Gly	Ala
	1190					1195					1200
Ala	Ala	Gly	Thr	Thr	Cys	Ala	Gly	Cys	Ala	Gly	Gly
	1205					1210					1215
Gly	Cys	Cys	Gly	Ala	Cys	Gly	Cys	Thr	Cys	Cys	Thr
	1220					1225					1230
Thr	Ala	Cys	Cys	Ala	Gly	Cys	Ala	Gly	Gly	Gly	Ala
	1235					1240					1245
Ala	Ala	Cys	Cys	Ala	Gly	Cys	Thr	Gly	Thr	Ala	Thr
	1250					1255					1260
Gly	Ala	Gly	Cys	Thr	Gly	Ala	Ala	Thr	Cys	Thr	Gly
	1265					1270					1275
Cys	Gly	Gly	Ala	Gly	Ala	Gly	Ala	Gly	Gly	Ala	Ala
	1280					1285					1290
Gly	Ala	Cys	Gly	Thr	Gly	Cys	Thr	Gly	Gly	Ala	Cys
	1295					1300					1305
Ala	Gly	Gly	Cys	Gly	Gly	Gly	Cys	Cys	Gly	Gly	Gly
	1310					1315					1320
Cys	Cys	Cys	Gly	Ala	Ala	Ala	Thr	Gly	Gly	Gly	Ala
	1325					1330					1335
Ala	Ala	Gly	Cys	Cys	Ala	Cys	Gly	Ala	Cys	Gly	Gly
	1340					1345					1350
Ala	Ala	Cys	Cys	Cys	Cys	Cys	Ala	Gly	Gly	Ala	Gly
	1355					1360					1365

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Cys Thr Gly Thr Ala Cys Ala Ala Thr Gly Ala Gly Cys Thr Gly	1370	1375	1380
Cys Ala Ala Ala Ala Gly Gly Ala Cys Ala Ala Ala Ala Thr Gly	1385	1390	1395
Gly Cys Cys Gly Ala Gly Gly Cys Thr Thr Ala Thr Thr Cys Thr	1400	1405	1410
Gly Ala Ala Ala Thr Cys Gly Gly Gly Ala Thr Gly Ala Ala Gly	1415	1420	1425
Gly Gly Ala Gly Ala Gly Ala Gly Ala Ala Gly Gly Cys Gly Cys	1430	1435	1440
Gly Gly Ala Ala Ala Ala Gly Gly Cys Cys Ala Cys Gly Ala Thr	1445	1450	1455
Gly Gly Cys Cys Thr Gly Thr Ala Cys Cys Ala Gly Gly Gly Gly	1460	1465	1470
Cys Thr Gly Ala Gly Cys Ala Cys Cys Gly Cys Thr Ala Cys Ala	1475	1480	1485
Ala Ala Gly Gly Ala Cys Ala Cys Cys Thr Ala Thr Gly Ala Thr	1490	1495	1500
Gly Cys Ala Cys Thr Gly Cys Ala Cys Ala Thr Gly Cys Ala Gly	1505	1510	1515
Gly Cys Cys Cys Thr Gly Cys Cys Cys Cys Thr Cys Gly Gly	1520	1525	1530
Gly Gly Cys Thr Cys Ala Gly Gly Thr Gly Ala Gly Gly Gly Gly	1535	1540	1545
Cys Gly Cys Gly Gly Gly Ala Gly Cys Cys Thr Gly Cys Thr Gly	1550	1555	1560
Ala Cys Thr Thr Gly Thr Gly Gly Gly Gly Ala Thr Gly Thr Ala	1565	1570	1575
Gly Ala Gly Gly Ala Ala Ala Ala Thr Cys Cys Thr Gly Gly Thr	1580	1585	1590
Cys Cys Thr Ala Thr Gly Ala Gly Ala Ala Thr Cys Ala Gly Cys	1595	1600	1605
Ala Ala Ala Cys Cys Ala Cys Ala Cys Cys Thr Cys Cys Gly Gly	1610	1615	1620
Ala Gly Cys Ala Thr Ala Thr Cys Ala Ala Thr Cys Cys Ala Gly	1625	1630	1635
Thr Gly Thr Thr Ala Cys Thr Thr Gly Thr Gly Cys Cys Thr Thr	1640	1645	1650
Cys Thr Thr Thr Thr Gly Ala Ala Cys Thr Cys Cys Cys Ala Thr	1655	1660	1665
Thr Thr Cys Cys Thr Cys Ala Cys Cys Gly Ala Gly Gly Cys Ala	1670	1675	1680
Gly Gly Cys Ala Thr Thr Cys Ala Thr Gly Thr Gly Thr Thr Cys	1685	1690	1695
Ala Thr Ala Thr Thr Gly Gly Gly Gly Thr Gly Cys Thr Thr Thr	1700	1705	1710
Ala Gly Thr Gly Cys Thr Gly Gly Gly Cys Thr Thr Cys Cys Gly	1715	1720	1725
Ala Ala Ala Ala Cys Gly Gly Ala Ala Gly Cys Thr Ala Ala Cys	1730	1735	1740

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Thr 1745	Gly	Gly	Gly	Thr	Ala	Ala 1750	Ala	Cys	Gly	Thr	Cys 1755	Ala	Thr	Cys
Ala 1760	Gly	Thr	Gly	Ala	Cys	Cys 1765	Thr	Thr	Ala	Ala	Ala 1770	Ala	Ala	Ala
Ala 1775	Thr	Thr	Gly	Ala	Gly	Gly 1780	Ala	Thr	Cys	Thr	Thr 1785	Ala	Thr	Cys
Cys 1790	Ala	Ala	Thr	Cys	Ala	Ala 1795	Thr	Gly	Cys	Ala	Cys 1800	Ala	Thr	Cys
Gly 1805	Ala	Cys	Gly	Cys	Gly	Ala 1810	Cys	Thr	Cys	Thr	Cys 1815	Thr	Ala	Cys
Ala 1820	Cys	Ala	Gly	Ala	Ala	Thr 1825	Cys	Thr	Gly	Ala	Cys 1830	Gly	Thr	Ala
Cys 1835	Ala	Cys	Cys	Cys	Gly	Thr 1840	Cys	Ala	Thr	Gly	Cys 1845	Ala	Ala	Ala
Gly 1850	Thr	Cys	Ala	Cys	Gly	Gly 1855	Cys	Ala	Ala	Thr	Gly 1860	Ala	Ala	Gly
Thr 1865	Gly	Thr	Thr	Thr	Thr	Cys 1870	Thr	Thr	Cys	Thr	Cys 1875	Gly	Ala	Gly
Cys 1880	Thr	Cys	Cys	Ala	Ala	Gly 1885	Thr	Ala	Ala	Thr	Thr 1890	Thr	Cys	Cys
Cys 1895	Thr	Gly	Gly	Ala	Gly	Thr 1900	Cys	Thr	Gly	Gly	Cys 1905	Gly	Ala	Thr
Gly 1910	Cys	Cys	Thr	Cys	Cys	Ala 1915	Thr	Cys	Cys	Ala	Cys 1920	Gly	Ala	Thr
Ala 1925	Cys	Gly	Gly	Thr	Thr	Gly 1930	Ala	Ala	Ala	Ala	Thr 1935	Cys	Thr	Gly
Ala 1940	Thr	Thr	Ala	Thr	Ala	Thr 1945	Thr	Gly	Gly	Cys	Cys 1950	Ala	Ala	Cys
Ala 1955	Ala	Thr	Ala	Gly	Cys	Cys 1960	Thr	Cys	Ala	Gly	Thr 1965	Thr	Cys	Thr
Ala 1970	Ala	Cys	Gly	Gly	Thr	Ala 1975	Ala	Cys	Gly	Thr	Gly 1980	Ala	Cys	Thr
Gly 1985	Ala	Ala	Ala	Gly	Thr	Gly 1990	Gly	Cys	Thr	Gly	Cys 1995	Ala	Ala	Ala
Gly 2000	Ala	Gly	Thr	Gly	Cys	Gly 2005	Ala	Ala	Gly	Ala	Gly 2010	Cys	Thr	Cys
Gly 2015	Ala	Ala	Gly	Ala	Ala	Ala 2020	Ala	Gly	Ala	Ala	Thr 2025	Ala	Thr	Cys
Ala 2030	Ala	Gly	Gly	Ala	Gly	Thr 2035	Thr	Cys	Cys	Thr	Cys 2040	Cys	Ala	Ala
Thr 2045	Cys	Ala	Thr	Thr	Thr	Gly 2050	Thr	Thr	Cys	Ala	Cys 2055	Ala	Thr	Thr
Gly 2060	Thr	Gly	Cys	Ala	Ala	Ala 2065	Thr	Gly	Thr	Thr	Thr 2070	Ala	Thr	Cys
Ala 2075	Ala	Cys	Ala	Cys	Cys	Thr 2080	Cys	Thr						

<210> SEQ ID NO 29

<211> LENGTH: 48

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: CD8 alpha hinge

-continued

<400> SEQUENCE: 29

Ala Lys Pro Thr Thr Thr Pro Ala Pro Arg Pro Pro Thr Pro Ala Pro
1 5 10 15
Thr Ile Ala Ser Gln Pro Leu Ser Leu Arg Pro Glu Ala Cys Arg Pro
20 25 30
Ala Ala Gly Gly Ala Val His Thr Arg Gly Leu Asp Phe Ala Cys Asp
35 40 45

<210> SEQ ID NO 30

<211> LENGTH: 144

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: CD8 alpha hinge

<400> SEQUENCE: 30

Gly Cys Gly Ala Ala Gly Cys Cys Cys Ala Cys Cys Ala Cys Gly Ala
1 5 10 15
Cys Gly Cys Cys Ala Gly Cys Gly Cys Cys Gly Cys Gly Ala Cys Cys
20 25 30
Ala Cys Cys Ala Ala Cys Ala Cys Cys Gly Gly Cys Gly Cys Cys Cys
35 40 45
Ala Cys Cys Ala Thr Cys Gly Cys Gly Thr Cys Gly Cys Ala Gly Cys
50 55 60
Cys Cys Cys Thr Gly Thr Cys Cys Cys Thr Gly Cys Gly Cys Cys Cys
65 70 75 80
Ala Gly Ala Gly Gly Cys Gly Thr Gly Cys Cys Gly Gly Cys Cys Ala
85 90 95
Gly Cys Gly Gly Cys Gly Gly Gly Gly Gly Cys Gly Cys Ala Gly
100 105 110
Thr Gly Cys Ala Cys Ala Cys Gly Ala Gly Gly Gly Gly Gly Cys Thr
115 120 125
Gly Gly Ala Cys Thr Thr Cys Gly Cys Cys Thr Gly Thr Gly Ala Thr
130 135 140

<210> SEQ ID NO 31

<211> LENGTH: 144

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Codon Optimized CD8 alpha hinge

<400> SEQUENCE: 31

Gly Cys Ala Ala Ala Ala Cys Cys Thr Ala Cys Cys Ala Cys Ala Ala
1 5 10 15
Cys Thr Cys Cys Thr Gly Cys Ala Cys Cys Ala Cys Gly Cys Cys Cys
20 25 30
Cys Cys Cys Thr Ala Cys Thr Cys Cys Ala Gly Cys Ala Cys Cys Thr
35 40 45
Ala Cys Cys Ala Thr Cys Gly Cys Ala Thr Cys Thr Cys Ala Gly Cys
50 55 60
Cys Ala Cys Thr Gly Ala Gly Thr Cys Thr Gly Cys Gly Ala Cys Cys
65 70 75 80
Ala Gly Ala Gly Gly Cys Cys Thr Gly Cys Cys Gly Gly Cys Cys Cys
85 90 95

-continued

Gly Cys Cys Gly Cys Cys Gly Gly Cys Gly Gly Gly Gly Cys Cys Gly
 100 105 110

Thr Cys Cys Ala Thr Ala Cys Cys Ala Gly Ala Gly Gly Gly Cys Thr
 115 120 125

Gly Gly Ala Cys Thr Thr Thr Gly Cys Cys Thr Gly Cys Gly Ala Thr
 130 135 140

<210> SEQ ID NO 32
 <211> LENGTH: 27
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CD28 transmembrane domain

<400> SEQUENCE: 32

Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu
 1 5 10 15

Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val
 20 25

<210> SEQ ID NO 33
 <211> LENGTH: 81
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CD28 transmembrane domain

<400> SEQUENCE: 33

Thr Thr Thr Thr Gly Gly Gly Thr Gly Cys Thr Gly Gly Thr Gly Gly
 1 5 10 15

Thr Gly Gly Thr Thr Gly Gly Thr Gly Gly Ala Gly Thr Cys Cys Thr
 20 25 30

Gly Gly Cys Thr Thr Gly Cys Thr Ala Thr Ala Gly Cys Thr Thr Gly
 35 40 45

Cys Thr Ala Gly Thr Ala Ala Cys Ala Gly Thr Gly Gly Cys Cys Thr
 50 55 60

Thr Thr Ala Thr Thr Ala Thr Thr Thr Thr Cys Thr Gly Gly Gly Thr
 65 70 75 80

Gly

<210> SEQ ID NO 34
 <211> LENGTH: 81
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Codon Optimized CD28 transmembrane domain

<400> SEQUENCE: 34

Thr Thr Thr Thr Gly Gly Gly Thr Cys Cys Thr Gly Gly Thr Gly Gly
 1 5 10 15

Thr Cys Gly Thr Gly Gly Gly Ala Gly Gly Gly Gly Thr Gly Cys Thr
 20 25 30

Gly Gly Cys Ala Thr Gly Thr Thr Ala Cys Thr Cys Ala Cys Thr Gly
 35 40 45

Cys Thr Gly Gly Thr Cys Ala Cys Cys Gly Thr Gly Gly Cys Cys Thr
 50 55 60

Thr Cys Ala Thr Cys Ala Thr Cys Thr Thr Cys Thr Gly Gly Gly Thr

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65	70	75	80
Gly			
<210> SEQ ID NO 35			
<211> LENGTH: 675			
<212> TYPE: PRT			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: CD19 CAR-MU-28			
<400> SEQUENCE: 35			
Asp Ile Gln Met Thr Gln Thr Thr Ser Ser Leu Ser Ala Ser Leu Gly			
1 5 10 15			
Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Asp Ile Ser Lys Tyr			
20 25 30			
Leu Asn Trp Tyr Gln Gln Lys Pro Asp Gly Thr Val Lys Leu Leu Ile			
35 40 45			
Tyr His Thr Ser Arg Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly			
50 55 60			
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Asn Leu Glu Gln			
65 70 75 80			
Glu Asp Ile Ala Thr Tyr Phe Cys Gln Gln Gly Asn Thr Leu Pro Tyr			
85 90 95			
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Thr Gly Gly Gly Gly Ser			
100 105 110			
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Val Lys Leu Gln Glu			
115 120 125			
Ser Gly Pro Gly Leu Val Ala Pro Ser Gln Ser Leu Ser Val Thr Cys			
130 135 140			
Thr Val Ser Gly Val Ser Leu Pro Asp Tyr Gly Val Ser Trp Ile Arg			
145 150 155 160			
Gln Pro Pro Arg Lys Gly Leu Glu Trp Leu Gly Val Ile Trp Gly Ser			
165 170 175			
Glu Thr Thr Tyr Tyr Asn Ser Ala Leu Lys Ser Arg Leu Thr Ile Ile			
180 185 190			
Lys Asp Asn Ser Lys Ser Gln Val Phe Leu Lys Met Asn Ser Leu Gln			
195 200 205			
Thr Asp Asp Thr Ala Ile Tyr Tyr Cys Ala Lys His Tyr Tyr Tyr Gly			
210 215 220			
Gly Ser Tyr Ala Met Asp Tyr Trp Gly Gln Gly Thr Ser Val Thr Val			
225 230 235 240			
Ser Ser Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys			
245 250 255			
Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys			
260 265 270			
Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val			
275 280 285			
Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr			
290 295 300			
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu			
305 310 315 320			
Gln Tyr Gln Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His			
325 330 335			

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Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 340 345 350
 Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
 355 360 365
 Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu
 370 375 380
 Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 385 390 395 400
 Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 405 410 415
 Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 420 425 430
 Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val
 435 440 445
 Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 450 455 460
 Lys Ser Leu Ser Leu Ser Pro Gly Lys Phe Trp Val Leu Val Val Val
 465 470 475 480
 Gly Gly Val Leu Ala Cys Tyr Ser Leu Leu Val Thr Val Ala Phe Ile
 485 490 495
 Ile Phe Trp Val Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr
 500 505 510
 Met Asn Met Thr Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln
 515 520 525
 Pro Tyr Ala Pro Pro Arg Asp Phe Ala Ala Tyr Arg Ser Glu Arg Val
 530 535 540
 Gln Pro Leu Glu Glu Asn Val Gly Asn Ala Ala Arg Pro Arg Phe Glu
 545 550 555 560
 Arg Asn Lys Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr
 565 570 575
 Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg
 580 585 590
 Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met
 595 600 605
 Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu
 610 615 620
 Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys
 625 630 635 640
 Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu
 645 650 655
 Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu
 660 665 670
 Pro Pro Arg
 675

<210> SEQ ID NO 36

<211> LENGTH: 879

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: CD19 CAR-MU-28 with T2A and IL-15

<400> SEQUENCE: 36

Met 1	Ala	Leu	Pro	Val 5	Thr	Ala	Leu	Leu	Leu 10	Pro	Leu	Ala	Leu	Leu 15	
His	Ala	Ala	Arg 20	Pro	Asp	Ile	Gln	Met 25	Thr	Gln	Thr	Thr	Ser 30	Ser	Leu
Ser	Ala	Ser	Leu 35	Gly	Asp	Arg	Val 40	Thr	Ile	Ser	Cys 45	Arg	Ala	Ser	Gln
Asp	Ile 50	Ser	Lys	Tyr	Leu	Asn 55	Trp	Tyr	Gln	Gln	Lys 60	Pro	Asp	Gly	Thr
Val 65	Lys	Leu	Leu	Ile	Tyr 70	His	Thr	Ser	Arg	Leu 75	His	Ser	Gly	Val	Pro 80
Ser	Arg	Phe	Ser	Gly 85	Ser	Gly	Ser	Gly	Thr 90	Asp	Tyr	Ser	Leu	Thr 95	Ile
Ser	Asn	Leu	Glu 100	Gln	Glu	Asp	Ile	Ala 105	Thr	Tyr	Phe	Cys 110	Gln	Gln	Gly
Asn	Thr 115	Leu	Pro	Tyr	Thr	Phe	Gly 120	Gly	Gly	Thr	Lys	Leu 125	Glu	Ile	Thr
Gly 130	Gly	Gly	Gly	Ser	Gly	Gly 135	Gly	Gly	Ser	Gly	Gly 140	Gly	Gly	Ser	Glu
Val 145	Lys	Leu	Gln	Glu 150	Ser	Gly	Pro	Gly	Leu	Val 155	Ala	Pro	Ser	Gln	Ser 160
Leu	Ser	Val	Thr	Cys 165	Thr	Val	Ser	Gly	Val 170	Ser	Leu	Pro	Asp	Tyr 175	Gly
Val	Ser	Trp	Ile 180	Arg	Gln	Pro	Pro	Arg 185	Lys	Gly	Leu	Glu	Trp 190	Leu	Gly
Val 195	Ile	Trp	Gly	Ser	Glu	Thr	Thr 200	Tyr	Tyr	Asn	Ser	Ala 205	Leu	Lys	Ser
Arg 210	Leu	Thr	Ile	Ile	Lys	Asp 215	Asn	Ser	Lys	Ser	Gln	Val	Phe	Leu	Lys
Met 225	Asn	Ser	Leu	Gln	Thr 230	Asp	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Lys 240
His	Tyr	Tyr	Tyr	Gly 245	Gly	Ser	Tyr	Ala	Met 250	Asp	Tyr	Trp	Gly	Gln	Gly 255
Thr	Ser	Val	Thr 260	Val	Ser	Ser	Glu	Pro 265	Lys	Ser	Cys	Asp	Lys	Thr	His
Thr	Cys 275	Pro	Pro	Cys	Pro	Ala	Pro 280	Pro	Val	Ala	Gly	Pro 285	Ser	Val	Phe
Leu 290	Phe	Pro	Pro	Lys	Pro	Lys 295	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro
Glu 305	Val	Thr	Cys	Val 310	Val	Val	Asp	Val	Ser	His 315	Glu	Asp	Pro	Glu	Val 320
Lys	Phe	Asn	Trp	Tyr 325	Val	Asp	Gly	Val	Glu 330	Val	His	Asn	Ala	Lys 335	Thr
Lys	Pro	Arg	Glu 340	Glu	Gln	Tyr	Gln	Ser 345	Thr	Tyr	Arg	Val	Val	Ser	Val
Leu 355	Thr	Val	Leu	His 360	Gln	Asp	Trp	Leu	Asn	Gly	Lys 365	Glu	Tyr	Lys	Cys
Lys 370	Val	Ser	Asn	Lys 375	Ala	Leu	Pro	Ala	Pro	Ile	Glu 380	Lys	Thr	Ile	Ser
Lys 385	Ala	Lys	Gly	Gln 390	Pro	Arg	Glu	Pro	Gln	Val	Tyr 395	Thr	Leu	Pro	Pro 400

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Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	405	410	415	
Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	420	425	430	
Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	435	440	445	
Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	450	455	460	
Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	465	470	475	480
Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Phe	Trp	485	490	495	
Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	500	505	510	
Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	515	520	525	
Leu	His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	530	535	540	
Arg	Lys	His	Tyr	Gln	Pro	Tyr	Ala	Pro	Pro	Arg	Asp	Phe	Ala	Ala	Tyr	545	550	555	560
Arg	Ser	Glu	Arg	Val	Gln	Pro	Leu	Glu	Glu	Asn	Val	Gly	Asn	Ala	Ala	565	570	575	
Arg	Pro	Arg	Phe	Glu	Arg	Asn	Lys	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	580	585	590	
Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	595	600	605	
Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	610	615	620	
Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	625	630	635	640
Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	645	650	655	
Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	660	665	670	
Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	675	680	685	
His	Met	Gln	Ala	Leu	Pro	Pro	Arg	Gly	Ser	Gly	Glu	Gly	Arg	Gly	Ser	690	695	700	
Leu	Leu	Thr	Cys	Gly	Asp	Val	Glu	Glu	Asn	Pro	Gly	Pro	Met	Arg	Ile	705	710	715	720
Ser	Lys	Pro	His	Leu	Arg	Ser	Ile	Ser	Ile	Gln	Cys	Tyr	Leu	Cys	Leu	725	730	735	
Leu	Leu	Asn	Ser	His	Phe	Leu	Thr	Glu	Ala	Gly	Ile	His	Val	Phe	Ile	740	745	750	
Leu	Gly	Cys	Phe	Ser	Ala	Gly	Leu	Pro	Lys	Thr	Glu	Ala	Asn	Trp	Val	755	760	765	
Asn	Val	Ile	Ser	Asp	Leu	Lys	Lys	Ile	Glu	Asp	Leu	Ile	Gln	Ser	Met	770	775	780	
His	Ile	Asp	Ala	Thr	Leu	Tyr	Thr	Glu	Ser	Asp	Val	His	Pro	Ser	Cys	785	790	795	800
Lys	Val	Thr	Ala	Met	Lys	Cys	Phe	Leu	Leu	Glu	Leu	Gln	Val	Ile	Ser				

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	805		810		815										
Leu	Glu	Ser	Gly	Asp	Ala	Ser	Ile	His	Asp	Thr	Val	Glu	Asn	Leu	Ile
			820					825					830		
Ile	Leu	Ala	Asn	Asn	Ser	Leu	Ser	Ser	Asn	Gly	Asn	Val	Thr	Glu	Ser
		835					840					845			
Gly	Cys	Lys	Glu	Cys	Glu	Glu	Leu	Glu	Glu	Lys	Asn	Ile	Lys	Glu	Phe
	850					855					860				
Leu	Gln	Ser	Phe	Val	His	Ile	Val	Gln	Met	Phe	Ile	Asn	Thr	Ser	
865				870						875					

<210> SEQ ID NO 37
 <211> LENGTH: 242
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: anti-CD19 scFv
 <400> SEQUENCE: 37

Asp	Ile	Gln	Met	Thr	Gln	Thr	Thr	Ser	Ser	Leu	Ser	Ala	Ser	Leu	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Ser	Cys	Arg	Ala	Ser	Gln	Asp	Ile	Ser	Lys	Tyr
		20					25						30		
Leu	Asn	Trp	Tyr	Gln	Gln	Lys	Pro	Asp	Gly	Thr	Val	Lys	Leu	Leu	Ile
	35					40						45			
Tyr	His	Thr	Ser	Arg	Leu	His	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55					60				
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Ser	Leu	Thr	Ile	Ser	Asn	Leu	Glu	Gln
65				70					75					80	
Glu	Asp	Ile	Ala	Thr	Tyr	Phe	Cys	Gln	Gln	Gly	Asn	Thr	Leu	Pro	Tyr
		85						90						95	
Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Thr	Gly	Gly	Gly	Gly	Ser
		100						105					110		
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	Lys	Leu	Gln	Glu
	115					120						125			
Ser	Gly	Pro	Gly	Leu	Val	Ala	Pro	Ser	Gln	Ser	Leu	Ser	Val	Thr	Cys
	130					135					140				
Thr	Val	Ser	Gly	Val	Ser	Leu	Pro	Asp	Tyr	Gly	Val	Ser	Trp	Ile	Arg
145				150						155				160	
Gln	Pro	Pro	Arg	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Gly	Ser
		165						170						175	
Glu	Thr	Thr	Tyr	Tyr	Asn	Ser	Ala	Leu	Lys	Ser	Arg	Leu	Thr	Ile	Ile
		180						185					190		
Lys	Asp	Asn	Ser	Lys	Ser	Gln	Val	Phe	Leu	Lys	Met	Asn	Ser	Leu	Gln
	195					200						205			
Thr	Asp	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Lys	His	Tyr	Tyr	Tyr	Gly
	210					215					220				
Gly	Ser	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Ser	Val	Thr	Val
225				230						235				240	

Ser Ser

<210> SEQ ID NO 38
 <211> LENGTH: 675
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: CD19 CAR-MU-28

<400> SEQUENCE: 38

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Asp Ile Gln Met Thr Gln Thr Thr Ser Ser Leu Ser Ala Ser Leu Gly
1      5      10      15
Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Asp Ile Ser Lys Tyr
20     25     30
Leu Asn Trp Tyr Gln Gln Lys Pro Asp Gly Thr Val Lys Leu Leu Ile
35     40     45
Tyr His Thr Ser Arg Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
50     55     60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Thr Ile Ser Asn Leu Glu Gln
65     70     75     80
Glu Asp Ile Ala Thr Tyr Phe Cys Gln Gln Gly Asn Thr Leu Pro Tyr
85     90     95
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Thr Gly Gly Gly Gly Ser
100    105    110
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Glu Val Lys Leu Gln Glu
115    120    125
Ser Gly Pro Gly Leu Val Ala Pro Ser Gln Ser Leu Ser Val Thr Cys
130    135    140
Thr Val Ser Gly Val Ser Leu Pro Asp Tyr Gly Val Ser Trp Ile Arg
145    150    155    160
Gln Pro Pro Arg Lys Gly Leu Glu Trp Leu Gly Val Ile Trp Gly Ser
165    170    175
Glu Thr Thr Tyr Tyr Asn Ser Ala Leu Lys Ser Arg Leu Thr Ile Ile
180    185    190
Lys Asp Asn Ser Lys Ser Gln Val Phe Leu Lys Met Asn Ser Leu Gln
195    200    205
Thr Asp Asp Thr Ala Ile Tyr Tyr Cys Ala Lys His Tyr Tyr Tyr Gly
210    215    220
Gly Ser Tyr Ala Met Asp Tyr Trp Gly Gln Gly Thr Ser Val Thr Val
225    230    235    240
Ser Ser Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
245    250    255
Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
260    265    270
Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
275    280    285
Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr
290    295    300
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
305    310    315    320
Gln Tyr Gln Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
325    330    335
Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
340    345    350
Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
355    360    365
Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu
370    375    380

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Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 385 390 395 400
 Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 405 410 415
 Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 420 425 430
 Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val
 435 440 445
 Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 450 455 460
 Lys Ser Leu Ser Leu Ser Pro Gly Lys Phe Trp Val Leu Val Val Val
 465 470 475 480
 Gly Gly Val Leu Ala Cys Tyr Ser Leu Leu Val Thr Val Ala Phe Ile
 485 490 495
 Ile Phe Trp Val Arg Ser Lys Arg Ser Arg Leu Leu His Ser Asp Tyr
 500 505 510
 Met Asn Met Thr Pro Arg Arg Pro Gly Pro Thr Arg Lys His Tyr Gln
 515 520 525
 Pro Tyr Ala Pro Pro Arg Asp Phe Ala Ala Tyr Arg Ser Glu Arg Val
 530 535 540
 Gln Pro Leu Glu Glu Asn Val Gly Asn Ala Ala Arg Pro Arg Phe Glu
 545 550 555 560
 Arg Asn Lys Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr
 565 570 575
 Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg
 580 585 590
 Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met
 595 600 605
 Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu
 610 615 620
 Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys
 625 630 635 640
 Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu
 645 650 655
 Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu
 660 665 670
 Pro Pro Arg
 675

<210> SEQ ID NO 39
 <211> LENGTH: 653
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Anti-CD19 CAR (CD28-CD3z-IL15)

<400> SEQUENCE: 39

Asp Ile Gln Met Thr Gln Thr Thr Ser Ser Leu Ser Ala Ser Leu Gly
 1 5 10 15
 Asp Arg Val Thr Ile Ser Cys Arg Ala Ser Gln Asp Ile Ser Lys Tyr
 20 25 30
 Leu Asn Trp Tyr Gln Gln Lys Pro Asp Gly Thr Val Lys Leu Leu Ile
 35 40 45

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Tyr	His	Thr	Ser	Arg	Leu	His	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	50	55	60	
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Ser	Leu	Thr	Ile	Ser	Asn	Leu	Glu	Gln	65	70	75	80
Glu	Asp	Ile	Ala	Thr	Tyr	Phe	Cys	Gln	Gln	Gly	Asn	Thr	Leu	Pro	Tyr	85	90	95	
Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Thr	Gly	Gly	Gly	Gly	Ser	100	105	110	
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Glu	Val	Lys	Leu	Gln	Glu	115	120	125	
Ser	Gly	Pro	Gly	Leu	Val	Ala	Pro	Ser	Gln	Ser	Leu	Ser	Val	Thr	Cys	130	135	140	
Thr	Val	Ser	Gly	Val	Ser	Leu	Pro	Asp	Tyr	Gly	Val	Ser	Trp	Ile	Arg	145	150	155	160
Gln	Pro	Pro	Arg	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Ile	Trp	Gly	Ser	165	170	175	
Glu	Thr	Thr	Tyr	Tyr	Asn	Ser	Ala	Leu	Lys	Ser	Arg	Leu	Thr	Ile	Ile	180	185	190	
Lys	Asp	Asn	Ser	Lys	Ser	Gln	Val	Phe	Leu	Lys	Met	Asn	Ser	Leu	Gln	195	200	205	
Thr	Asp	Asp	Thr	Ala	Ile	Tyr	Tyr	Cys	Ala	Lys	His	Tyr	Tyr	Tyr	Gly	210	215	220	
Gly	Ser	Tyr	Ala	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Ser	Val	Thr	Val	225	230	235	240
Ser	Ser	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	245	250	255	
Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	260	265	270	
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	275	280	285	
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	290	295	300	
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	305	310	315	320
Gln	Tyr	Gln	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	325	330	335	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	340	345	350	
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	355	360	365	
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	370	375	380	
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	385	390	395	400
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	405	410	415	

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Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
			420					425					430		
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
		435					440					445			
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
	450					455					460				
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Phe	Trp	Val	Leu	Val	Val	Val
465					470					475				480	
Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr	Val	Ala	Phe	Ile
			485						490					495	
Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu	His	Ser	Asp	Tyr
			500					505					510		
Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg	Lys	His	Tyr	Gln
		515					520					525			
Pro	Tyr	Ala	Pro	Pro	Arg	Asp	Phe	Ala	Ala	Tyr	Arg	Ser	Arg	Val	Lys
		530				535					540				
Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln
545					550					555					560
Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu
			565						570					575	
Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg
			580					585					590		
Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met
		595					600					605			
Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly
	610					615					620				
Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp
625					630					635					640
Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala	Leu	Pro	Pro	Arg			
				645					650						

1. A polynucleotide comprising:
 - a) a nucleic acid encoding a chimeric antigen receptor (CAR) comprising an intracellular signaling region comprising an OX40L intracellular signaling domain; and
 - b) a nucleic acid encoding an IL-15.
2. (canceled)
3. The polynucleotide of claim 1, wherein the OX40L intracellular signaling domain comprises an amino acid sequence set forth in SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.
4. The polynucleotide of claim 1, wherein the OX40L intracellular signaling domain comprises an amino acid sequence having at least 99% identity to SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.
- 5.-13. (canceled)
14. The polynucleotide of claim 1, wherein the IL-15 comprises the amino acid sequence set forth in SEQ ID NO: 22.
15. The polynucleotide of claim 1, wherein the IL-15 comprises an amino acid sequence having at least 99% identity to SEQ ID NO: 22.
16. The polynucleotide of claim 14, wherein the IL-15 is encoded by a nucleic acid comprising SEQ ID NO: 17 or SEQ ID NO: 18.
17. The polynucleotide of claim 1, wherein the polynucleotide encodes a polypeptide comprising the CAR and the IL-15.
18. The polynucleotide of claim 1, further comprising a nucleic acid encoding a self-cleaving peptide.
19. The polynucleotide of claim 18, wherein the CAR is joined to the IL-15 by the self-cleaving peptide.
20. The polynucleotide of claim 19, wherein the self-cleaving peptide is a T2A self-cleaving peptide.
21. (canceled)
22. The polynucleotide of claim 19, wherein the T2A self-cleaving peptide is capable of inducing ribosomal skipping between the CAR and the IL-15.
- 23.-38. (canceled)
39. A vector comprising the polynucleotide of claim 1.
- 40.-41. (canceled)
42. A cell comprising the polynucleotide of claim 1.
43. A cell expressing the chimeric antigen receptor and the IL-15 encoded by the polynucleotide of claim 1.
44. The cell of claim 42, wherein the cell is a lymphocyte.

45. The cell of claim **44**, wherein the lymphocyte is a natural killer (NK) cell.

46.-51. (canceled)

52. A population of cells comprising a plurality of the cells of claim **42**.

53.-54. (canceled)

55. A pharmaceutical composition comprising the population of cells of claim **52**.

56.-72. (canceled)

73. A frozen vial comprising the composition of claim **55**.

74. A method of treatment comprising administering the cell of claim **42** to a subject having a disease or condition.

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